Process Journal

By Nefeli Lemessiou

Unit 1: Locating Practise

Seven Objects

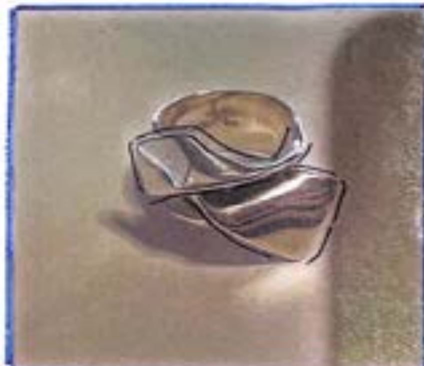
A THING/OBJECT THAT SHOULDN'T EXIST.

CUSHION TOILET SEAT

- Unhygienic → does not get cleaned easily and shouldn't be placed on a toilet
- Stains very easily → not inviting for users to want to sit on and needs frequent replacements.

RING

- Emotional connection → given to me by my mother that she got it from a patient of her.
- Handmade → not a lot of similar pieces exist + its value its impossible to replace.
- Replica of a historical artefact from a museum I like.



THE MOST IMPORTANT OBJECT YOU OWN.

AN OBJECT THAT MAKES YOU ANGRY.

DISPOSABLE WOODEN SPOONS

- USELESS: when placed inside a hot liquid, they flatten, so they cannot be used anymore.
- TEXTURE: their texture is very bad which is something very important for an item that goes inside the mouth

THE MOST BEAUTIFUL THING/OBJECT IN THE WORLD.

HYBRID BY SELETTI

- WEIRD → two different plates being merged is unusual and there is a beauty in the unusual, different.
- KINTSUGI: putting back together broken items signifying that just because it broke, it doesn't mean its not beautiful anymore.

DUREX PLEASUREMAX

- Function: protect so people can have safe-sex
- Texture: have a specific texture that improve the sex experience - allow users to feel more, giving them an experience similar to not using a condom.

THE MOST USEFUL THING IN THE WORLD THAT SERVES A SPECIFIC FUNCTION.

ONE THING/OBJECT THAT INSPIRES YOU.

FREESTYLE LIBRE CGM DEVICE

- IMPACT: changed lives for people with diabetes - improved their experiences
- DESIGN: very discreet - fits nicely with people's skin, does not cause discomfort.

EVIL-EYE BRACELET

- CULTURAL CONNECTION: a Greek jewelry that signifies our beliefs + traditions.
- ABSURDITY → thinking that a bracelet will protect you from bad energies.
- STYLE: I always wore bracelets since I was very young.

IF YOU WERE AN OBJECT WHICH OBJECT WOULD BEST ENCAPSULATE YOU?

Peer Review: Collecting Keywords

Keywords:

personal connection.

Function.

complex patterns.

health care

cultural connection.

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health

experience

emotion

practical

women

From Clare:

- everyday
- personalisation
- Functionality
- Design
- Portability

From Annie:

- Emotion
- Practical
- Future
- Experimental Design
- Meaningful

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healthcare

accessories

cutural communication

eco-friendly tableware

personal hygiene products

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function

emotion

appearance

culture

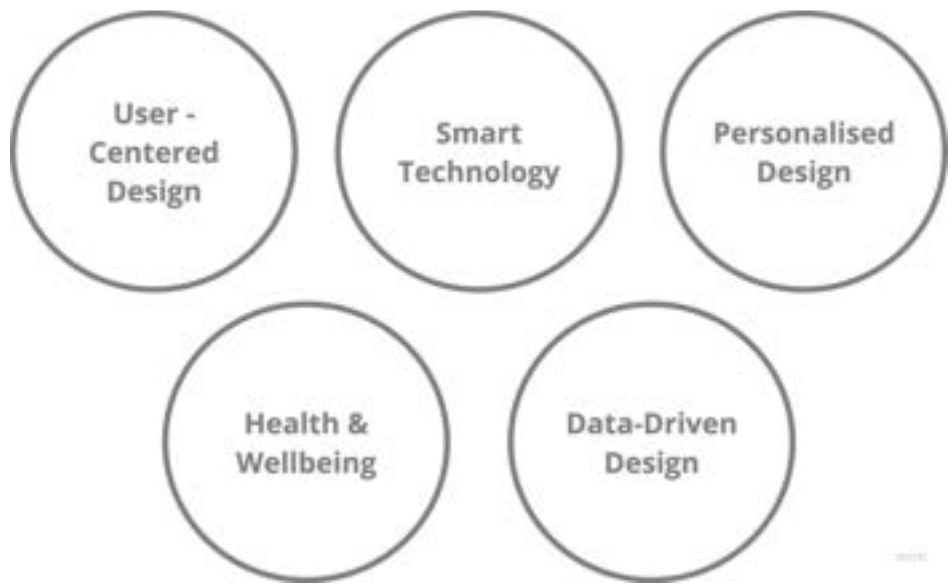
material

Initial Keywords

Based on the words I collected from the peer review and my personal interests; I came up with some initial keywords that I believe could describe my design practise and could lead me towards identifying design opportunities worthy of exploration.

Reflection

I categorised my keywords into two categories: those that describe my practise and those that could serve as design opportunities. Most of them fell into the first category and by the end of the week I had the five keywords describing my principles.

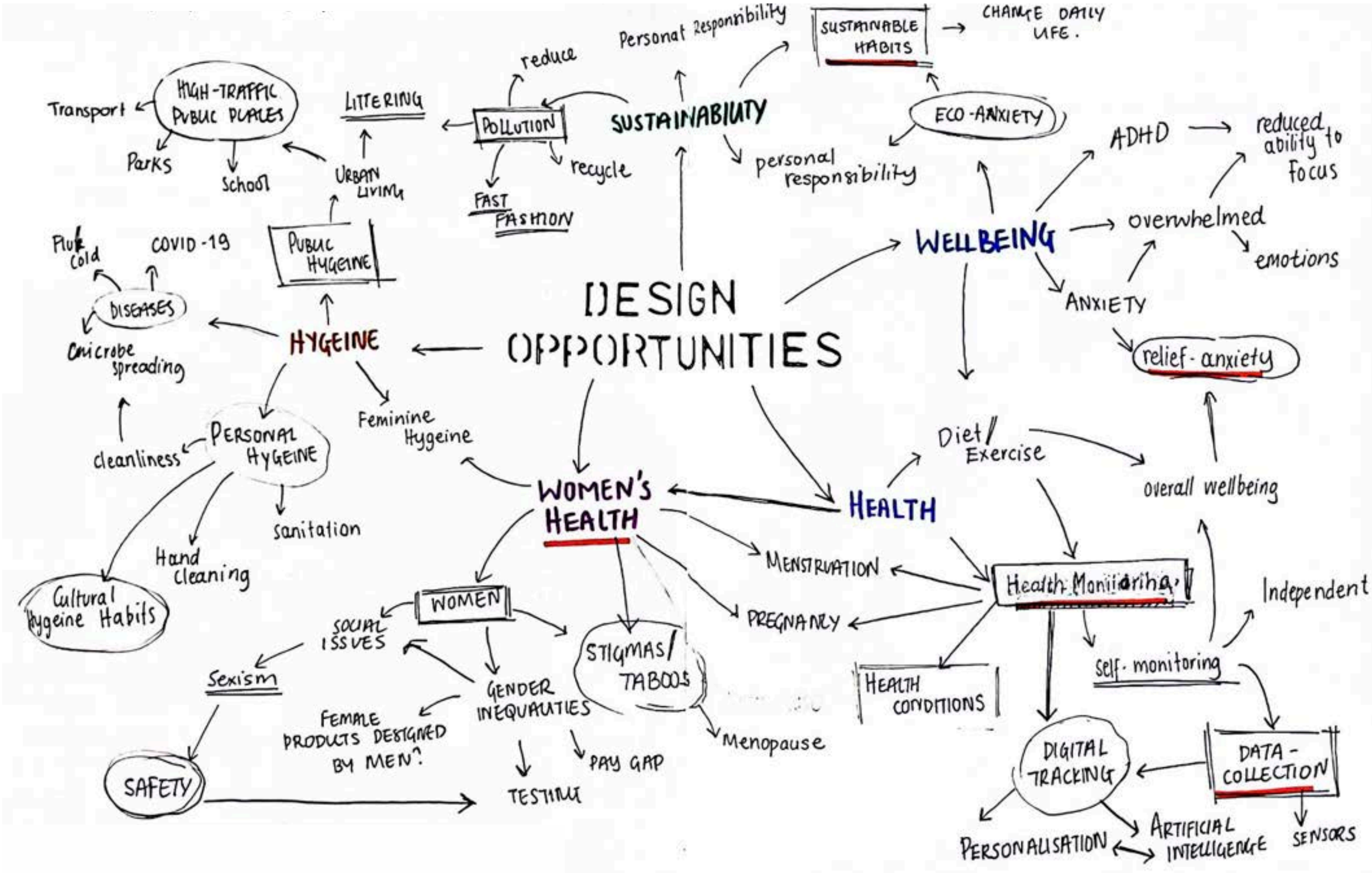


Next week

- Use Keywords to identify design opportunities
- Start doing research

Brainstorming

After coming up with some initial keywords, I used them as the starting point for exploration before settling on five keywords that could develop into design opportunities for unit 2.



Opportunities & Research:

The mind map enables me to produce five more specific keywords that I could do some research around and test their potential as design opportunities. I decided on these five categories based on my interests and goals as a designer.

Self-Health Monitoring:

Self-health monitoring

Gaps in my knowledge

What you need to know and why?

- Identify a **specific medical condition** such as gastrointestinal disorders and fully understand the conditions - its complications, what is causing it and how it should be monitored. *It could also be a more general health tracker that tracks diet, exercise etc.*
- **Engagement**: Understanding how users interact with health-monitoring tools and what keeps them engaged (long-term use is key)
- **User Struggles**: Understanding users health experiences - what makes their conditions challenging and what exactly they need to monitor to improve their experiences
- **User Specific Needs:** Understanding what kind of feedback users needs and in what ways it can be achieved -> how the experience can be personalised to each user to meet specific needs -> What questions patients have around their conditions and what kind of assistance do they require?
- **Mental Health**: Understand how to design devices that account for users mental health in the sense that make the monitoring experience as stress-free as possible (avoid making a device that creates anxiety to the user over his condition)
- **Health data and privacy**: With the rise of wearables and apps, it's important to understand how to protect sensitive health data and design for transparency.
- **Integration**: how can self-tracking can become seamlessly integrated into the user's everyday life- and how can habits be created

How are you going to find the unknown?

- **Interviews with health professionals:** Interview doctors and other experts in the condition I identified and ask questions to get a better understanding of the condition and how it should be monitored
- **Interviews/Focus Groups with people affected from the condition:** Discuss the experiences of these people. How is their everyday lives impacted and how they can be improved through self monitoring. Understand their specific needs.
- **Design Workshops:** Ask people to design their own solutions to the issue, how their ideal devices would be and what they would monitor.
- **Probe Kits:** Ask people to keep daily diaries of their experiences, in what points of the day they are most affected. Identify their struggles throughout the day more.
- **Research on GDPR Regulations in the UK:** Look into the data collection laws to ensure data protection and privacy and understand what kind of data i allowed to collect
- **Case Studies**: Look into existing products that tackle similar issues and study users interacting with them. Explore their limitations and how improves products can be developed

Inspiring work in the field

- **Dexcom G7 Continuous Glucose Monitoring Device:** A continuous glucose monitor that tracks blood sugar levels in real time via a sensor, helping people with diabetes manage their glucose.
- **FoodMarble AIRE**: A portable breath analyser that tracks digestion and offers feedback to help users manage IBS symptoms by understanding food reactions.
- **CaraCare**: A health app for managing digestive disorders like IBS, offering symptom tracking and personalised dietary plans.
- **Apple Health:** A health app that collects data from fitness apps and devices to provide a dashboard for tracking things like sleep, diet etc.

Who is being impacted by it?

- People that are **suffering from a specific chronic condition**- usually it is a relatively specific demographic *(people struggling with gastrointestinal conditions such as IBS, Crohn's etc)*
- Could also be athletes, elderly population or people interested in wellness and health tracking

Women's Health:

Women's Health

Gaps in my knowledge

What you need to know and why?

- How can female experiences around their health can be improved?
- **Women's Experiences:** Understanding difficulties women face around menstruation, menopause and pregnancy and how can these be addressed through design?
- **Engagement**: Identify key characteristics that make women engage with a product - what kind of products women like to use and why?
- **Stigma and taboos**: Understanding cultural and societal barriers that affect women's health and well-being, and how they create limitations when designing for women's health
- **Inclusivity - Woman-centered design**: get a deep understanding of the specific physiological needs of a woman and be able to design for them. Understand women's diverse experiences to make sure that the design outcome is inclusive
- **Women's Needs:** What kind of feedback women need? -> what are questions that women have around their health and how can they be addressed? - What kind of assistance women want around women's health
- **Education and Awareness:** Identify what kind of knowledge women are lacking around their health and how that can be tackled through design

How are you going to find the unknown?

- **Interviews with health professionals:** Interview gynaecologists and other experts to understand the technical aspects of female health, including how the reproductive system work and how we can safely design around it.
- **Inventories:** Ask women to bring items that they think are successfully design for women and items that are not to identify aspects of good woman-centered design
- **Menstruation/ Pregnancy/ Menopause Diaries:** Ask women across different stages to record struggles they face and questions they have around their feminine health in order to identify specific areas to design for.
- **Redesigning Workshops**: Ask women to redesign existing products that respond to women's health in order to identify these products limitations and explore how they can be done better
- **Visit woman's wellness clinics:** Opportunity to meet with professionals and explore first hand struggles women face. Record experiences and find my research audience

Inspiring work in the field

- **Flo Health App:** A period and fertility tracking app that helps users monitor their menstrual cycles, ovulation, and reproductive health, offering personalised insights and health tips.
- **Sukha by Matter**: A home HPV screening kit designed to empower women to easily and safely test for HPV, aiming to make cervical cancer prevention more accessible.
- **Looncup**: A smart menstrual cup that tracks menstrual flow and provides real-time data on volume and cycles through an app, helping users monitor their menstrual health.
- **Momma's Kit by Leadoff Studio**: A postpartum care kit designed to support new mothers with essential items for recovery and self-care after childbirth.

Who is being impacted by it?

- Women/ Menstruators of all ages and across all stages of life, including menstruation, pregnancy, menopause, and beyond.
- Additionally, individuals seeking gender-affirming healthcare.

Sustainable Habits:

Sustainable Habits

Gaps in my knowledge

What you need to know and why?

- **Habit Creation & Engagement:** Understanding how habits can be naturally created and maintained. Understand the theory behind in order to be able to design that are engaging and create habit loops that are not easily broken.
- **Environmental Concerns and Flexibility :** Understanding how people might have different environmental concern or how they might have limitations on what they can do. Explore how can habit-making products can provide flexibility and personalisation for creating a specific habit that works for the user
- **Environmental Impact:** Understand why it is important to create sustainable habits -> explore climate change and environmental damage - what are areas of everyday life that cause the most pollution/impact? - identify human habits that are unsustainable and need changing.
- **User Motivations:** Understand how to motivate users to adapt specific sustainable habits -> Different users might need different incentives - identify how we can motivate most of the users
- **Limitations:** what is usually stopping users for maintaining habits and how it can be tackled?
- **Integration:** How can these habits be integrated into user's everyday life seamlessly -> How do we account for users with different cultural backgrounds that have different habits
-

How are you going to find the unknown?

- **Probe Kit:** A probe kit that asks users to document sustainable and unsustainable they make throughout a week, and then analysing their reflections and motivations.
- **Workshop:** Ask users to map out their everyday lives and try to identify habits that they could integrate in their routines that are sustainable
- **Focus Groups:** Discuss what peoples motivations and concerns are when it comes to the environment -> explore what people are willing to change and how aware are of the result that their current habits have on the environment.
- **Journey Maps:** Give people specific tasks - ask them to do a sustainable practise for a week- and make a journey map of their experience. Explore how they integrate it their routine
- **Interviews:** Talk with experts in sustainable behaviour change and on habit theory to get a better understanding and ideas.

Inspiring work in the field

- **Patagonia:** A clothing brand that encourage sustainable fashion habits by focusing on the repair and reuse of clothing instead of replacement.
- **Too Good To Go:** A mobile app that connects users with local restaurants and stores to buy leftover food at a reduced price, helping to reduce food waste.
- **The Loop by TerraCycle:** A system for purchasing products in reusable containers that are returned, cleaned, and refilled to minimise single-use packaging.
- **TetraBIN:** An interactive recycling bin that gamifies waste disposal, encouraging proper recycling through engaging feedback and rewards.
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Who is being impacted by it?

- Everyone including consumers, environmental activists/advocates, and future generations that will be heavily impacted by climate change and resource depletion if habits are not changed.

Anxiety Relief:

WEEK 3

Anxiety Relief

Gaps in my knowledge

What you need to know and why?

- **Types of Anxiety:** Understanding the different types of anxiety -> not everyone's experience with anxiety is the same and there are different categories for anxiety that need to be examined separately.
- **Anxiety Triggers:** Understanding what triggers anxiety for people- this will be different for each person but possible trends can be identified- what environments, thoughts or situation can trigger anxiety
- **Vulnerability:** Explore which people are more vulnerable to anxiety - age groups, stage in life, employment status could affect people's vulnerability- if we identify this maybe we can prevent it?
- **Coping Mechanisms:** What ways do people currently use to relieve anxiety? what do they prefer? - could be breathing exercises, yoga -> how this can be integrated in products or be enhanced?
- **User Specific Needs:** Understanding how each user's specific experience with anxiety can be unique and how a personalised design response can be achieved. -> what kind of feedback/support each user might need?
- **Integration:** how can anxiety relief products/services can seamlessly integrated into the user's everyday life- how to ensure that the process of interacting with a device wont cause more anxiety for the user or make them feel stigmatised.

How are you going to find the unknown?

- **Interviews with experts:** Talk with psychologists/psychiatrists to better understand the different types of anxiety and how they can be safely tackled through design- explore coping mechanisms
- **Interviews with people with anxiety** : Talk with people with anxiety in order to understand them and allow them to share their personal experiences in a private and safe space
- **Anxiety Diaries:** Ask people to record their daily experiences with anxiety- specifically to record what triggers them, what environment they are in and what they are doing when the trigger comes and what coping mechanism they use during their anxiety attacks.
- **Journey Maps:** Ask people to map down how they experience an anxiety attack and how it feels for them specifically.
- **Inventories:** Ask people to bring objects that help them relax and are comforting and objects that make them anxious.
- **Co-design Workshop:** Ask people to design comforting items that they feel like they would help them with anxiety

Inspiring work in the field

- **Calm App:** A mental wellness app thats offers guided meditations, sleep stories, breathing exercises, and relaxing music to reduce anxiety and stress.
- **Breathe:** An Apple Watch feature that guides users through short breathing exercises, using haptic feedback to encourage deep breathing and relaxation.
- **Fidget Cube by Antsy Labs:** A small, handheld cube designed with multiple tactile surfaces (buttons, switches, dials) to help users relieve stress and anxiety through fidgeting.
- **TrustMe Pebbles by Clir (Rob Middleton):** A tactile, palm-sized device designed to help soothe anxiety and stress through smooth, weighted stones that provide a calming sensory experience when handled.

Who is being impacted by it?

- People that anxiety disorders or people that experience acute stress or anxiety
- Everyday people- students, working professionals that might experience heavy stress for a period of time. People going through stressful periods.

Custom/Data-Driven Design:

Custom/Data-Driven Design

Gaps in my knowledge

What you need to know and why?

- Identify a specific area where using data collection to provide feedback would be useful-> It is a good method for making personal assistants for health monitoring or for planning
- **Data Collection:** Understanding how to collect data- would that be a sensor, will the user input the data themselves, will it be a wearable? What kind of data I want to collect and what is the appropriate method of collecting them.
- **Data Security and privacy**: With this kind of design projects, it's important to understand how to protect sensitive data and design for transparency-> Data laws should be explored
- **Artificial Intelligence:** How AI can be used to analyse and interpret data -> Also how can AI be utilised to give a personalised approach and real-time responses - AI could make users experience interactive
- **Customisation/Personalisation:** How these two are different and how both of them can be incorporated into a designed products-> Which one users appreciate more and how they can be applied in the area I am designing for.
- **Feedback:** what kind of feedback/output should the device give based on the data its collecting and how will it be doing it? Could it be a voice assistant or a chat assistant?
- **Ethical Considerations:** Understanding the ethical implications of collecting using personal data, ensuring designs do not manipulate users and understand the implications of the device giving feedback that may negatively impact them.

How are you going to find the unknown?

- **Data Diaries:** Ask people to keep a diary of their daily use of a smart home device like the nest thermostat, ask them to note down if their preferences/needs are met by the product's data-driven actions and what they would change.
- **User Studies:** Tracking users' interaction with a personalised app over a period of time to explore how well the app's recommendations will fit the users changing habits.
- **Focus Groups:** Talk with potential users to understand what they want to be tracking - their preferred methods of collecting data and what kind of feedback they want to receive based on them.
- **Probe Kit:** Ask people to document their interaction with personalised recommendations from a specific online platform such as a clothing site, and record how they feel about the suggestions they get and if they would like something different.

Inspiring work in the field

- **Spotify**: A music streaming service that offers personalized playlists and recommendations based on users' listening habits, preferences, and data-driven algorithms.
- **Tonal**: A smart home gym system that with the help AI and data-driven technology to offer personalised strength training workouts, adjusting resistance based on real time user performance .
- **Amazon Alexa**: A voice virtual assistant that utilised user data to perform tasks, control smart home devices, and provide personalised recommendations.
- **Fitbit**: A wearable fitness tracker that monitors activity, heart rate, sleep, and other health metrics, providing personalised insights and recommendations to improve well-being.

Who is being impacted by it?

- Users seeking personalised/customised 'smart' products, could be around fitness, health, or fashion.
- People that are interested in utilising AI and technology to make their everyday lives easier and people that like smart/digital assistants.

Not really a lot of design methods. Maybe it is not a great keyword / could be changed to something else?

However data driven design can work in the combination with one of the other categories

More of a way to respond to a design are rather than an area itself

Tutorial Feedback:

- Does sustainable habits need to mean eco-friendly habits? ~ could it be the idea of creating a habit and sustaining it/maintaining it?

  Would work well with health monitoring idea that requires users to have sustainable tracking habits

- Is data-driven design a design opportunity? What exactly is the design problem?

  More of a method that can be used to respond to other design opportunities.

Finding the 5th opportunity:

Observation 1: Lack of engagement/reduced attention spam (especially amongst younger people) -> possibly a result of social media

Observation 2: Unhygienic Public Environments (especially in high-traffic areas) -> caused by overpopulation, apathy etc.

Final Design Opportunities:

Final Opportunities

Following the tutorials and peer reviews, I realised that my keywords need to be a little refined in order to become **Design Problems** that I can explore.

Self-Health Monitoring/Management

→ Lack of integrated, engaging products on the market for health management that cater to unique needs.
→ Most are generalised and lack personalisation → could investigate chronic diseases – **Gastrointestinal conditions**

Women's Health

→ Not many solution that educate women around their bodies – mostly tracking apps that lack personalisation and engaging features
→ Taboos around the topic and gender inequalities still make it hard to design for women's health efficiently
→ Could explore menopause, pregnancy and menstruation

Anxiety Relief

→ Most people nowadays struggle with anxiety → Popular topic that many designers want to tackle but none of the solutions seem to work yet.
→ It is within the realm of wellbeing, also I struggle with anxiety myself.

Sustainable Habits

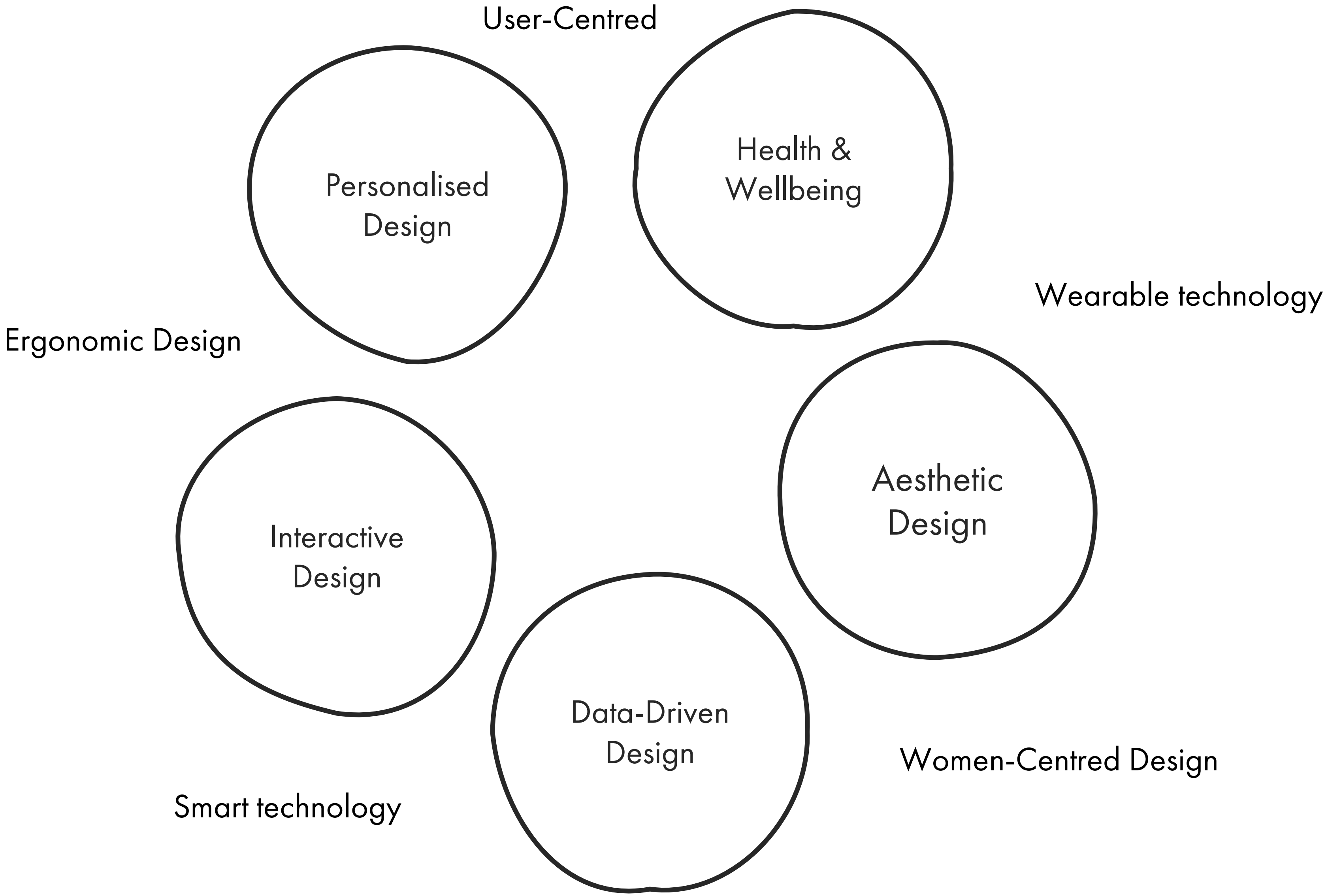
→ Sustainability is also popular → people mostly focus on making sustainable products and only some focus on encouraging sustainable habits
→ Involves everyone and focuses on changing people's lifestyles by altering their habits.

Public Hygiene

→ Following COVID-19 is clear that public hygiene is important → Littering, Dirty surfaces and insufficient hygiene can lead to the spread of diseases.
→ Current solutions are insufficient as they often fall short on extremely high–traffic areas.

Final Design Principles

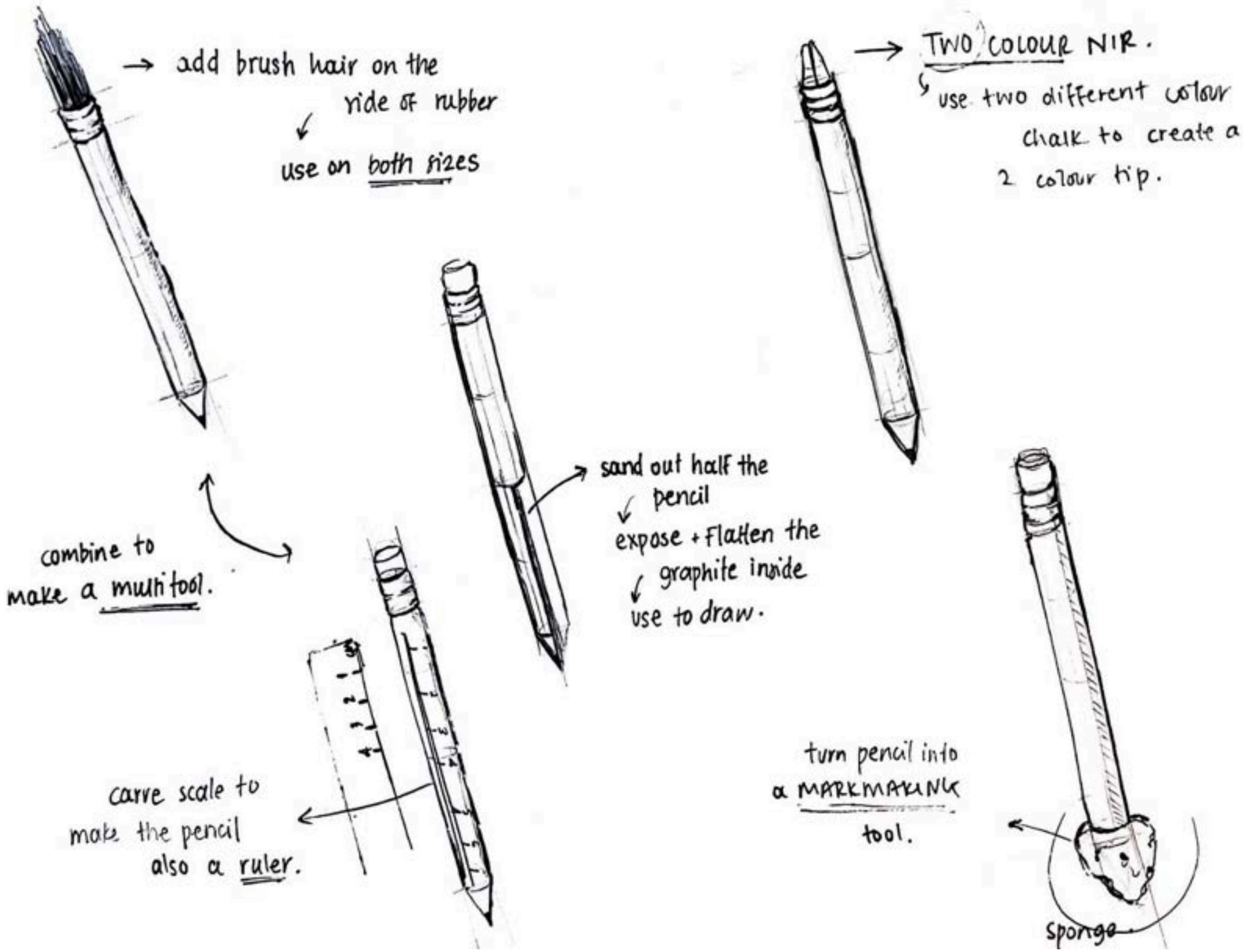
At this point I decided to settle on my final keywords that describe my design practice



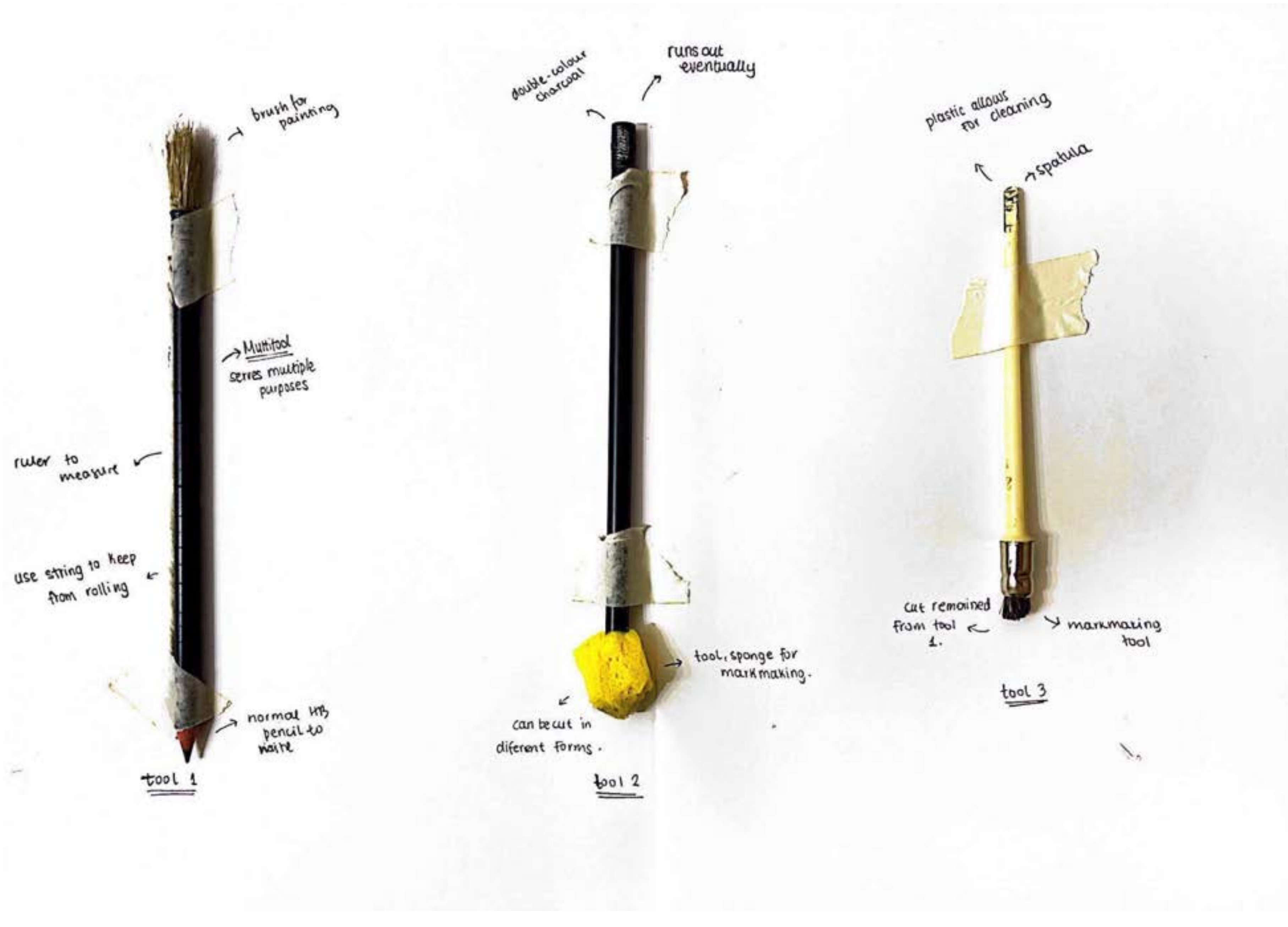
Unit 2: Experimental Design

Well-Prepared Pen

In this brief we were asked to modify one of our primary tools – the pen and make it produce a different output. Then we were asked to create work to demonstrate the new function of our made tool. My tool of choice was a **pencil**.



Modified Tools



Results & Reflection:

The approach I took in repurposing the pencil focused primarily on turning it into a multitool that could provide various ways of writing or marking. Most of my alterations—except for turning it into a ruler—involved transforming the pencil into different types of mark-making tools that produced a variety of results. Although the outcomes were interesting, I realized I could have pushed my ideas further. Instead of just making another type of marking tool, I could have turned the pencil into a more unconventional and unexpected tool that serves a completely different purpose.

Out of Context

For this brief, we were asked to explore alternative functions of one chosen tool by placing it in different contexts. We had to give our tool a new name and place it in a new situation to produce at least 3 photographs of the tool with 3 different names and situations. My tool of choice was a **Sharpie.**

WEEK 6

SharPin

For my first experiment with repurposing my tool, I chose to transform a Sharpie into a hair clip. This idea was inspired by the marker's cap clip, which is typically used to attach it to notebooks or clothing. By leveraging this feature in an unconventional way, I discovered that the cap could also serve as a functional hair clip to keep hair out of one's face. This adaptation prompted discussions about the types of people who might use a marker as a hair clip and what it might communicate about their identity. For instance, it could appeal to those who identify with alternative styles or are artists, reflecting a creative and practical use of everyday items.

WEEK 6

StabPen

For this experiment, I adopted a more playful and unconventional approach with my tool. I decided to repurpose the Sharpie as a stabbing tool, inspired by its form, which could allow it to be used as a sort of piercing device. In real-life scenarios, its stabbing functionality would likely be more practical for tasks such as opening parcels or creating holes in cardboard. Its lack of a blade means that it wouldn't be lethal for humans but could still cause harm. This unexpected take on a Sharpie sparked interesting conversations about how everyday objects can be repurposed as weapons in situations of danger, highlighting the adaptability of mundane tools.

WEEK 6

Vandalism Tool

For the third experiment, I explored not the repurposing of the pen itself but the context in which it is used. Sharpies are often associated with graffiti and are widely used to write messages in public spaces such as bathrooms and walls. Their permanence makes them ideal for such expressions. With this in mind, I depicted the Sharpie as a tool for vandalism, thereby igniting discussions about the context of a tool's use and how it can alter the tool's perceived meaning. This approach challenges perceptions, illustrating how some view the Sharpie as an artist's pen, others as an all-purpose pen, and some as a tool for vandalism.

WEEK 6

Sharpie pegs

For my final experiment, I utilised the cap of the Sharpie in a practical way by repurposing it as a tool for hanging clothes to dry. This approach deviates from the more conceptual uses explored in previous experiments. Instead, it focuses on a straightforward, functional adaptation of the Sharpie. By examining the physical properties, or affordances, of the Sharpie, I identified an alternative practical use: hanging clothes when traditional tools are unavailable. This experiment represents an inventive but more realistic approach to repurposing a Sharpie, showcasing the versatility of seemingly single-purpose items.

Tool Meets Matter Meets Tool

For the TMMMT brief, we were asked to conduct a lateral exploration of the capacity of our chosen tool through inventive 3D experiments. We were required to investigate the visual and material language that emerges when our tool interacts with matter—whether a material yields to the tool or resists it. Our task involved exploring how our tool can be used to manipulate matter by incorporating materials and techniques beyond its original intended purpose. At the same time, we had to consider the principles underlying the function of our tool and the potential within its morphology, context, or associative envelope. For this brief, we were required to produce 20 tangible examples of material manipulation.

Reflection:

I tried to explore the different aspects of my tool, starting with its form, affordances, properties, conceptual meanings, and finally, its functionality. I began by using it with clay to make anaglyphs and dipped it in paint to create stamps. Some of my favourite results came when I cut the marker open to access the ink inside. I experimented by dipping the ink in water and acetone and even exposing it to fire to observe how it would react. I then used that ink to create various mark-making patterns. Additionally, dipping the marker into water and acetone and applying it to different surfaces produced unexpected results, such as bright yellow stains, which I found particularly interesting.

WEEK 7

cut out circles
use to create textures.
Foil forms
cling film
nail polish.
roller/flatten tool
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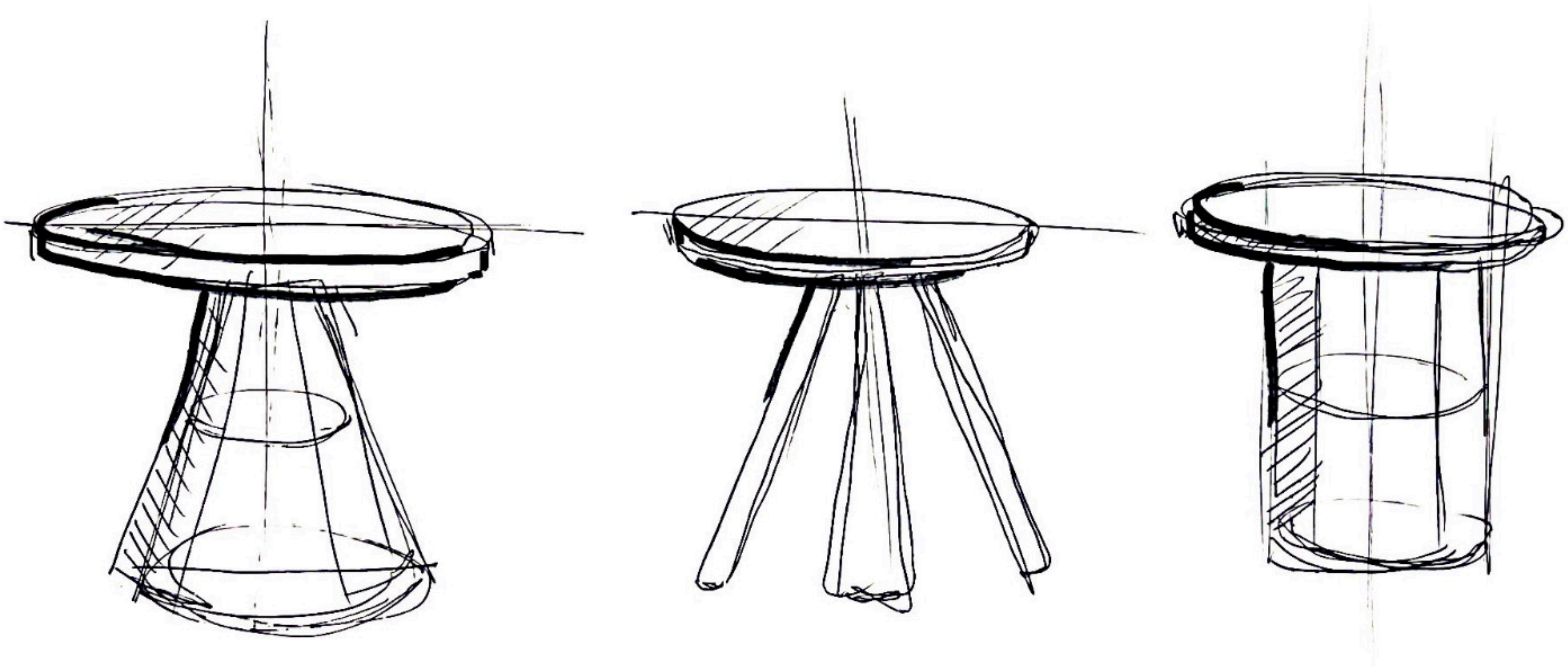
used to erase [CENSOR]
Tattoo maker
Stirring tool
soak in water
dip in aceton
burn
FELTS
water
aceton
COTTON PADS
added water
added aceton
Tip in water
Tip in aceton
set on fire & crumble
marked the other side of paper
TISSUE

One Tool One Stool:

For the fourth and final brief, we were tasked with producing a fully functional stool, approximately 40 cm high, that could safely support the weight of an adult, using only our tool of choice and one material. With the knowledge gained from previous explorations of our tool, we were required to identify a material that worked well with it. Using only these two elements, we had to create a stool, with the use of any fasteners strictly prohibited.

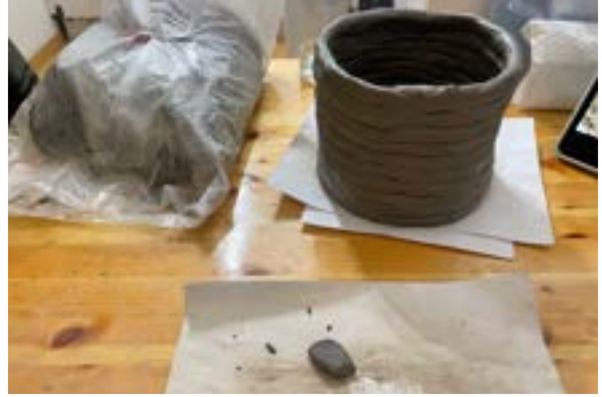


I began my exploration by trying to figure out what is the most stable form for my stool so it can withstand maximum weight while needing the least amount of material possible. I opted for a singular leg that can be hollow in the middle.

Making:

For the making of the stool, I had to use the sharpie as the tool and paired it with a single material. After much consideration and inspired by the experiments I conducted with the sharpie in the previous activity (TMMMT), I decided clay was the best option. I believed it would allow me to create a stable structure capable of withstanding human weight without the need for screws, glue, or other fasteners. Initially, I planned to fire the clay, but due to the long waiting time for the kilns, I opted for air-drying clay. While less durable, I assumed it would be strong enough for a stool.

Through my exploration, I discovered that the sharpie could be used to flatten the clay, roll it into shapes, cut it, and create decorative stamps. Therefore, I used it to shape strips of clay, which I stacked together to form the leg of the stool. To increase stability and ensure the structure could bear weight, I added an additional layer in the middle of the leg. Next, I used the sharpie to flatten three layers of clay, which I stacked on top of each other to create the seat of the stool. Finally, I used the sharpie to press decorative stamps onto the seat for added visual appeal.



WEEK 7 -10

Gastrointestinal Conditions: Definitions

How do we define Gastrointestinal (GI)?

"Gastrointestinal: of, relating to, or affecting the stomach and the intestines"

(Source: Merriam-Webster Dictionary)

"Having to do with the gastrointestinal (GI) tract or GI system. The GI tract includes the mouth, throat, esophagus, stomach, small intestine, large intestine, rectum, and anus. Food and liquids travel through the GI tract as they are swallowed, digested, absorbed, and leave the body as feces. The GI system includes these organs as well as the salivary glands, liver, gallbladder, and pancreas, which make digestive juices and enzymes that help the body digest food and liquids. Also called GI."

(Source: National Cancer Institute)

"The gastrointestinal (GI) tract is a series of hollow organs that extends from the mouth to the anus. It includes the esophagus, stomach, small intestine, and large intestine (colon), which processes food, extracts nutrients, and expels waste."

(Source: National Institute of Diabetes and Digestive and Kidney Diseases)

How do we define GI Conditions?

"Gastrointestinal conditions encompass a wide range of disorders, including irritable bowel syndrome (IBS), inflammatory bowel disease (IBD), gastroesophageal reflux disease (GERD), and more. These conditions can cause symptoms such as abdominal pain, bloating, constipation, diarrhea, and changes in bowel habits"

(Source: Mayo Clinic)

"Gastrointestinal conditions refer to diseases or disorders that affect the digestive tract and its organs. Common conditions include GERD, ulcerative colitis, Crohn's disease, and peptic ulcers, which can lead to discomfort and affect digestion and absorption of nutrients"

(Source: Cleveland Clinic)

"Gastrointestinal diseases include a variety of disorders that can affect the organs of the GI tract, leading to symptoms ranging from mild discomfort to severe pain and complications requiring medical treatment."

(Source: National Institute of Diabetes and Digestive and Kidney Diseases)

Common GI chronic conditions

There are many gastrointestinal conditions, ranging from those that can be cured to chronic ones where the patient must manage the condition for the rest of their life. For this project, I am focusing on chronic conditions that are typically lifelong, characterized by periods of calm and flare-ups. These types of conditions often require long-term management through tracking and monitoring, making them ideal for the scope of my brief.

Irritable Bowel Syndrome (IBS)

A chronic disorder affecting the large intestine, characterised by symptoms such as abdominal pain, bloating, and changes in bowel habits (diarrhea, constipation, or alternating between the two). It is a functional disorder, meaning it affects how the gut functions but does not cause structural damage to the intestines.

Affects 10-20% of UK population

Gastroesophageal Reflux Disease (GERD)

A chronic condition where stomach acid or bile irritates the food pipe (esophagus), causing symptoms like heartburn, regurgitation, chest pain, and difficulty swallowing. GERD occurs when the lower esophageal sphincter, which normally prevents acid from flowing back into the esophagus, becomes weakened or relaxes inappropriately.

Affects 20% of UK population weekly

Irritable Bowel Diseases (IBD)

An umbrella term for chronic inflammatory conditions of GI track. IBD is characterised by periods of flare ups and remission and can affect any part of the GI track. IBD includes:

Crohn's: Inflammation in any part of the GI track often affects the small intestine
Ulcerative Colitis: Chronic condition that causes inflammation and ulcers in the lining of the colon and rectum

Affects 0.5% of UK population

Peptic Ulcer Disease

Sores that develop on the lining of the stomach, small intestine, or esophagus, caused by the erosion of the mucosal lining. The condition is often related to infection with Helicobacter pylori bacteria or long-term use of nonsteroidal anti-inflammatory drugs (NSAIDs). Symptoms include burning stomach pain, bloating, and nausea.

Affects 15.5 million people in UK

Initial Design Framework: Refining my design opportunity

After a tutorial with Alon regarding my design opportunity, I decided to take a step back from focusing solely on designing self-monitoring devices for GI conditions. Instead, I chose to **explore gastrointestinal conditions more holistically by examining their social, economic, medical, and emotional aspects.** Taking a broader approach to the topic will allow me to gain a deeper understanding of all facets of gastrointestinal diseases. This, in turn, will enable me to develop **a management tool for people suffering from these conditions that addresses their challenges in a more holistic manner, moving away from tools that are overly medical or technological.**

How can design products be developed to support individuals with chronic gastrointestinal conditions in effectively managing their health, fostering a deeper understanding of their condition, and improving their quality of life?

Who?

People that are suffering from chronic gastrointestinal conditions such as IBS and IBD and are looking for solutions to manage the effects of their conditions. They can be of any age and background.

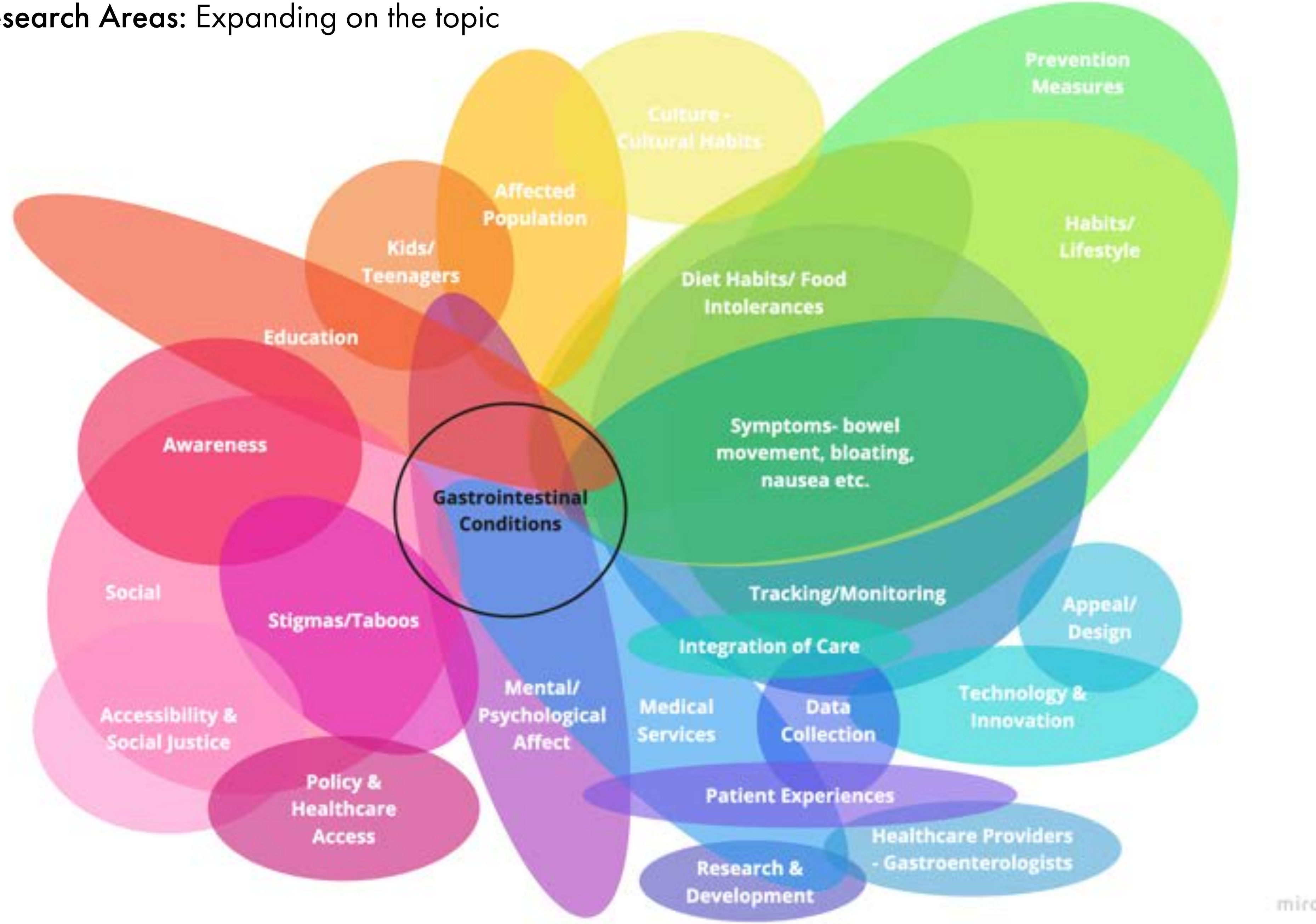
Why?

Chronic gastrointestinal conditions affect a significant portion of the global population and are highly relevant in the UK. People who suffer from these conditions face numerous daily challenges that impact their physical and mental health, as well as many aspects of their lives. More solutions are needed to address these challenges effectively.

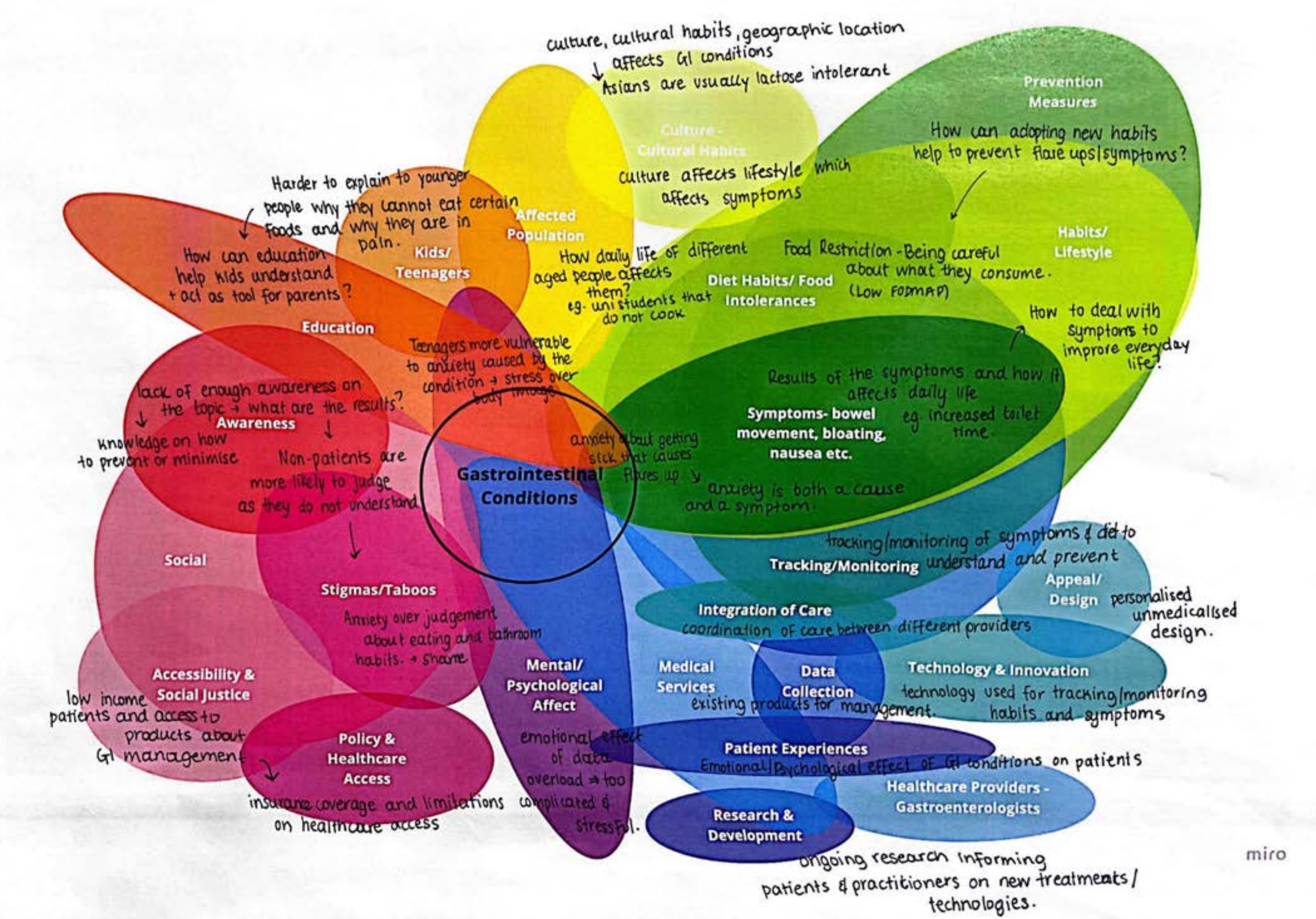
Next step:

Identify the subtopics within my theme and produce areas of research. This approach will help me gain a holistic understanding of the topic, broaden my perspective, and develop design questions that will be explored through my experiments.

Research Areas: Expanding on the topic



WEEK 7



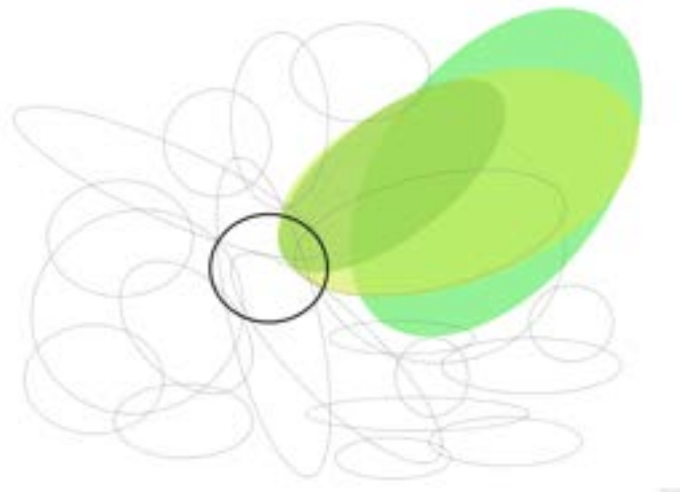
culture, cultural habits, geographic location
affects GI conditions
Asians are usually lactose intolerant
Culture - Cultural Habits
culture affects lifestyle which affects symptoms
Prevention Measures
How can adopting new habits help to prevent flare ups/symptoms?
Habits/ Lifestyle
Harder to explain to younger people why they cannot eat certain foods and why they are in pain.
Affected Population
Kids/ Teenagers
How can education help kids understand + act as tool for parents?
Education
How daily life of different aged people affects them?
eg. uni students that do not cook
Diet Habits/ Food Intolerances
Food Restriction - Being careful about what they consume. (Low FODMAP)
How to deal with symptoms to improve everyday life?
Teenagers more vulnerable to anxiety caused by the condition → stress over body image
Results of the symptoms and how it affects daily life
eg. increased toilet time
Symptoms- bowel movement, bloating, nausea etc.
lack of enough awareness on the topic → what are the results?
Awareness
knowledge on how to prevent or minimise
Non-patients are more likely to judge as they do not understand
Gastrointestinal Conditions
anxiety about getting sick that causes flares up
anxiety is both a cause and a symptom.
tracking/monitoring of symptoms & diet to understand and prevent
Tracking/Monitoring
Social
Stigmas/Taboos
Appeal/ Design
personalised unmedicalised design.
Integration of Care
coordination of care between different providers
Anxiety over judgement about eating and bathroom habits. → shame
Mental/ Psychological Affect
Medical Services
Data Collection
existing products for management.
Technology & Innovation
technology used for tracking/monitoring habits and symptoms
Accessibility & Social Justice
low income patients and access to products about GI management
Policy & Healthcare Access
emotional effect of data overload → too complicated & stressful.
Patient Experiences
Emotional/Psychological effect of GI conditions on patients
Healthcare Providers - Gastroenterologists
insurance coverage and limitations on healthcare access
Research & Development
ongoing research informing patients & practitioners on new treatments/ technologies.
miro

10 Areas of Research

I narrowed down my initial map into a simpler, more comprehensible one with more clear areas of research. This enabled me to identify **10 specific subtopics within GI conditions that I can explore**. The next step will be **to form research questions around these categories that I will then explore through my experiments**

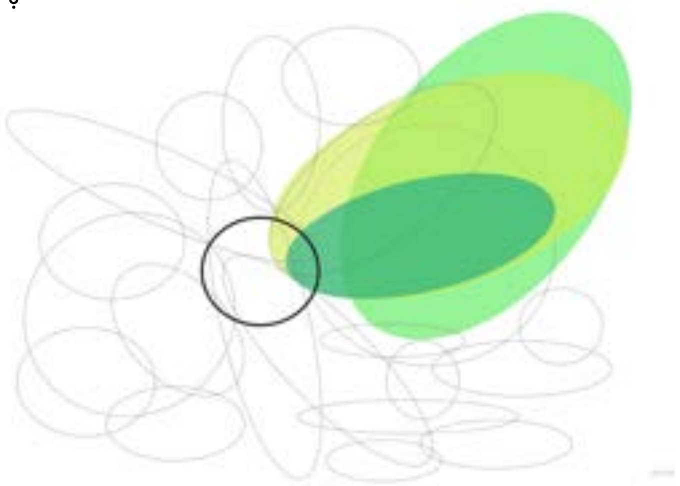
Diet Management
(Diet Habits, Food Intolerances etc.)

- What kind of food should GI patients usually avoid? – exploration of nutrients and how they affect the GI track
- How consumption of processed foods impacts GI health in patients and what alternative food choices could reduce GI symptoms.
- What kind of foods are usually recommended for GI patients and what is their accessibility?
- How important is the consumption of fibre for GI patients and how it improves their conditions
- Exploration of supplements and intervention that can enhance diet and improve GI track health and digestion. - Probiotics
- How do food restrictions differ based on the GI condition of the patient? - People with celiac disease should not eat gluten
- How can personalised diets lead to the prevention of symptoms?
- How do GI patients identify intolerances and what foods bother them?
- How psychological and emotional factors affect the food choices of GI patients?
- What are existing methods/products that allow GI patients to identify food sensitivities?
- What are existing products responding to this subtopic?



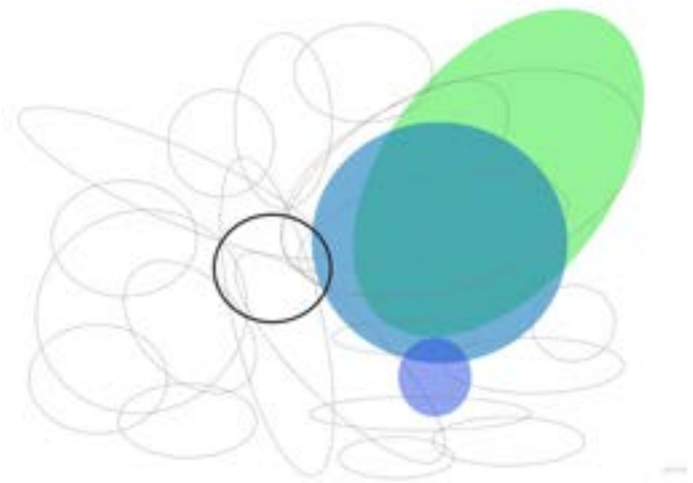
Symptom Management
(Pain Relief, Symptom Prevention etc.)

- What are the most common symptoms that GI patients experience? – *bloating, abdominal pain, cramps*
- How are symptoms different for each patient?
- How do GI conditions affect bowel movement? – *Diarrhoea, Constipation etc.*
- How does it affect bathroom use? – *might be more often for many patients*
- How do these symptoms affect people's everyday life*? – have different habits, plan their lives around their symptoms*
- Does it affect their plans, ability to do public activities or work?
- Explore how fear of experiencing symptoms can lead to anxiety which can then result to flare ups
- How do experiencing certain symptoms can affect patient's mental health?
- How to help GI patients deal with symptoms caused by their diseases like stomach pain, bloating, inability to sleep and reflux?
- What kind of methods do GI patients currently use to relief pains, bloating and other symptoms?
- What are existing products that help with pain relief and improves the experience for GI patients ?
- How can symptoms be predicted and prevented?



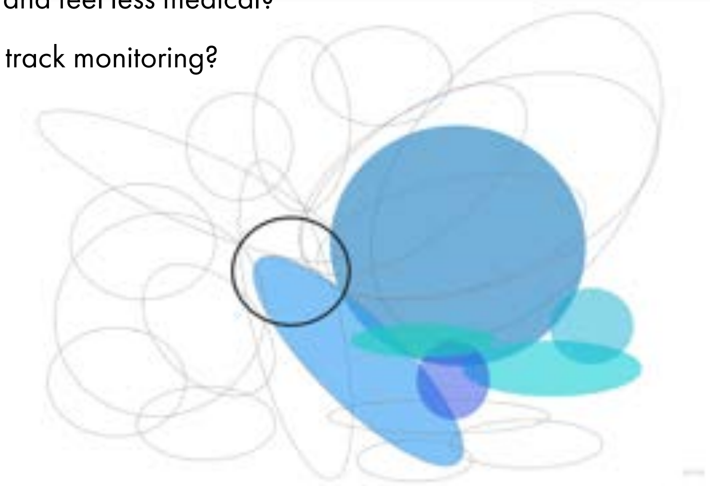
Tracking & Monitoring
(Keeping records of diet and symptoms)

- Explore how can tracking their diets help patients identify food sensitivities/intolerances and how specific symptoms associate with specific foods.
- How does symptom tracking improve GI patients experiences by allowing them to identify and prevent flare ups?
- What are the specific aspects of the GI conditions that needs to be tracked?
- How food databases can be integrated into tracking systems to automatically log meals, nutrients, and potential allergens that impact GI conditions
- Explore symptom and diet tracking- What scales, terminologies are appropriate? *- Bowel Movement uses Bristol Stool Form Scale*
- How data should be portrayed to not be overwhelming and not be too scientific? - *Limit medicalisation of tracking products*
- Explore the use of AI to offer personalised suggestions based on a patient's history, symptoms, and tracking data.
- How can symptom and diet tracking enable GI patients have more informed conversations with healthcare providers?
- How can it help healthcare providers give better feedback knowing all the information that was tracked by the patient?
- How might tracking all this data cause anxiety for GI patients instead of helping them?
- Explore the idea of data overload and how it can negatively affect users' experiences



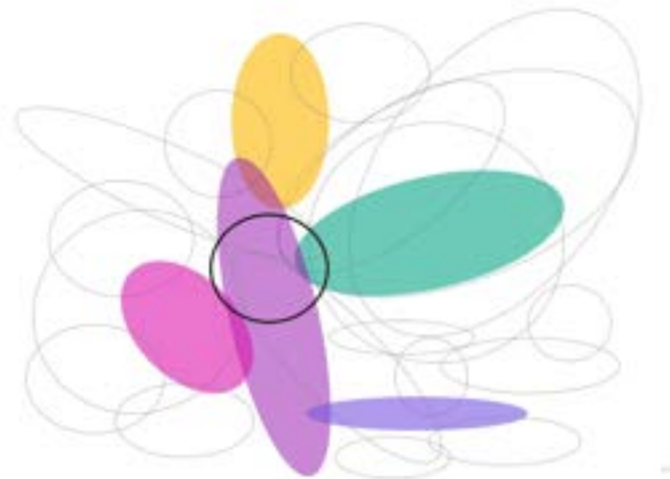
Technology & Innovation
(Artificial Intelligence and digital interventions)

- How technology can be implemented to improve the experience of GI patients?
- What can be designed that improves healthcare for GI patients? – tools for both patients and healthcare providers
- How can AI and data analysis be employed to provide GI patients with personalised feedback?-Identify patterns, predict symptoms and suggest diets
- How can voice assistants or AI-powered chatbots be integrated into GI care to provide patients with immediate advice and help manage symptoms?
- How digital interventions, such as diet and symptom tracking tools, influence long-term behaviour change and disease management in GI patients?
- Explore the possibility of wearable devices that don't just track activity but also digestive health, like stomach acid levels and reflux.
- How can data privacy and security concerns be addressed in the development of GI tracking devices or apps?
- How to implement technology for better integration of care by making systems that coordinate care given by different providers?
- How can we create systems that can be used from both healthcare providers and patients simultaneously?
- What are existing technologies/products that respond to the issue?
- How can the design of existing technologies be improved to be more appealing, user-friendly and feel less medical?
- What are limitations of existing technologies and can features be added to them to include GI track monitoring?



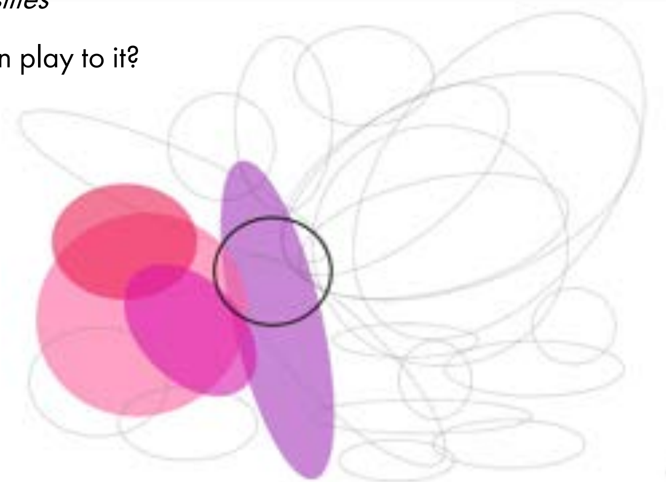
Mental Health
(Anxiety, Overwhelmedness and Fear)

- How does GI conditions can affect the patient mental health?
- How does doing daily activities be different GI patients in ways that may cause them anxiety?
- Explore anxiety and fear of experiencing GI related symptoms and how can that affect mental health
- How can anxiety be the cause of flare ups and GI symptoms to develop?
- Explore anxiety and poor mental health as both a cause and a symptom. – *how it makes patients gets stuck in a cycle and how to escape?*
- How mental health in relation to GI conditions affects different age groups differently?
- How can symptoms like gaining and losing weight negatively affect teenagers who struggle with body image?
- How can GI symptoms make patients feel excluded and unable to do things everybody else can? –*how is that harder for younger people?*
- Explore anxiety experienced by the fear of people judging the patient about their symptom. *- Judgement for needing to go to the bathroom often*
- Do patients need to know exactly how everything works and possible symptoms in medical terms? –*knowing possible symptoms might cause them*
- How may medicalised terms and complicated explanations cause more anxiety to GI patients
- Explore how data and feedback overload can have negative effects on GI patients.



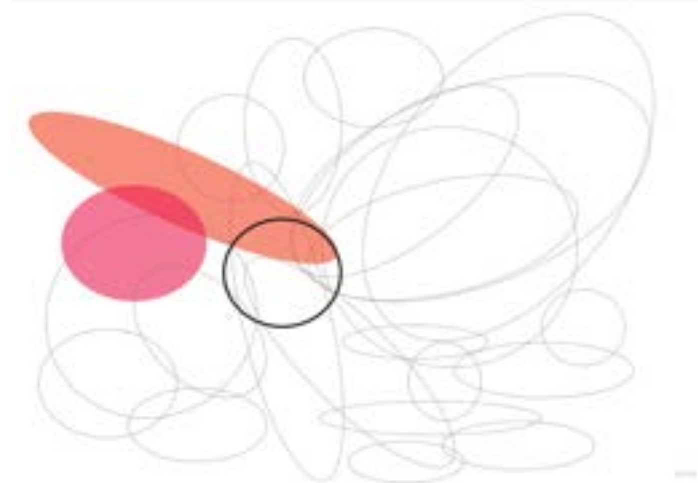
Social Impact and Stigma
(Taboos and Social Implication)

- What are the stigmas/taboos around the topic of GI conditions?
- How do this stigmas/taboos affect people experiencing GI conditions?
- How do misconceptions about GI conditions contribute to the stigma? – *How do people misunderstand conditions like IBS or Crohn's disease*
- Are GI patients constantly anxious that they might get judged from certain activities that are associated to their issue? – *bathroom time*
- How does the fear of judgment affect the willingness of GI patients to seek help or be open about their symptoms?
- How do people with GI conditions deal with dietary restrictions in social settings? – *trying to hide their diet limitation*
- How can symptoms like bloating cause judgement making patients more self-aware?
- Explore symptoms like gaining and losing weight and how can they lead to social judgement?
- How does social media influence the stigma around GI conditions? – *Are GI patients more likely to experience judgment due to online discussions*
- Are teenagers more vulnerable to this social judgement caused by stigmas and taboos?
- In what environments is this judgmental behaviour more prevailing? – *schools, colleges, universities*
- What steps should be taken to eliminate these stigmas from the society and what role can design play to it?



Education & Awareness
(Taboos and Social Implication)

- How much do people without GI conditions know about them?
- How much do people with Gi conditions know and understand about their condition?
- How can more awareness be raised around the topic? - *Social media campaigns be leveraged to raise awareness about the importance of gut health*
- How can education help kids better understand and accept with their conditions? – *Understand how their diets and habits might need to change*
- What role can gamify learning play in helping individuals with GI conditions better understand their bodies and the impact of their condition?
- How can we ensure that people with GI conditions can easily access reliable information about their condition?
- How can digital tools be used to educate individuals about the psychological and emotional aspects of GI conditions, not just the physical symptoms?
- What role can healthcare providers play in educating patients about the long-term management of GI conditions,?
- How can we make information about GI conditions more accessible? - *to non-English speakers or those with low literacy levels*
- How to education families and caregivers to ensure they understand the daily challenges of living with a GI condition?



Affected Population
(Age, Life Stage, Culture, Geographic Location)

- What is culture's role in GI patients' experiences?
- How do different cultural habits affect GI patients? – differences in cuisine, food availability and eating practises might affect
- How does geographical location affect the kind of GI conditions someone might develop? – lactose intolerance prevailing in Asian countries.
- How does age can make the experience of having a GI condition different for the individual?
- Explore difference obstacles that teenagers experience in comparison with older people. - Having to navigate school/college with a GI condition?
- What are the long-term effects on aging populations with chronic GI conditions, and how does this impact their quality of life?
- How do gender and identity influence the experience of GI patients? – women may be more vulnerable due to their reproductive system.
- What role does lifestyle, that is usually associated with stage of life, plays? – Are university students that do not cook more vulnerable?
- How are children's experiences different due to their inability to comprehend that they have a condition?
- How do you explain to a kid why they shouldn't eat specific things and why they experience specific symptoms?
- How do the families of people with GI conditions are affected? – how are they required to assist and what are their experiences?
- Explore the experiences of parents of kids' suffering with GI condition. – What challenges do they face?

Research Methodologies

Interviews with Professionals:

Interview gastroenterologists and other experts to get a better understanding of the condition in terms of diet requirements and symptoms, as well as identify the difficulties they face in providing care to people with these conditions. Gain insights on what worsen the conditions and how it can be approached.

Interview with Patients:

Engage to guided conversations with GI patients. Use these conversations to get insights on their experiences, what aspects of them they would like improved, their daily struggles, and the methods/rituals they currently use to relief symptoms and improve their situations.

Secondary Research:

Conduct extensive research to comprehensively understand GI conditions and identify how design can respond to it. Look into regulations around collecting medical data, the concept of gut – brain and how it can create opportunities for HCI design and visualisation of data.

Case Studies:

Explore existing products in the market that respond to different aspects of GI conditions. Examine how effective they are, their limitation and even ask some people to use them for a period and give feedback on their experienced using these products.

Probe:

Design a probe to explore GI patients experiences in relationship with tracking. Ask them to daily record symptoms and what they eat to examine if this practise causes them anxiety, how the process of manual tracking can be improved and what are the most intuitive and fast methods of manual tracking

Prototyping:

Use electronics in the form of sensors such as pulse sensor to test how we can collect data directly from the body instead of manually logging it. Explore how to turn this data into comprehensible outputs that communicate to users' information about their condition.

Existing Work: Products about GI Self Management

WEEK 9

Gut Health Tracking Apps

Apps designed for tracking symptoms, food intake, stress levels, and bowel habits, helping individuals identify triggers and monitor their GI health over time.

Cara Care

Cara Care is an app designed for people suffering from IBS, IBD, and other similar GI issues, helping them manage their digestive health. It allows users to log and monitor their symptoms, food intake, bowel movements, and stress levels. In combination with providing dietary guidance and educational resources, it serves as a comprehensive tool for managing gut health and improving the experiences of GI patients.

mySymptoms Food Diary

mySymptoms Food Diary is an app designed to help users with food intolerances, allergies, and IBS by enabling them to track and identify links between their diet and symptoms. By logging food intake and symptoms, the app collects and analyses the data to provide personalised insights into how their diet impacts their health, serving as a self management tool that allows the users to keep track of their diet.

Bowel Movement Tracker

These trackers help people monitor the frequency, consistency, and appearance of bowel movements, which is important for those with conditions like IBS, IBD, or chronic constipation.

PoopLog

Poop Log is an app designed for people with digestive concerns, such as constipation or diarrhea, often caused by GI conditions. By asking users to log their bowel habits, it helps them track bowel movements and better understand their digestive health. The app uses the Bristol Stool Chart to classify stools into seven types, allowing users to describe their bowel movements accurately.

GI Monitor

GI Monitor is an app designed to help individuals with GI conditions like Crohn's and IBS track and manage their symptoms. By asking users to log key aspects of their condition, such as abdominal pain, diarrhea, blood in stool, and constipation, it provides personalised feedback to help users better understand fluctuations in their condition.

Food Sensitivity & Intolerance Tests

At-home test kits that help people identify potential food sensitivities and intolerances that may be exacerbating GI symptoms like bloating, gas, diarrhea, or cramps.

Everlywell Food Sensitivity Test

Everlywell Food Sensitivity Test is a home testing kit designed for individuals with digestive problems who want to identify potential food intolerances. The kit includes everything needed to collect a simple finger prick blood sample at home, which is then sent to a designated lab for analysis. After the lab processes the sample, they send the results in a report that includes dietary guidance based on the findings.

YorkTest Food Sensitivity Test

The YorkTest Food Sensitivity Test is a home testing kit designed for individuals looking to identify food intolerances that may affect their digestive health. The kit provides all necessary materials to collect a simple finger prick blood sample at home, which is then sent to a lab for analysis. After processing, a detailed report is provided with results and personalised dietary recommendations based on the findings.

Breathe Test Kits

Breath test kits measure gases like hydrogen and methane in the breath to diagnose gastrointestinal conditions such as SIBO, lactose intolerance, and food sensitivities.

FoodMarble AIRE

FoodMarble AIRE is a portable breath test device that helps identify food intolerances by measuring hydrogen in your breath after eating or drinking. The device tracks your digestive response in real-time, providing insights into how your body reacts to foods like lactose or fructose. This information can help you manage GI conditions and personalize your diet for improved digestive health.

Gastrolyzer SIBO Breath Test

The Gastrolyzer SIBO Breath Test is a diagnostic tool designed to detect small intestinal bacterial overgrowth (SIBO). It measures hydrogen and methane levels in your breath after consuming a sugar solution. Elevated gas levels can indicate an imbalance of bacteria in the small intestine, which is a common cause of digestive issues. The test provides valuable insights for managing GI conditions like bloating, pain, and irregular bowel movements.

Acupressure Bands

Acupressure bands are wearable devices that apply gentle pressure to specific points on the wrist, helping to alleviate symptoms associated with gastrointestinal conditions.

Sea-Band

Sea-Band is an acupressure wristband designed to relieve nausea and vomiting associated with gastrointestinal issues. By applying gentle pressure to the P6 acupressure point on the wrist, it helps reduce symptoms of nausea caused by conditions like motion sickness, morning sickness, and GI discomfort. Non-invasive and drug-free, Sea-Band provides a convenient and natural solution for managing nausea during daily activities or travel.

Acuband

Acuband is an acupressure wristband designed to alleviate nausea and digestive discomfort. It works by applying consistent pressure to the P6 acupressure point on the wrist, helping to reduce symptoms like bloating, nausea, and indigestion. Ideal for those looking for a non-drug solution, Acuband is an easy-to-use, comfortable option for managing nausea related to GI conditions, offering a natural alternative to medications.

Experiment 1: Interview with a Gastroenterologists

What I want to learn?

1. What is the process of diagnosing these conditions?

2. What is the patient's role in diagnosis?

3. Recommended ways for managing these conditions

4. Common symptoms and role of stress

5. What needs to be tracked and suggestions on how

6. Opinion about utilising technology to improve GI patient's experiences

7. Gaps in the market for GI conditions support

How will I explore it?

Contact a gastroenterologists (a doctor that specialises in GI Conditions)

Conduct an interview with the gastroenterologists to gather information about chronic GI conditions and what kind of approach would be the most appropriate one for this project

What questions will I ask?

- Brief intro into GI condition
- How do you diagnose these conditions? What can patients do to help the diagnosis method?
- What kind of features would a product for tracking GI conditions have?
- Would it be possible to link the product to a healthcare system like NHS?
- How anxiety links to GI conditions, and how to manage it?
- Tips on how to relief symptoms and manage diet.
- Thoughts about existing products and how they affect patients.
- What products would they recommend.
- Thoughts about gaps in the market and what would be a right approach on the topic

Interview Insights

Insight 1

- → Mapping out the process of diagnosis:
  - → First step is to **ask questions about the symptoms** → clear answers are vital as exact nature of symptoms affect diagnosis, while many symptoms can be overlapping
- → PROBLEM: **Patients struggle to describe symptoms clearly**
  - → Might not mention certain things like stool habits because they are embarrassed.
  - → Symptoms come and go making it hard to identify trends
- → Encourage patients to keep a **symptom's diary** to write down when symptoms occur, what they eat and how they are feeling emotionally

Insight 3

- → Role of **Anxiety and Stress** in worsening the conditions:
  - → Anxiety can exacerbate GI conditions and GI issues can trigger more anxiety, making the patient be stuck in a loop
- → It is **usually overlooked** by existing products that do not tackle this directly and sometimes even by healthcare providers themselves.
- → Suggested ways for breaking the anxiety cycle:
  - → **Cognitive Behavioural Therapy** - teaching the brain to not panic every time there's a little discomfort.
  - → **Mindfulness and Relaxation** Techniques – deep breathing and medication
  - → Exercise- get moving, even a little

Insight 2

- → Positive towards the idea of using technology to improve GI patient's experiences
  - → Suggestion of using Artificial Intelligence as a means
- → Important to ensure that **technology is not replacing professional advice**
  - → Should emphasise **on empowerment over diagnosis** to avoid dangers of misdiagnosis
  - → Should ensure that user will be encouraged to communicate the data with their healthcare provider to get professional support when its needed

Insight 4

- → Still room for more products – but its all about finding the right angle.
  - → Lack of products that **provide a holistic, patient-centered approach.**
  - → Need products that integrate multiple aspects of care
- → Potential of products that focus on the mental health aspect of living with a GI condition.
- → Gap lies in creating **comprehensive, personalised, and engaging** solutions that address the full spectrum of managing GI conditions – physical, emotional, behavioral.

Insight 5

- Aspects of monitoring:
  - **Simplicity is key** – intuitive, easy to use and non overwhelming tools that take minimum time but provide valuable data
  - **Tracking key symptoms** such as bloating and bowel movement – scales for severity, frequency, duration and time it appears.
  - **Logging food and drink** – easy logging that could connect with symptoms experienced after
  - **Stress Tracker** – stress levels through mood ratings or sensors
- Tips about monitoring:
  - Features for **easy sharing of data** with healthcare providers
  - **Voice commands** or **voice-to-text** features – more natural, less like a chore
  - **Reminders** and **Notification** to enable consistency

Insight 6

- Product should feel like a **companion**
  - Not just another task on their to-do list.
- **Automated or low-effort**
  - Simplifying the tracking process as much as possible by providing lists etc.
- **Gradual, Customisable** tracking
  - Users choose what they want to track starting with a few aspects and add more are they feel comfortable

Conclusions

Tracking is important for managing the condition after diagnosis but also for making the process of diagnosis easier.

Using some sort of technology is a good approach.

Mental Health and specifically anxiety plays a very important role to the management GI conditions.

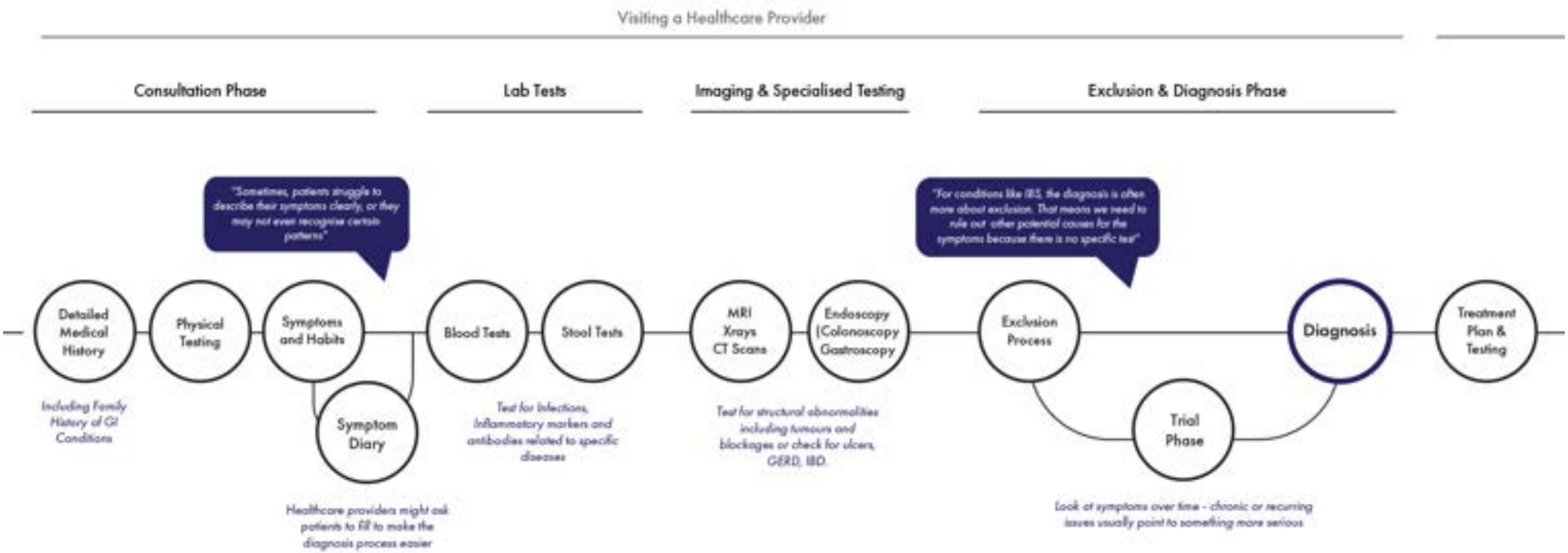
New devices are needed that take new approached and focus more on providing stress-free and holistic assistance.

Next Steps

- Look into the patients' experiences – what are the challenges of living with these conditions
- Create journey maps of their lives with the conditions – identify what point in their daily life the condition affects them and where will a product fit.
- Explore If they use any tracking tools and what is their thoughts around them.
- **Move back into the idea of some sort of tracking device but find a new approach for it that moves away from existing products**

Diagnosis Journey Map

During the interview with the gastroenterologists, I gathered information around the diagnosis process from the doctor's perspective. Based on these insights, I made a journey map of the process of getting diagnosed. The goal is to collect more insights from patients' perspective and add on this map to build a longer one that includes the whole experience of a patient from when they experience the first symptoms to their life after diagnosis.



Experiment 2: Interview with People struggling with these conditions

What do I want to explore?

1. How does the condition affect their daily life?

2. What makes the condition worse?

3. What products/ tools they currently use?

4. What are ways they currently use to manage with the condition?

5. Opinion about existing tracking apps

6. How would an ideal tracking product be and what point of the day will fit into?

7. Daily Life of people with conditions

How will I explore it?

Part 1: Interviews

Ask general questions about the condition – diagnosis, experience, manage methods

Ask questions around existing apps/ tools for managing their condition

Understand people's experiences and what they are looking for in products

Part 2: Mapping Exercise

Ask participants to create a **journey map a standard day** from when they wake up to when they go to sleep.

Ask participants to create a **journey map of a non-standard day** from when they wake up to when they go to sleep.

Identify the points during the day that the condition affects them the most and differences between the two maps

Identify at which point of their daily schedule would a management tool fit the best/

Part 1 – Notes from Interviews

Participant (1): (male, mid/late 20s, IBS)

part 1:

- condition length: since high school
  - ↳ flare-up periods gets worse
- effect in daily life: gas, frequent diarrhea, pain, cramps
  - → needing to go to the toilet at uncomfortable times
  - ↳ anxiety caused by fear of needing to go to the toilet when out.
  - ↳ frequently needing to use public toilets that are in bad condition
  - ↳ fear of getting dirty & feeling chafe after using the toilet.
  - ↳ affects confidence, especially in public settings.
- makes it worse: (1) food consumption (dairy, spice)
  - (2) alcohol consumption
  - (3) major event that causes anxiety (date, live show)
- managing method: having pain relief pills → don't always stop the toilet need.
- experience of communicating symptoms:
  - easy / straight forward on generic questions
  - more complicated in specific
  - hassle + draining.

part 2:

- products used: currently nothing
  - ↳ used apps for a while but stopped.
- Why?:
  - tracking apps are too much commitment
  - don't feel engaged + quickly lose interest.
- How would you want it to be different:
  - NOT TIRING TO USE – AS LESS HASSLE AS POSSIBLE
  - ONLY NEED TO USE ONCE - TWICE A DAY
  - TAKE NO MORE THAN 2 MINS. + BE DONE IN 4-5 click with no typing.
  - FREQUENT REMINDERS

Participant (2): (female, early 20s, IBD)

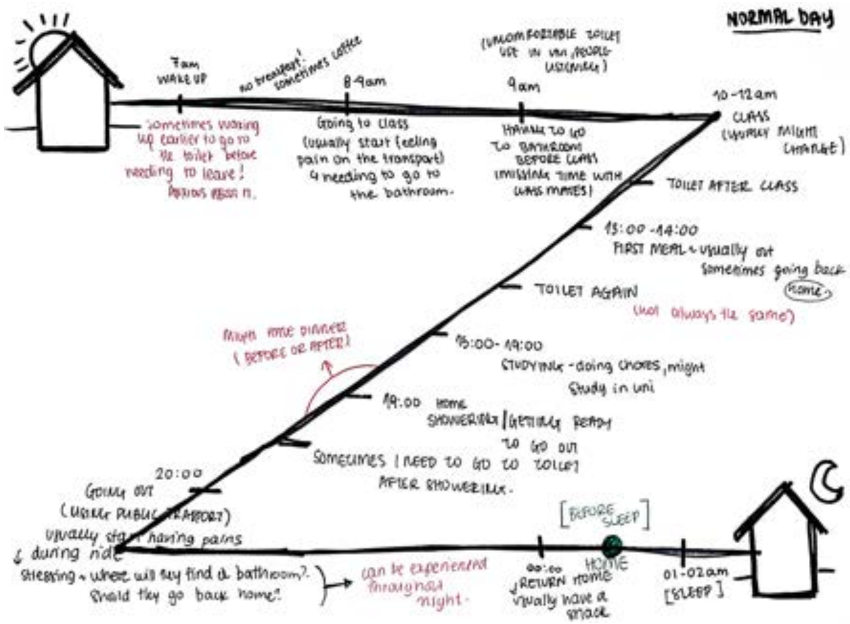
part 1:

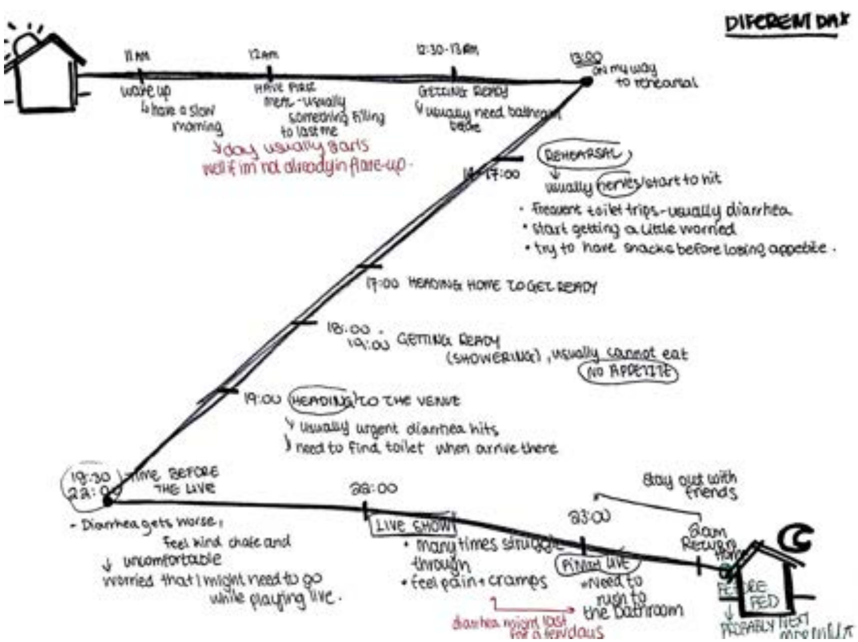
- condition length: diagnosed last year after symptoms got bad, but had stomach problems in the past
- effect in daily life: changes between diarrhea / constipation, bloating, cramps, nausea, reflux, backpain, pain, gas
  - ↓ needing to use the toilet on uncomfortable places / times
  - ↳ bloating + clothes feel uncomfortable / too tight
  - ↳ gaining + losing frequently weight
  - ↳ needing to be too mindful of what I eat + very conscious in public settings / usually scared to eat out.
  - ↳ anxious of doing public activities out of fear of pain. can't have alcohol with friends.
- makes it worse: (1) not getting enough sleep
  - (2) specific food ~ fast foods, spicy food
  - (3) anxiety → gets worse around exams
- managing methods:
  - Prescripted pills
  - not getting worked up when a symptom comes
  - CONTROLLED LIFESTYLE → very mindful of what I eat. → not enjoying this at all.
    - ↳ trying to understand what bothers me
- experience of getting diagnosed:
  - TROUBLE ANSWERING QUESTIONS → VERY CONFUSING IN MY HEAD
    - ↳ IMPOSSIBLE TO REMEMBER + COMMUNICATE WELL.
  - ANNOYING + DISHEARTENING
    - ↳ TOOK TOO LONG TO IDENTIFY WHAT IT WAS
  - TIRING + CAUSED ANXIETY WHICH MADE EVERYTHING WORSE.

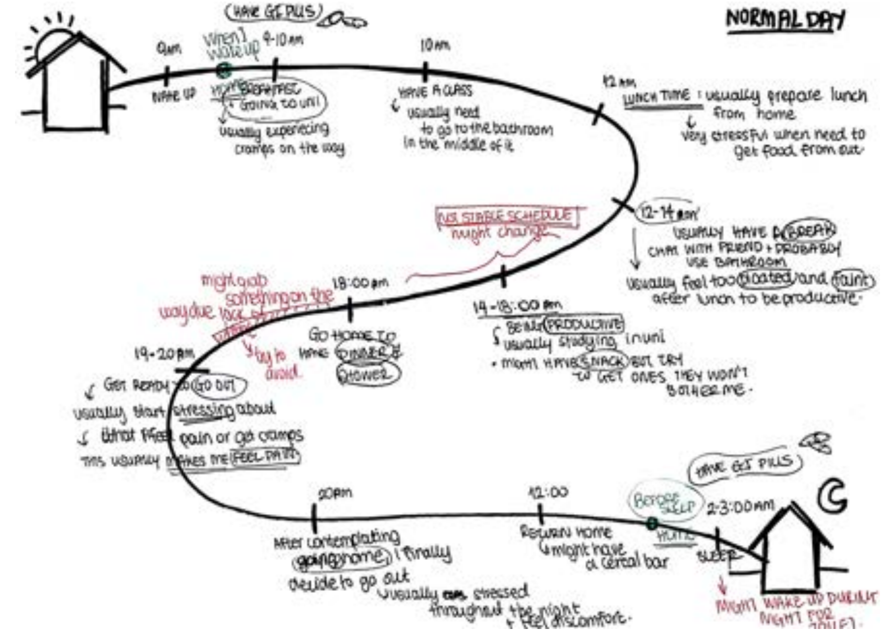
part 2:

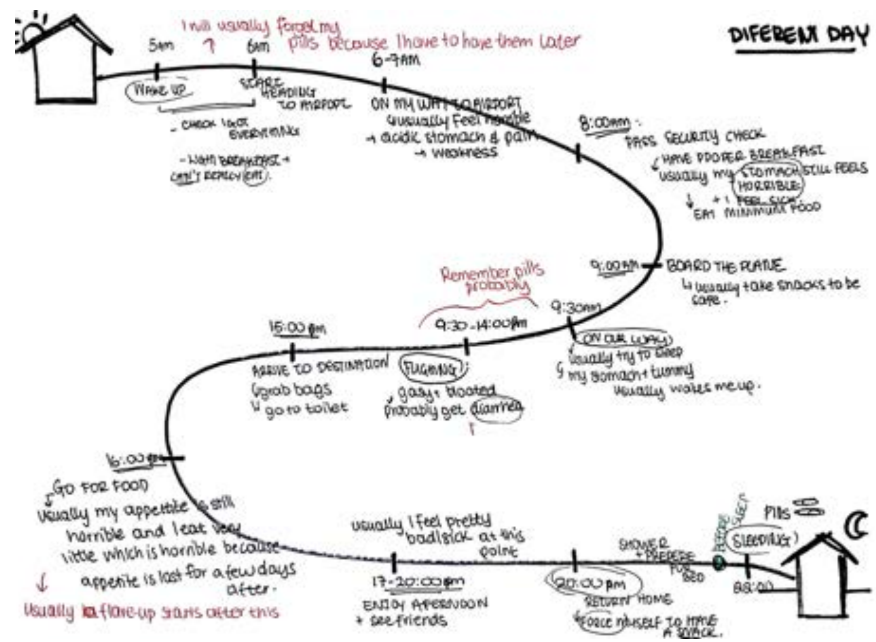
- products used: NOTHING
  - EVERYTHING ALWAYS FELT TOO STRESSFUL
- Why?: DIDN'T FEEL LIKE IT
  - ↳ they all felt too complicated + didn't want to think about the condition more than I already do.
  - ↳ get bored too easily.
- HOW WOULD YOU WANT A DEVICE TO BE:
  - STRESS-FREE → something that is relaxing to use
  - FAST + ON POINT.
  - ENGAGING → maybe like a game?

Part 2 – Notes from Mapping Exercise









Insights

Insight 1

- → Constantly affecting patient' daily life, causing anxiety, overthinking and reduced confidence
  - → Increased toilet trip in uncomfortable settings that cause anxiety
  - → Uncertainty about what to eat
  - → Stress about going out and doing daily activities
- → Affects patients' social life and it is a problem they experience throughout a day
- → **Will the design product need to be something that users have always with them?**

Insight 2

- → Anxiety is a big factor that contributes to this conditions and causes a **never-ending cycle**
  - → Mental Health affects people's condition more than they expected

Insight 3

- → Diagnosis process is usually stressful and overwhelming
  - → Struggle to remember and communicate specific symptoms clearly
  - → Difficult to make connections between habits and symptoms
  - → Causes anxiety which worsens the condition and makes it harder to identify patterns and connections

Insight 4

- → Participants feel like existing solutions need too much effort and commitment
  - → **Causes them more stress than relief**
  - → Not engaging enough because of how much time they need and how complicated they are.
- → They want something fast, seamless, stress-free that doesn't take too much time of their life.

Insight 5

- → If they had to interact with a product, they would probably **want to do it at home before going to sleep**
  - → More research on more people needs to be conducted in this case and depends on product

Insight 5

- → During days that the patient does not follow the normal routine, the condition tends to become worse
  - → The body reacts badly when the routine is broken
- → How social, everyday life plays a role in improving or worsening the condition?
- → Will it be useful if the product was tracking more than mood, and food but it was also **tracking everyday life like activities and reflect on how changes in everyday habits that might not necessarily have a direct connection to the condition might still affect it?**

Literature Review: Gut – Brain Axis

Article 1: The gut-brain axis: historical reflections by Miller I.

Article 5: The gut-brain axis: historical reflections

CONTEXT:

- Gut-Brain axis significance: HIGHLIGHTS CONNECTION BETWEEN GUT AND BRAIN. EMPHASIZING HOW GUT BACTERIA INFLUENCE EMOTIONAL WELL-BEING AND BEHAVIOUR
- Doctors and patients recognised the relationship between gut and emotions, but framed it through models like "nervous sympathy" or psychology rather than microbiology.

What is it exactly? a network of nerves and biochemical signals, that connect the brain and the gastrointestinal track

- GUT-BRAIN AXIS MAINTAIN HOMEOSTASIS AND IS IMPORTANT FOR REGULATING THE FUNCTION OF THE CENTRAL AND ENTERIC NERVOUS SYSTEMS. IT CAN IMPACT DISORDERS SUCH AS IBS & MOOD DISORDERS. → MENTAL HEALTH CAN ALSO IMPACT THE GUT, AND DIGESTIVE CONDITIONS CAN IMPACT MOOD

HISTORICAL INSIGHT: Gut-Brain connection has shifted between reductionist (focusing on isolated organs) and holistic (considering the whole body) approaches

↓

- Gut health has often been linked to broader societal issues, including industrialisation, diet and mental health crisis.
- Introduction of "psychobiotics" (bacteria with mental health benefits)

Insights from historical approaches to the gut-mind connection can inform contemporary practices

HOW TO BALANCE REDUCTIONIST ADVANCES SUCH AS PHARMACEUTICALS WITH HOLISTIC, PATIENT-CENTERED CARE?

Trend towards intergrative medicine, combining biological, psychologica[l] and social perspectives.

Value of addressing the gut-brain axis holistically, intergrating emotional, dietary, and societal factors.

Article 2: Mind-Gut Computer Interaction: Research Areas and Opportunities by Majrashi K. and Uitdenbogerd A.

Article 6: Mind-Gut Computer Interaction: Research Areas and Opportunitie[s]

① HCI (Human-Computer Interaction) & ② Mind-Gut (Gut-Brain) communication.

- TWO MAIN MOTIVATIONS TO STUDY MIND-GUT COMMUNICATION in relation to HCI: ① advancement of the understanding of the mind-gut communication for people in health domain
  ② usage of mind-gut activities as data for design and evaluation of interactive systems in HCI fields.

DEFINE AS "MIND GUT COMPUTER INTERACTION" (MGCI)

SUGGEST FOUR RESEARCH AREAS: a) ubiquitous, wearable and mobile computing
b) visualisation
c) user experience design
d) HCI evaluation

→ Reaching a good level of understanding of the mind-gut communication by using HCI techniques, models and methods

LIMITED STUDIES IN THIS FIELD, PROPOSAL FOR FURTHER RESEARCH

ⓐ Use of ubiquitous, wearable, or mobile computing tools and systems for detecting and recording the mind-gut activities.

ⓑ use of visualisation techniques, systems and tools to enhance the understanding of mind-gut communication

ⓒ Development of food recommender systems that can help to provide tailored food intake advice to individuals according to their mind-gut activity data for optimising their mind-gut communications, and in turn provide an optimal user-experience of the system. Thinking of a scope for making communication from the gut more explicit, maybe in the form of visual dialogue

GUT BEING LIKE AN EXTRA PERSON AT RESTAURANT TABLE SAYING "don't eat that!" → HOW WOULD THE USER EXPERIENCE BE DESIGNED?

ⓓ The extend to which the mind-gut activities used as a kind of measurements for the usability and user experience of user interfaces and in HCI empirical studies (eg. is there unique patterns of mind-gut activities when users interact with a bad design compare to a good design?)

Scientific Framework: Gut – Brain Axis

What is the gut-brain axis?

The gut-brain axis is a bidirectional communication system between the gastrointestinal (GI) tract and the central nervous system (CNS), mediated by the nervous system, hormones, immune signals, and the gut microbiome. It links gut health to brain function, showing how changes in the gut (bacterial composition, inflammation) can influence mental health (mood, cognition) and vice versa.

How does it relate to my research?

My research focuses on creating self-tracking tools for managing chronic gastrointestinal conditions like IBS and IBD. The gut-brain axis is directly relevant because:

- These conditions are influenced by both physical factors (inflammation, gut motility) and mental states (stress, anxiety).
- Tracking and understanding the gut-brain connection can help patients identify patterns between food intake, symptoms, and emotional states, offering insights into their condition.
- Addressing the gut-brain relationship enhances the holistic management of these conditions, improving both physical and mental well-being

How can it create opportunities for new self-tracking technologies?

Integrate Mental and Physical Health

Track mood, stress, and symptoms together to reveal connections.

Provide feedback loops showing how mental states affect gut health and vice versa.

Incorporate Novel Metrics

Use physiological signals (heart rate) and gut-related biomarkers (breath tests) to provide insights.

Foster Awareness and Reflection

Design tools that help users reflect on how lifestyle, diet, and emotions interact to influence their condition.

Encourage Holistic Interventions

Combine data visualisation with actionable advice (relaxation techniques, dietary suggestions) personalised to the user's tracked patterns.

By leveraging the gut-brain axis, we can create new tools that offer patients more meaningful and comprehensive ways to understand and manage their conditions by providing a holistic approach.

Literature Review: Applications on GI chronic conditions

Article 1: Digital Health for functional gastrointestinal disorders by Pathipati M.P., Shah E.D., Kuo B. and Staller K.D

Journal Research - Secondary

Article 1 → Digital health for functional gastrointestinal disorders

- Functional gastrointestinal disorders (FGIDs) - disorders of the gut-brain interaction → amongst the most common medical conditions worldwide
- GI digital health apps fall into four categories:
  (1) symptom tracking platforms
  (2) digitally connected devices.
  (3) telemedicine
  (4) patient support groups.

(1) Symptom Tracking Platforms:

- TRACKING ALONE IS UNLIKELY TO RESULT IN IMPROVED OUTCOMES, UNLESS IT IMPACTS BEHAVIOR AND TREATMENTS. → Apps that draw correlations between stimuli and symptoms. ORGANISING SYMPTOM DATA
- Food and bowel diaries can overwhelm clinicians with a high volume of marginally useful data → digital tracking should identify and report relevant metrics.

  product example is Foodprint → "pre-visit notes"
  patients synthesise their data + questions before appointments

1. OPPORTUNITY: build on this by linking symptoms more accurately to objective data such as stool form and validated symptom severity scores, synthesising this information for efficient provider review.

(2) Digitally Connected Devices:

- Wearables linking symptom tracking apps to a health metric-enabled smartphone and correlating information such as step count, sleep, or vital signs to symptom triggers.
- KEY QUESTION - HOW THESE ABSTRACT MEASUREMENTS TRANSLATE INTO ADDRESSABLE PATHOLOGY?

(3) Telemedicine:

- Treatment of FGIDs requires extensive contact between patients and caregivers
  IBS: 2.4-3.5 million physician office visits per year.
- Remote care delivery can fill this gap. → offer the ability to access medical services.
  ↳ virtual treatment for FGIDs, slightly less effective.
  60% of patients had a clinically meaningful improvement in IBS symptoms using telemedicine.
- Cara Care Germany → symptoms improved 40.4% in first 100 days of the program.

(4) Patient Support:

- Functional GI disorders frequently cause isolation or embarrassment, leading to underreporting of symptoms or avoiding care
- Peer-to-peer support networks and group education programs improve symptoms → importance of a safe community to discuss difficult topics.

  DIGITAL HEALTH offers a way to provide patient support across time and space.
  ↳ carry RISK of spreading medically inaccurate information.

OPPORTUNITY: *Need for effective content curation to help digital support networks to realise their potential.*

FUTURE DIRECTIONS:

DIGITAL HEALTH offers promise in treatment of FGIDs but it has seen minimal adoption in GI practise. → GI had lowest telemedicine rate of adoption.

Three Major Steps for wider adaptation:

1. Clinical Studies to evaluate the impact of these interventions.
2. Business models needs to come into sharper focus → who is the customer? patient or clinical providers.
   ↳ sustainable models that will last.
3. Combine multiple applications under one roof → unified program that can customise use of tracking platforms, wearables, virtual care, and patient support based on their specific needs → fit into providers workflow both with regards to reporting relevant metrics and integrating with medical records. DIGITAL TOOLS CANNOT FUNCTION IN ISOLATION.

Limitation - widening healthcare disparities - building them on widely available

Article 2: Can Smartphones help deliver smarter care for patients with IBS by Kelso M., Feagins L.A.,

Article 2: Can Smartphones help deliver smarter care for patients with IBD?

- 70% of American Internet users have search for health-related information within the past year.
- younger patients and those with higher education levels were most likely to use the internet to gather IBD-related information.
  - searches include: information on treatmeant options and their side effects.
    - available and new diagnostic procedures
    - support groups
    - nutrition recommendations
    - complementary/alternative therapies
    - guidelines from scientific societies.
  - these patients also prefer to communicate with their physician by web-based secure messaging platforms.
- 2/3 of Americans reported interest in using mobile apps to manage health-related issues.
- more than 165 000 mHealth apps available - less than 10%. of focus on providing diagnostic tools, remote consultation, and chronic disease management
  - even fewer allow for real-time transmission of data & interaction between the patient and provider.
- Mobile applications represent a substantial opportunity for increasing health literacy, patient engagement, and compliance, particularly among patients with IBD given the young age of onset, usually 20s and 30s.

→ THINGS APPS CAN DO:

- patient education on disease & management
- remote disease monitoring: symptom tracking, medication adherence tracking, dietary logs
- earlier interventions based on tracked data: alert to medical team if symptom not on track
- Improved adherence (alarms/reminders)
- Improved Self-management/patient empowerment
- Online support network

→ Features app have:

- providing disease-related information
- symptom diaries
- nutrition diaries
- Weight tracking
- physical activity tracking
- medical administration logs & reminders
- personal health record
- remote monitoring
- less common:
  - geo-location to find restroom facilities
  - social media for IBD patients

- Majority of apps are not evidenced based, lack clinical validation, have limited professional medical involvement.
  - social media/community forums → 55% respondents were interested in obtaining IBD-related education through social media.
- Research on Telemedicine randomized controlled trial including a patient-facing app.
  - reduced outpatient visits and hospital admissions but no differences in quality of life, steroid use, flares, emergency room visits or surgeries.
- Research on "if a patient-centric self-monitoring and collaborative decision support platform will lead to sustainable improvement in overall quality of life".
  - THOSE USING THE MOBILE HEALTH APP HAD SIGNIFICANTLY HIGHER QUALITY OF LIFE.
- 1/4 of IBD patients are unsatisfied by the information they are given during diagnosis → amend by using supplementary (online) material.
  - MOBILE APPS CAN PROVIDE REMINDERS OR PROMPTS THAT WILL MOTIVATE PATIENTS TO PARTICIPATE IN BEHAVIORAL CHANGE.
  - MOBILE APPS ARE EASILY ACCESSIBLE AND BE TAILORED TO PATIENT'S CHARACTERISTICS.

Experiment 3: Case Studies on exiting apps about GI conditions

What do I want to explore?

1. Efficiency of existing apps for GI tracking
2. Explore their interfaces and interactions
3. How apps gather data?
4. How apps use the collected data and how users interact with them?
5. How users feel using tracking data and interacting with them?
6. Do apps manage to make the users form better relationships with their bodies?
7. What works in apps and what doesn't.

Apps explored in this experiment

MyHealthyGut

Tracks food and symptoms for managing celiac and gut health, with meal plans and educational resources.

IBS Coach

Tracks symptoms and habits to manage IBS through personalized plans.

My IBD Care

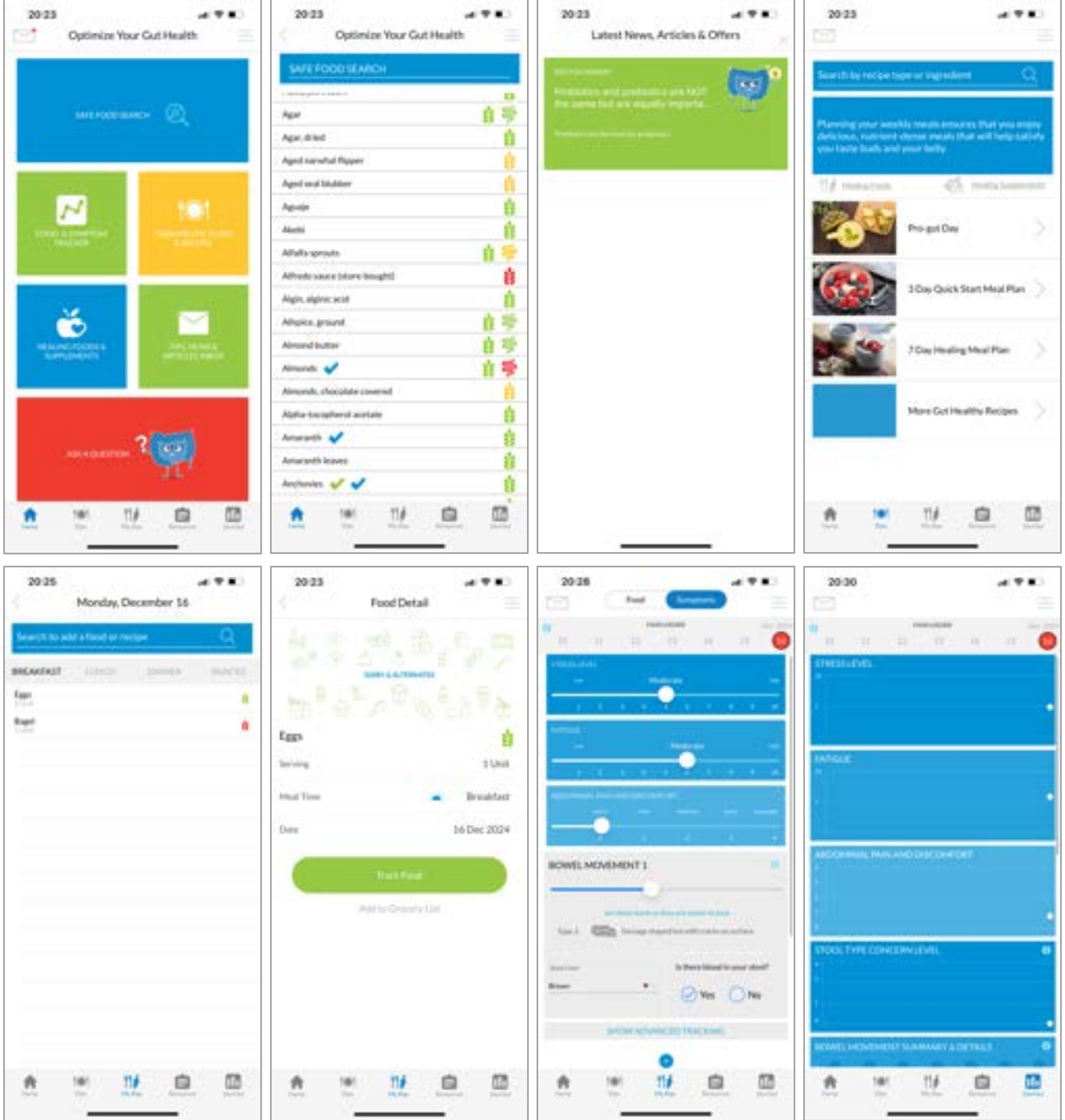
NHS-compatible app for Crohn's and colitis management, offering symptom tracking, medication reminders, secure messaging with healthcare teams, and expert courses.

How will I study them?

Step 1:
Ask 5 people to download and use two of the apps in question for 5 days

Step 2:
Make notes on their experience of using the two apps, what they think is working, what is not and what they would like different

Step 3:
Meet after the 5 days and discuss the notes to produce insights around the app's effectiveness



WEEK 12

MyHealthyGut

Interface Design:

- Not the nicest or most appealing interface.
- Navigation is not very difficult, but the system feels slow.

Food Tracking:

- Food tracking through lists is annoying and time-consuming.

Symptom Tracking:

- Symptom tracking is faster and uses helpful scales.
- Difficult to track items that change throughout the day, requiring multiple inputs.

Data Interpretation:

- Unclear how to use collected data or reports.
- Users struggle to understand what the data means and its relation to health.

Additional Resources:

- Offers recent articles and news related to conditions.
- Provides recipes, meal ideas, and healthy recipes, which are very useful.

WEEK 12

IBS Coach

Interface Design:
- Overwhelming and messy.
- Hard to navigate and not very intuitive.

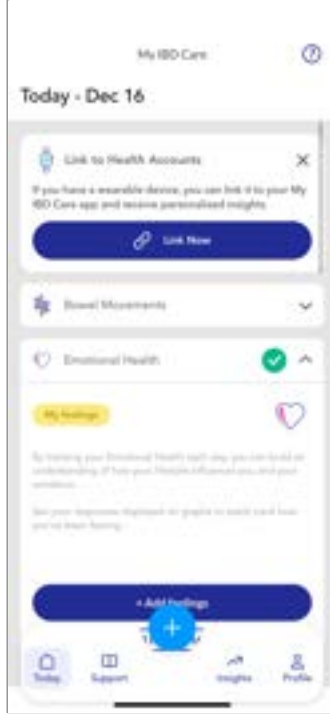
Usability:
- Feels very medical and information-heavy.

Tracking Features:
- Holistic tracking approach appreciated.
- Questionnaire format makes it engaging and helps organize thoughts.

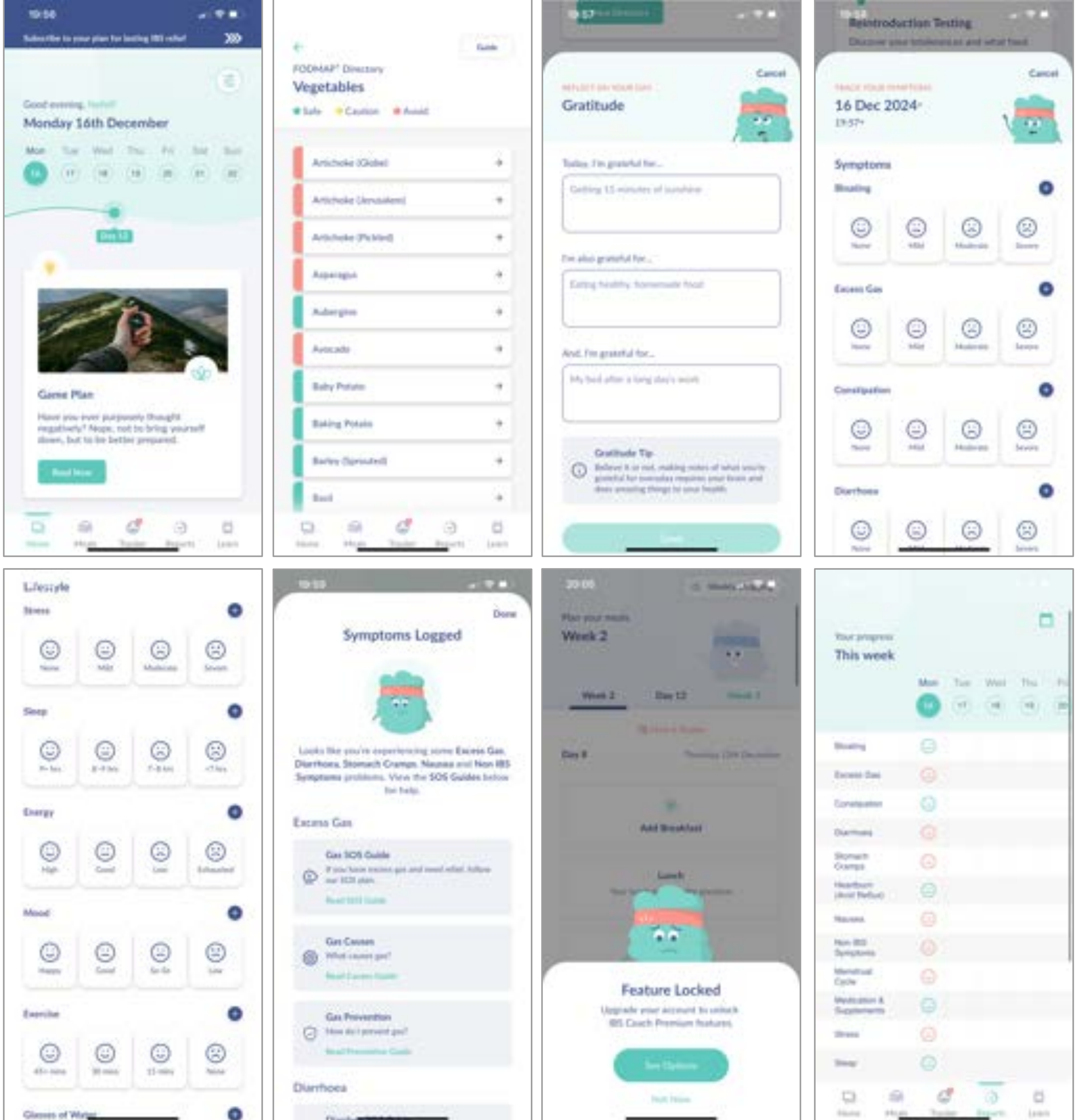
Integration:
- Integrated with NHS systems.
- Allows for tracking appointments, prescriptions, and doctor comments.

Educational Tools:
- Includes courses to educate users about their conditions.
- Provides techniques for managing and improving conditions — considered very useful.

Impact on User:
- Tracking all this data can create hyper-awareness and anxiety about health and body state



WEEK 12



My IBD Care

FODMAP Directory:
- Helpful directory outlining foods to avoid.
- Customisable through tests where users try different foods over time.

Ease of Use:
- Fast tracking is very easy.
- Overall interface is nice and fairly easy to navigate.

Symptom Advice:
- Provides advice based on symptoms, which is appreciated.

Data Interpretation:
- Unclear how to use collected data or what reports mean for the body.
- Users are unsure of the practical application of the insights.

Educational Tools:
- Offers useful educational material.
- Could be improved with more interactivity or personalized content instead of requiring users to read lengthy text.

Framework: Refining my Design Question & Research

Considering all the insights and information I have gathered from my experiments so far, I started refining my project and narrowing its scope. This lead me into refining my Design Question to make it more specific

How can a holistic tracking and monitoring tool be designed for individuals with chronic gastrointestinal conditions that integrates multiple aspects of their health, minimizes user stress, and presents data in an accessible and actionable way to enhance self-awareness and understanding of their condition?

I decided to focus on developing a tracking and monitoring tool, as I identified that tracking food, emotions, and symptoms is crucial for helping patients make connections and better understand their bodies. This includes understanding how their emotions affect their GI health and vice versa, as well as how social aspects , diet and other unexpected factors can impact their condition.

Based on the insights gathered so far, the challenges in developing this tool will be:

1. Making the collection and interaction with data as stress-free and engaging as possible. This will allow users to form meaningful relationships with their bodies and truly understand them—something that existing products fail to achieve.

2. Ensuring the tool gathers holistic data that focuses not on individual organs but on the entire system as a whole.

3. Communicating data in a meaningful and accessible way.

To address these challenges, I need to conduct further research into how data is collected, used, and communicated back to the user. I will divide my research into two parts:

Data Input – Exploring methods for collecting and inputting data.

Data Output – Investigating effective ways to communicate data to the user.

Experiment 4: Exploration on collection data interactions

What do I want to explore?

1. What is the **fastest** interaction for manually inputting tracking data

2. What is the **most intuitive and least stressful** interaction?

3. What would make the inputting data **more engaging**?

4. What methods of inputting and tracking data people prefer? Why?

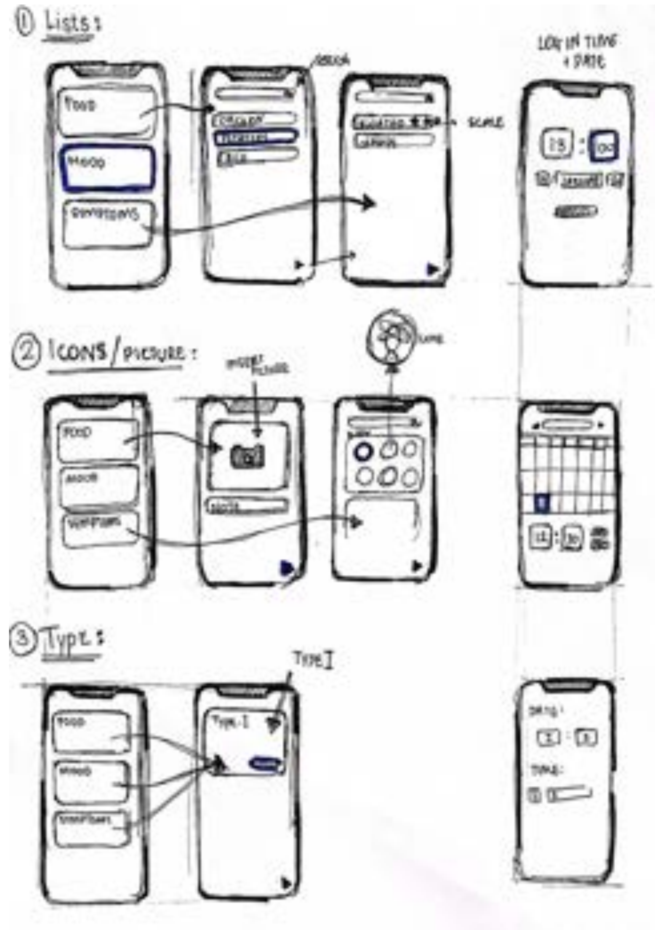
5. How people feel about **voice commands**?

How will I explore it?

1. Create **interactive interfaces** that participants can explore for a few days to determine which way of tracking data they prefer

2. Ask participants **to share their thoughts** around the interactions and see if there are other interactions that they might prefer and were not considered

Interaction I am exploring:



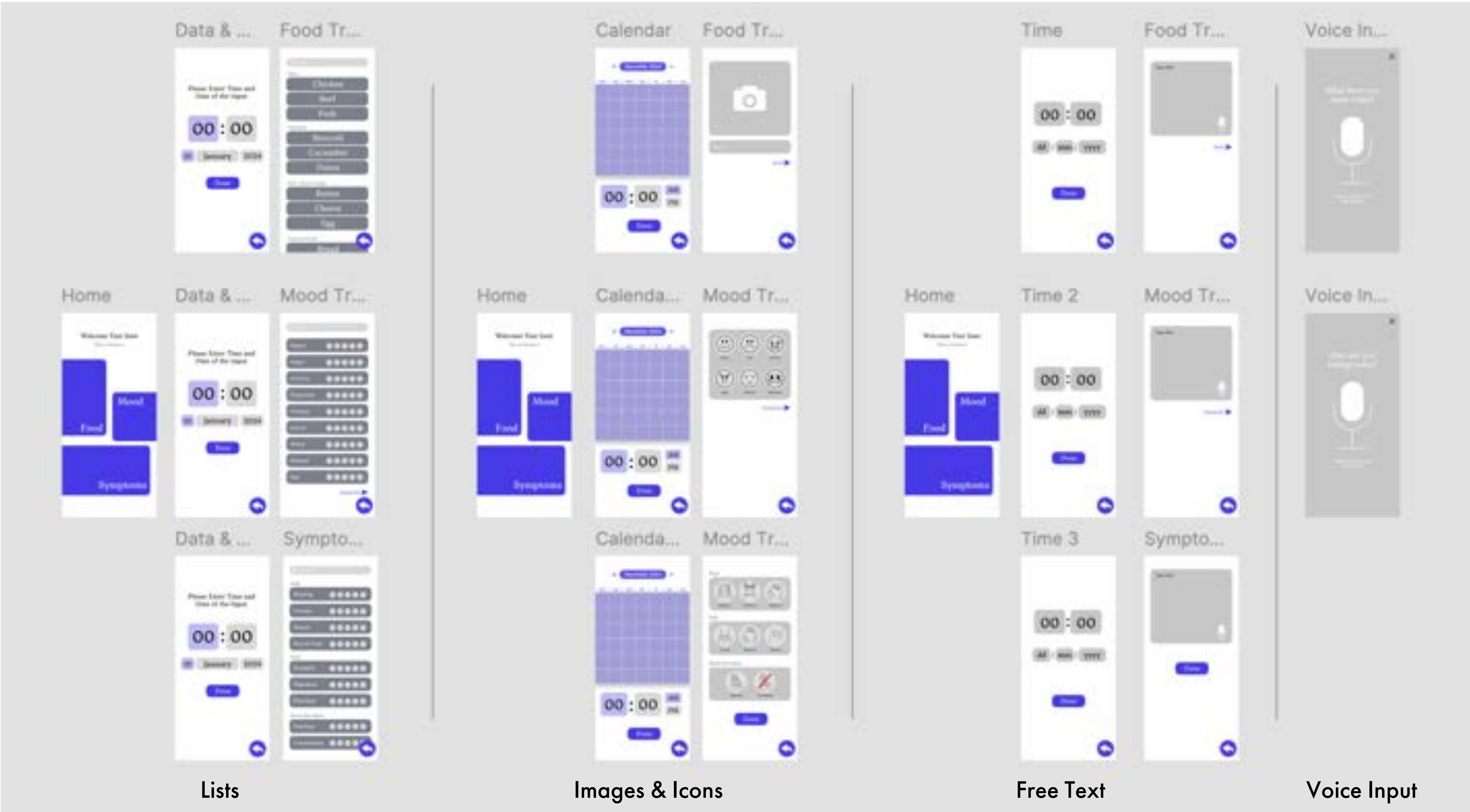
Providing **lists of foods, symptoms and mood**, that the user can go through and choose what it applies to them for a particular time in their day.

Providing the option of **uploading images of meals** consumed, while **illustrated icons** are provided to choose from for symptoms and mood.

Providing only a box where the user has the **freedom to note down/type** food, symptoms and mood as they please.

Providing the option of inputting food, symptoms and mood using **voice commands**.

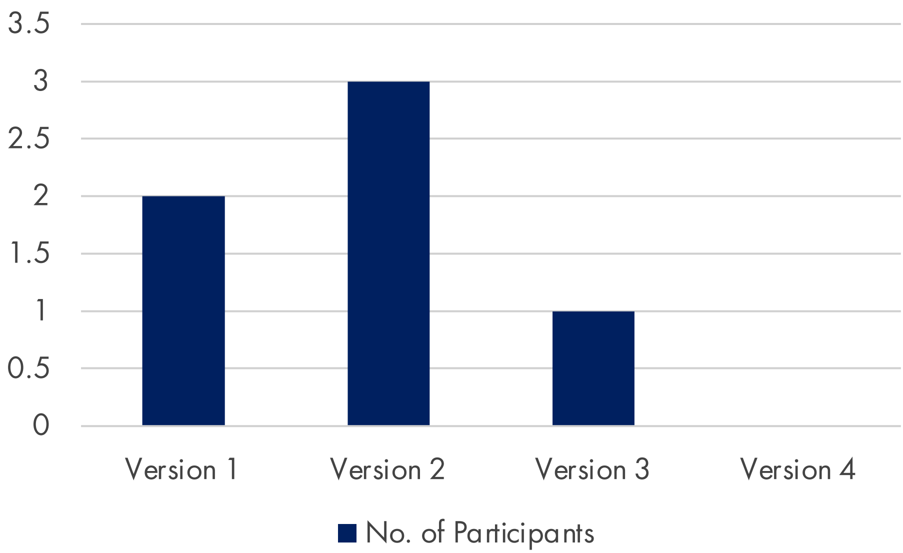
Prototyping: Producing Interactive Interfaces



I used Figma to create the different interactive interfaces in the form of an app that I can share with the participants of the experiment. It is important to mention **that I am not planning to create an app, I just designed this as it was an easy and fast way to test interactions and logging of data that can later translate into a physical product.** I shared the link for these interfaces with participants that were asked to use them for a few day and asked them to reflect on their experiences by answering a few questions.

Results & Insights

What version participants prefer?



How to make it more engaging:

- **Notifications**-to ensure that people won't forget
- **Incentives**- some kind of streak or reward mechanism to make people keep tracking
- **Gamification**- gamify the tracking process to make people more like to keep doing it.

Other recommendations:

- Some sort of portable buttons that can be carried around and pressed when using the toilet or experience pain in order to make the process of logging data even faster and hassle-free
- Use questions for tracking to make it easier and get more specific input- might make the interaction even faster

Insights on the different Versions:

Version 1: Lists
- Fastest option, requiring the least detail.
- Simple and quick; breaking data into steps makes it more digestible.
- Ensures required data is provided.
- Ready-set formats are ideal and preferred.
- Taking a picture feels like an extra, unnecessary step.
- Pictures may lack visibility or clarity of food items.

Version 2: Icons/Images
- Interactive and engaging — "like creating a library of food."
- Fastest and most sustainable method with minimal effort.
- Provides quick results.
- Less overwhelming than free text or lists.
- Food lists are complicated due to too many options.
- Users feel forced to use other input methods eventually.
- Seen as the most sustainable long-term option.

Version 3: Free Text Box
- Provides complete freedom to input any level of detail.
- Recognizes that lists can’t cover all foods, and pictures may lack clarity.
- Ideal in combination with icon-based options.
- Not preferred for symptoms and mood tracking.

Version 4: Voice Commands
- Generally liked as a feature, but not as the only input option.
- Users don’t want to rely on voice commands for every data input.
- Prefer voice commands for feedback rather than inputting data

Existing Work: Wearables and Sensor-based tools for health management

Whoop 4.0 by Aruliden

The Whoop 4.0 is a screen-free wearable fitness tracker designed by Aruliden in collaboration with Whoop. It continuously monitors key health metrics, including sleep, heart rate, skin temperature, and heart rate variability, using advanced sensors like an oximeter and photodiodes. The device tracks data 24/7, seamlessly integrating into daily life without the need for a display. Its innovative design allows users to wear it at all times, offering personalized insights via the companion app, making it a comprehensive fitness coach.

Galaxy Ring by Samsung

The Samsung Galaxy Ring is a cutting-edge wearable designed to track a wide range of health metrics 24/7. It monitors sleep quality, including signs of sleep apnea, and supports period and fertility tracking through a partnership with Natural Cycles. The ring also integrates with Samsung Health to provide a personalised "My Vitality Score," combining data from sleep, activity levels, and heart rate variability. AI-driven insights help users optimize their wellness, making it a comprehensive tool for health management.

Momma's Kit by Leadoff Studio

The Momma's Kit, created by Spora Health and Leadoff Studio, is a health monitoring system designed to reduce Black maternal mortality rates. The kit includes a blood pressure cuff and pulse oximeter to help mothers track vital signs during pregnancy and the "fourth trimester." Featuring an intuitive design with bold, graphic instructions, it provides a supportive, inclusive experience for users. The kit aims to promote daily monitoring, empowering women with vital health insights while fostering trust in healthcare.

Experiment 5: Exploration on Collecting Data through Sensors

What do I want to explore?

1. How can sensors be used to gather data about the body ?

2. How sensors work and what data they collect?

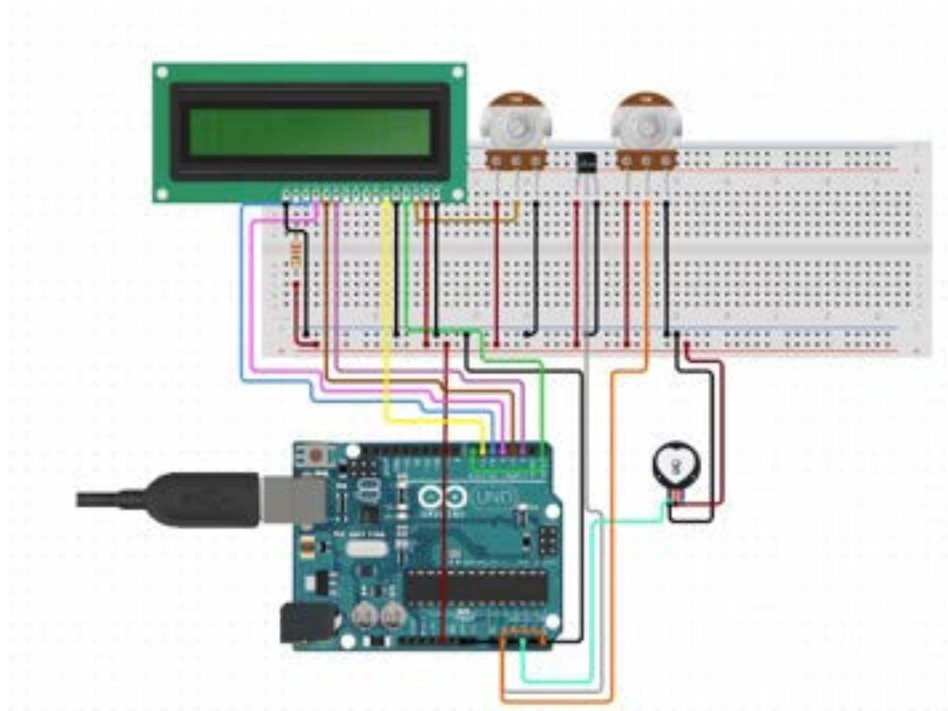
3. How people interact with the sensor?

4. What kind of sensor can be used for GI conditions?

How will I explore it?

1. **Use an Arduino Uno, a pulse sensor and a temperature sensor and code a prototype that tracks heart rate and body temperature.**

2. Explore the process of making something like that, how accurate it might be, what output it gives and how people will interact with it



Why tracking vitals is important for GI management?

- **Heart rate** monitoring in chronic GI conditions helps detect stress, autonomic dysfunction, dehydration, and inflammation, providing early signs of flare-ups.
- **Body temperature** tracking helps identify fever, infection, or inflammation, signaling flare-ups or complications in GI conditions like IBD or IBS.

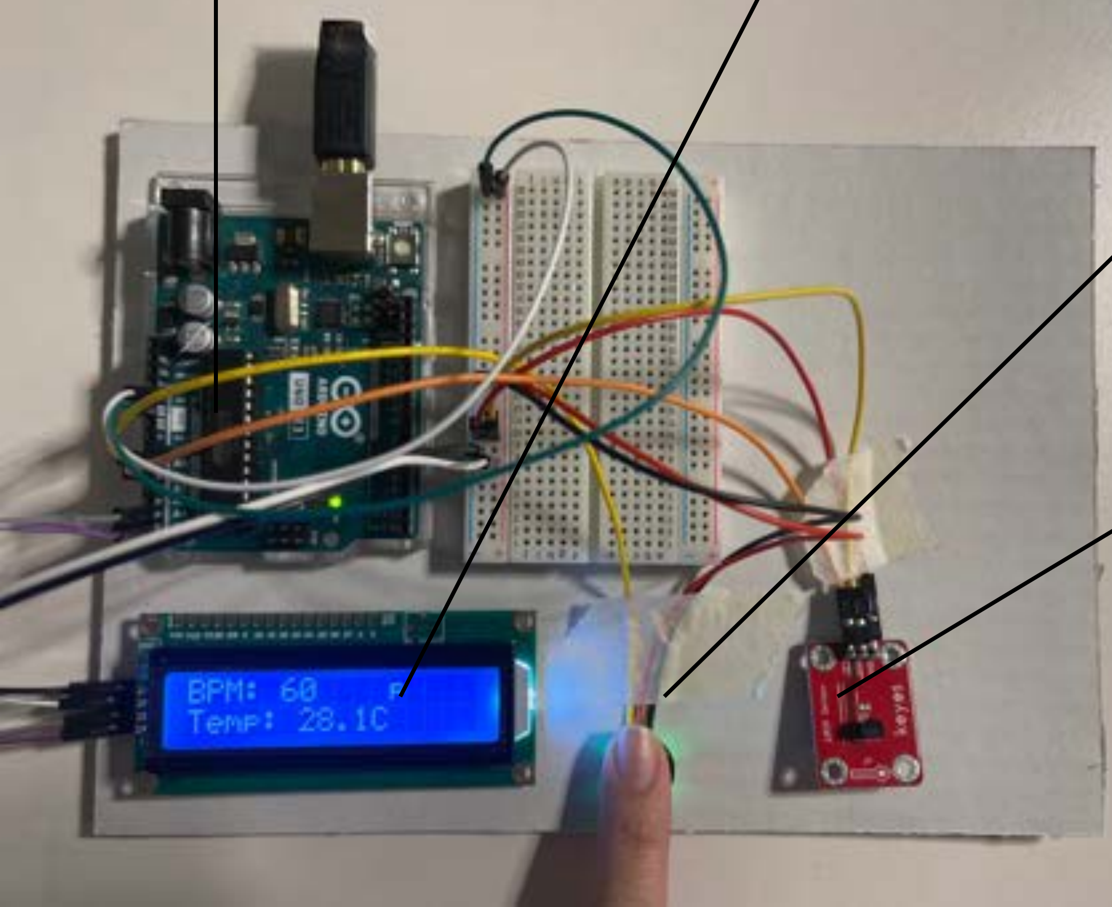
Prototyping: Calculating heart rate and body temperature

Arduino Uno
Used as the main processor of the system that analyses the sensor input and produces outputs. It is coded in C++ using Arduino IDE

LCD Display Screen
The screen is used to display the input of the sensors in a readable manner. This can be replaced with other sensors such as light that maybe portray the data collected in a less literal manner

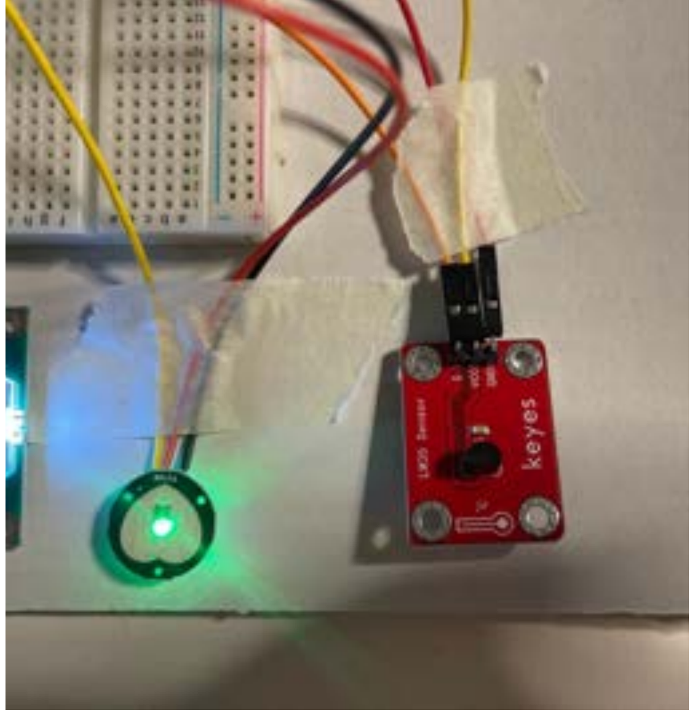
Pulse Sensor
The pulse sensor is used to calculate the heart rate of the user in beats per minute (BMP) by placing a finger on the sensor. It can also produce graphs, but it usually needs some time to calibrate.

LM35 Sensor
The **LM35** sensor is a temperature sensor that can be used to calculate body temperature in Celsius. However, measuring from the finger or wrist will not be accurate, meaning that the only accurate readings would be produced from the armpit.

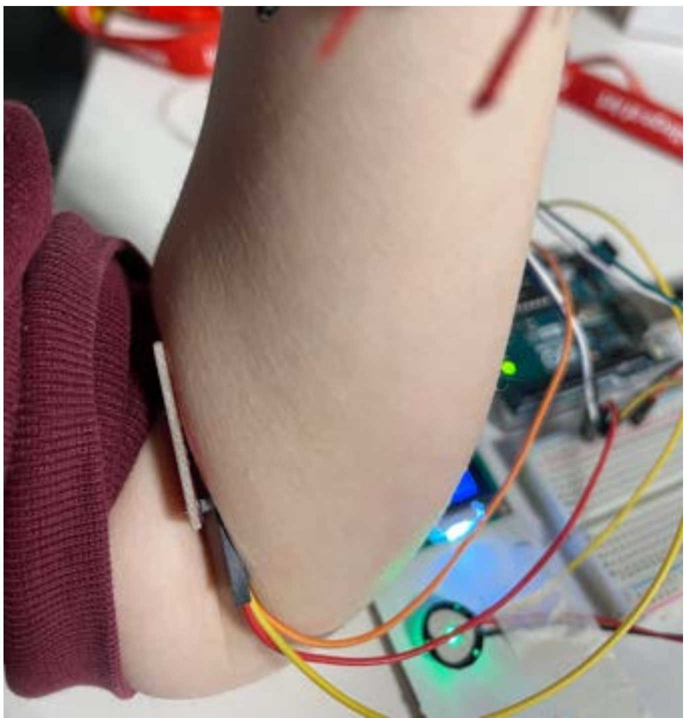


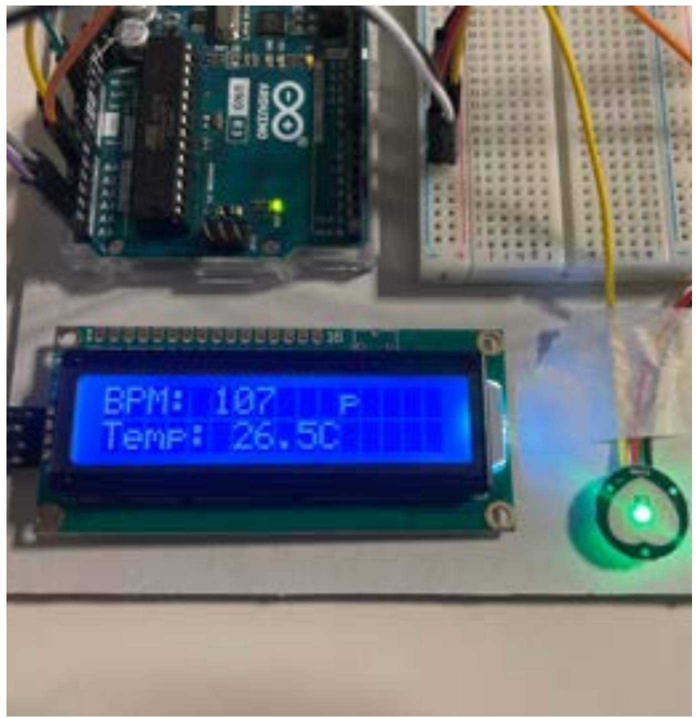
WEEK 13

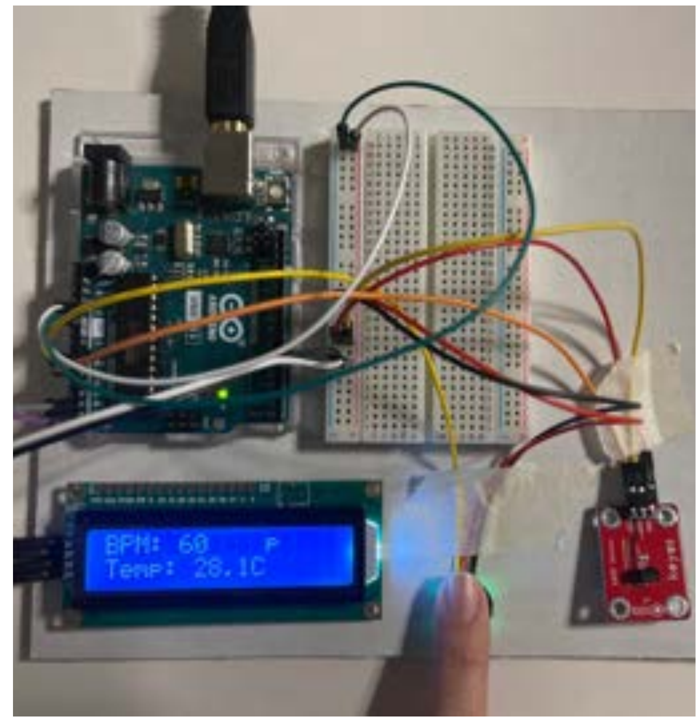

BPM: 59 P
Temp: 26.8C


keyes


BPM: 4 P
Temp: 29.9C




BPM: 107 P
Temp: 26.5C


BPM: 60 P
Temp: 28.1C

Prototyping: Inputting Data using Buttons

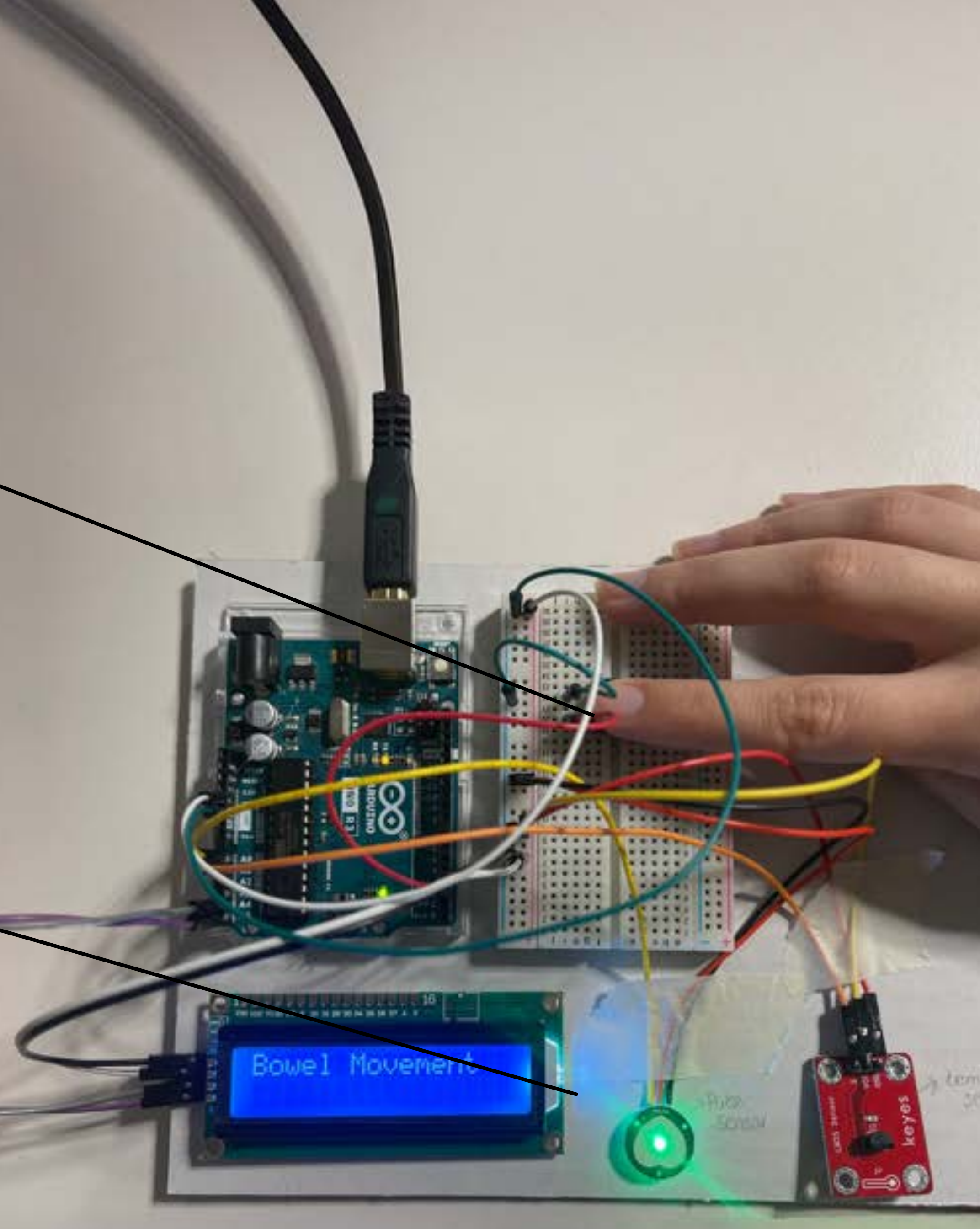
I decided to **take on board some of the participant feedback** I gained in the previous experiment and explore the idea of using buttons for inputting/gathering data

Buttons

Adding buttons that are **used to input/gather data in a manual manner but make the process faster.** Specific buttons can be colour coded and mean different things such as bowel movement or pain.
In a real-life scenario, these buttons can be carried around in the form of a wearable and users can press them when they go to the bathroom or when they experience pain, to log in the information easily without needing to go through the interactions that were explored in experiment 4.

LCD Display Screen

The screen here is just used to display what pressing the button means and, in this case, confirm that the input was received. This can be replaced with other sensors such as light or just by sending this input to an app or a device.



Conclusions: Gathering/Inputting Data

Conclusions:

The collection of data needs to be as **fast and hassle free** as possible.

Participants believe that less interaction make the product more engaging

Having **choices to select** from for the manual data input seems to be preferable

Participants like voice output but not always voice inputs

Visual options are generally most liked

Sensors **for automatic data tracking are preferable.**

Still there is **data that need to be collected in a manually**

New Research Questions Created:

1. How do we portray the data that are collected manually and through the sensors?
2. **What does the collected data mean and what does it tell people about their condition and their body?**
3. How does BPM, temperature or inputting of bowel movement communicate what is happening to the body?
4. How can this data be portrayed in a non-overwhelming manner that stresses people more than helping them
5. **How complex systems can be reduced to a single metric?**
6. **How to turn those numbers into comprehensible, simple feedback/ advice that allows people to understand their condition holistically and enhances the relationship between the user and their bodies?**
7. **How do we move away from numbers, graphs and over medicalised output that employ a reductionist data – driven approach ?**
8. Does it need to become less literal, become more philosophical or more reflective, conceptual?

Literature Review:

Article 1: Designing with Emerging Science by Jenkins T., Boer L., Homewood S., Almeida T., and Vallgårda A.

Article 4: Designing with Emerging Science: Developing an Alternative frame for Self-Tracking

- Self-tracking technologies facilitate the production of knowledge about the body through the collection of data: → DOCUMENT THE DATA OVER TIME IN RELATION TO INNER BODY PROCESSES SUCH AS... DIET.

- Rather than redesigning existing devices or building upon already existing ways of tracking the body → start with emerging knowledge about the body being brought into new design spaces, BRINGING NEW OPPORTUNITIES FOR DESIGN.

RELATION BETWEEN GUT AND BRAIN → GUT-BRAIN AXIS | new finding / medical science.

refers to biochemical signals sent between the GI track and the central nervous system. Link composition, diversity & activity of gut bacteria to particular moods & emotions as well as mental diseases and disorders

Our mental state are influenced by the bacteria in our gut, and vice versa.

INTERACTION DESIGN RESEARCH → engage with emerging science on gut-brain axis

"what might be a device that tracks the relationship between the gut and mental health looks like?" "how gut-brain science can be materialised in self-tracking technologies" "how self-tracking can be productively reframed"

- SIGNALS FROM THE GUT CAN REACH DIFFERENT PARTS OF THE BRAIN USING NEUROTRANSMITTERS. NEUROTRANSMITTERS REGULATE MOODS, EMOTIONS, MENTAL HEALTH.

- EXISTING APPS + PRODUCTS: CaraCare, Poop tracker, Atlas Health, Viome, FoodMarble, Mady

doing speculation on how this emerging science should be applied. assumption about how the body should be imagined, what tracking the body is for, and how data from self-tracking should be utilised.

GUT FEELING PROJECT:

Explore potential applications for self-tracking the gut-brain axis and aims to reflect on what it means to live with a technology that tracks the gut-brain connection.

- Gut Feeling 300 Probe

challenge the reductionist data-driven approach of commercial self-tracking devices that reduce bodily complexity to a single numerical value.

ABSURDITY OF REPRESENTING THE GUT-BRAIN CONNECTION THROUGH A SINGLE NUMERICAL NUMBER.

GAS SENSOR

DISPLAY

GIVE A VALUE FROM 0-254 that REPRESENTS GUT HEALTH.

REFLECT ON DEVICE'S OUTPUT + HOW IT AFFECTS THEIR MENTAL + PHYSICAL STATE.

5 Experts lived with this probe for 2 weeks → REFLECTED ON 17

- resisted probe's reductionism
- expressed discomfort with self-surveillance aspect + added unnecessary responsibility for their bodies.

CONCERNS: Hyper-awareness and anxiety about health, SELF-SURVEILLANCE

PROPOSED: ① more philosophical devices for exploring GUT HEALTH
② reflective tools that promote introspection and mindfulness
③ device that visualises internal biochemical processes, reducing stigma / taboos.

DESIGN CONCEPTS DERIVED:

① PHILOSOPHER'S MACHINE: mechanical device that combines environmental + gut data to generate reflective prompts (eg. words) that encourage philosophical engagement with gut-brain complexity.

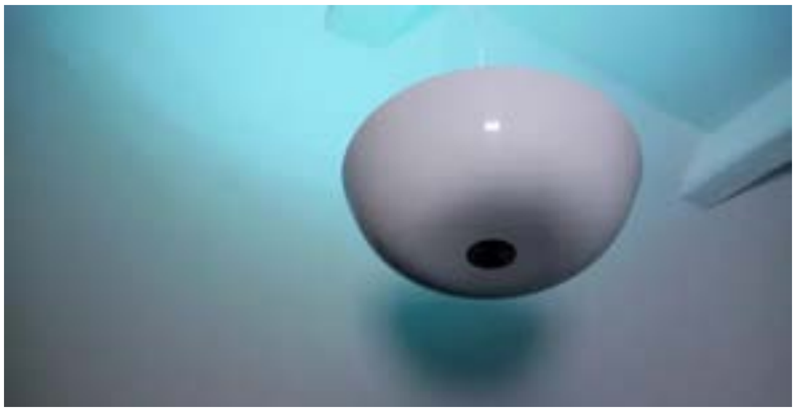
② REFLECTIVE FIDDLING MOMENT: a tactile device that changes based on gut health, reflecting the user's emotional or mental state to encourage embodied interaction with gut-brain data.

③ EXTERNAL MICROBIOME ACCESSORY: wearable or displayable device that incubates bacteria from the user's breath and visualises them, fostering curiosity and breaking taboos around gut health.

Existing Work: Data Visualisation/Communication

After some initial research on how to use gathered data to help GI patients form better relationships with their bodies, I started looking into more unconventional ways of communicating data that may create new opportunities. I started looking into existing project that take new approaches in portraying collected data that moves away from number, and graphs and create more reflective, conceptual, stress-free experiences.

Ambient Cycle by Sarah Homewood



"Ambient Cycle" by Sarah Homewood serves as a great example of visualising data in relation to the menstrual cycle. It uses an app and a Wi-Fi connected LED strip, gradually shifting colours in accordance with the user's menstrual cycle data. This project looks into **conceptualising one's body by constructing a data-based representation of oneself.** It utilises external environmental changes to mirror internal bodily changes, **aiming to explore how self-tracking affects the way people relate and experience their bodies.** Additionally, the project challenges taboos surrounding menstruation by transforming a sensitive and private aspect such as the menstrual cycle into a public display accessible to anyone.

Nowatch by Hylke Muntinga & Timothée Manscho

Nowatch by Muntinga and Manscho is a great example of data communication through a screen-free health tracker designed to **reduce stress and improve overall wellbeing**. Unlike traditional smartwatches, it **uses gentle vibrations to communicate key health information, preventing overstimulation**. By measuring physical metrics like heart rate, movement, and stress levels through electrodermal sensors, it offers personalised insights without the distraction of constant notifications. The device **promotes mindfulness and helps users manage stress through supportive vibrations and guided breathing exercises**, allowing for a calm and focused approach to health monitoring.

Peer Review Reflections

Peer review sessions throughout the semester provided essential feedback that helped refine my research around making self-tracking tools for managing chronic gastrointestinal (GI) conditions.

Early feedback encouraged me to move beyond a medicalised, data-heavy perspective and adopt a more holistic approach, integrating mental health into my product as everyone believed that it is particularly relevant

Peers kept stressing the risk of overwhelming users with too much data. We discussed their experiences with tracking apps, which gave me insight into what they typically dislike. This feedback led me to experiment with intuitive input methods such as lists, icons, and voice commands to ensure tracking remains simple and stress-free. Through these conversations, we consistently returned to the importance of creating engaging tools that feel supportive, not burdensome, and do not take up much time.

An interesting piece of advice I received while exploring interfaces was to look into sensors and their usefulness, as they play a significant role in health management. A peer who had just purchased a Whoop band—a product I was already referencing—explained how beneficial sensor-based tracking has been for their health management, providing me with valuable insights into the practical applications of this technology.

Another key topic we discussed was the importance of accessible, less medicalised data outputs. At that point, I was already exploring sensors and wearables, and a classmate recommended the Nowatch as an example of a product that communicates data in a stress-free, user-friendly manner. This suggestion was particularly insightful and aligned well with my project goals.

Unit 3: Advanced Practice

Idea Generation

Based on the research conducted and insights gathered from both experts and users, I began developing potential product ideas that could respond to **user needs and effectively address my design question**. It was important to generate concepts that focus not only on how data is collected, but also on how it is presented, ensuring that the entire experience remains as **stress-free as possible.**
I initially explored a range of vague, broad design solutions and gradually refined the ones that showed the most promise. I focused on the **feedback and ideas I had gathered from my interviews as inspiration.**

IDEA GENERATION

Idea 1

Interactive Light Panel

Wall-mounted or bedside light panel that changes colour and brightness to reflect gut health trends passively. Different hues could indicate digestion quality, stress level, or inflammation trends

Idea 2

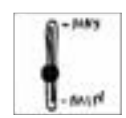
Adaptive Fabric/Haptic Feedback

- A scarf, blanket or waistband that subtly shifts texture or warmth based on digestion and stress-levels -> maybe watch in this case (wristband)
- Wearable belt that provides gentle feedback based on gut stress levels (warmth, light, pressure)

Idea 3

Analogue Dialogue/Physical Tracker

A physical dial or barometer like device that slowly shifts over time to reflect digestion and stress patterns



GUT HEALTH: What does it mean?

Idea 4

Bio-Responsive Tableware

Plate, bowl or utensils that give real-time feedback on food choices and digestive reaction by utilising Artificial Intelligence. *It connects with an app to showcase the feedback* and also uses vibrations to communicate it through the product.

Idea 5

Tactile Mood and Symptom Object

Small, handheld object (pebble) that **allow users to log their symptoms through touch.** It then subtly changes temperature, texture, colour or vibration patterns over time to communicate health patterns. *Connect with app to save input and process feedback.*

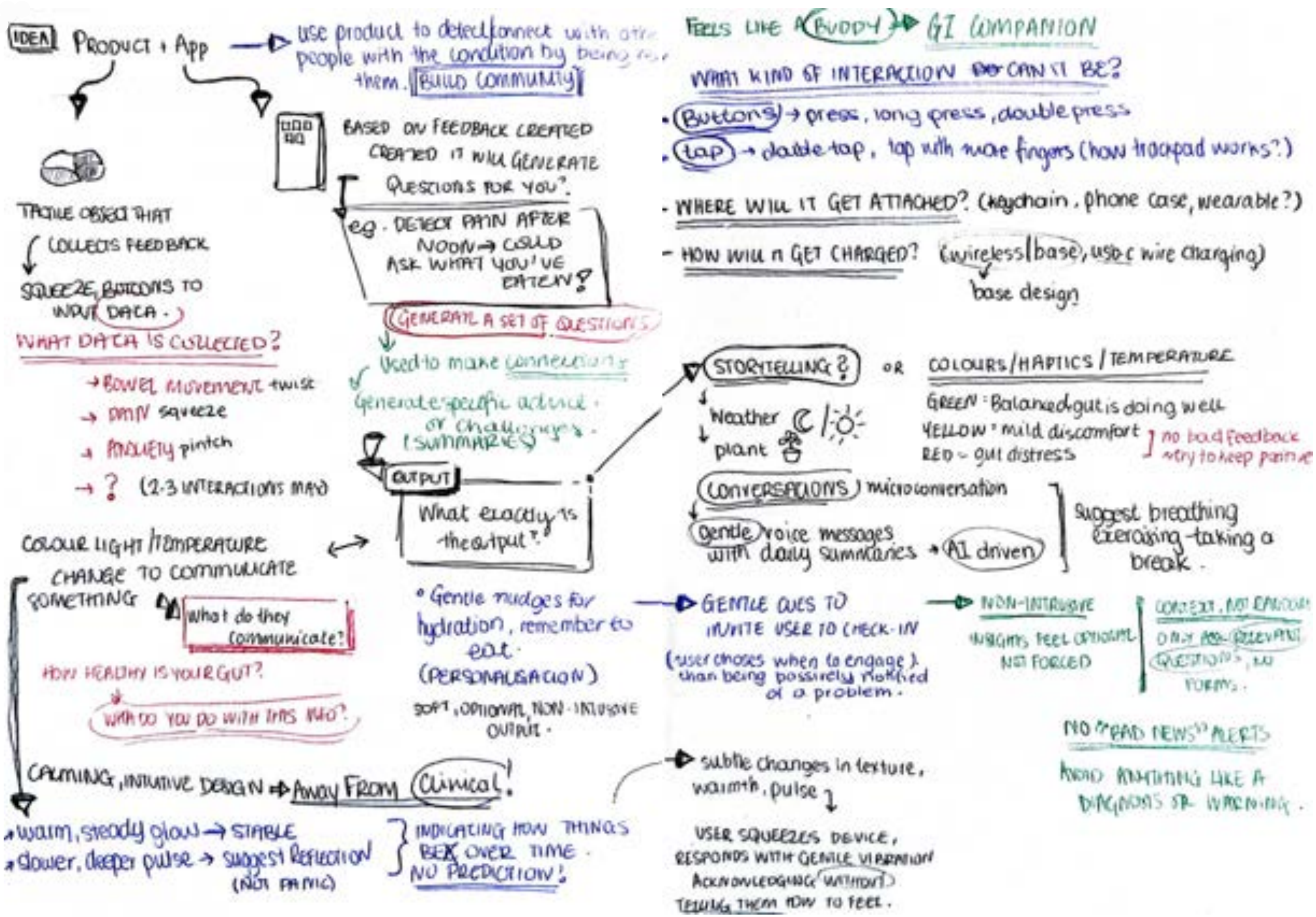
Idea 6

Interactive Mirror Feedback

A bathroom bedroom mirror that subtly overlays feedback about gut health without overwhelming visuals
-> How do we collect the data that it is portrayed?

Idea Development: Figuring out Functionality

Having chosen the concept of a handheld device connected to an app as my starting point, I continued developing the idea by making detailed notes. I explored key aspects such as **how the device might function, how specifically the data would be collected, and what the output would look like—whether it would be displayed on the device itself or within the app.**



WHAT?

A portable, tactile device that enables users to log symptoms through simple, natural gestures. Users can squeeze, twist, pinch, push, or tap the device to record experiences such as pain, bowel movements, or anxiety.

When symptoms like anxiety are detected, the device will respond in real time by providing light pulsing feedback to help the user relax.

The device connects to an app that processes and stores the collected data. At the end of the day, the app will prompt the user with a series of reflective questions, based only on the symptoms that were logged earlier.

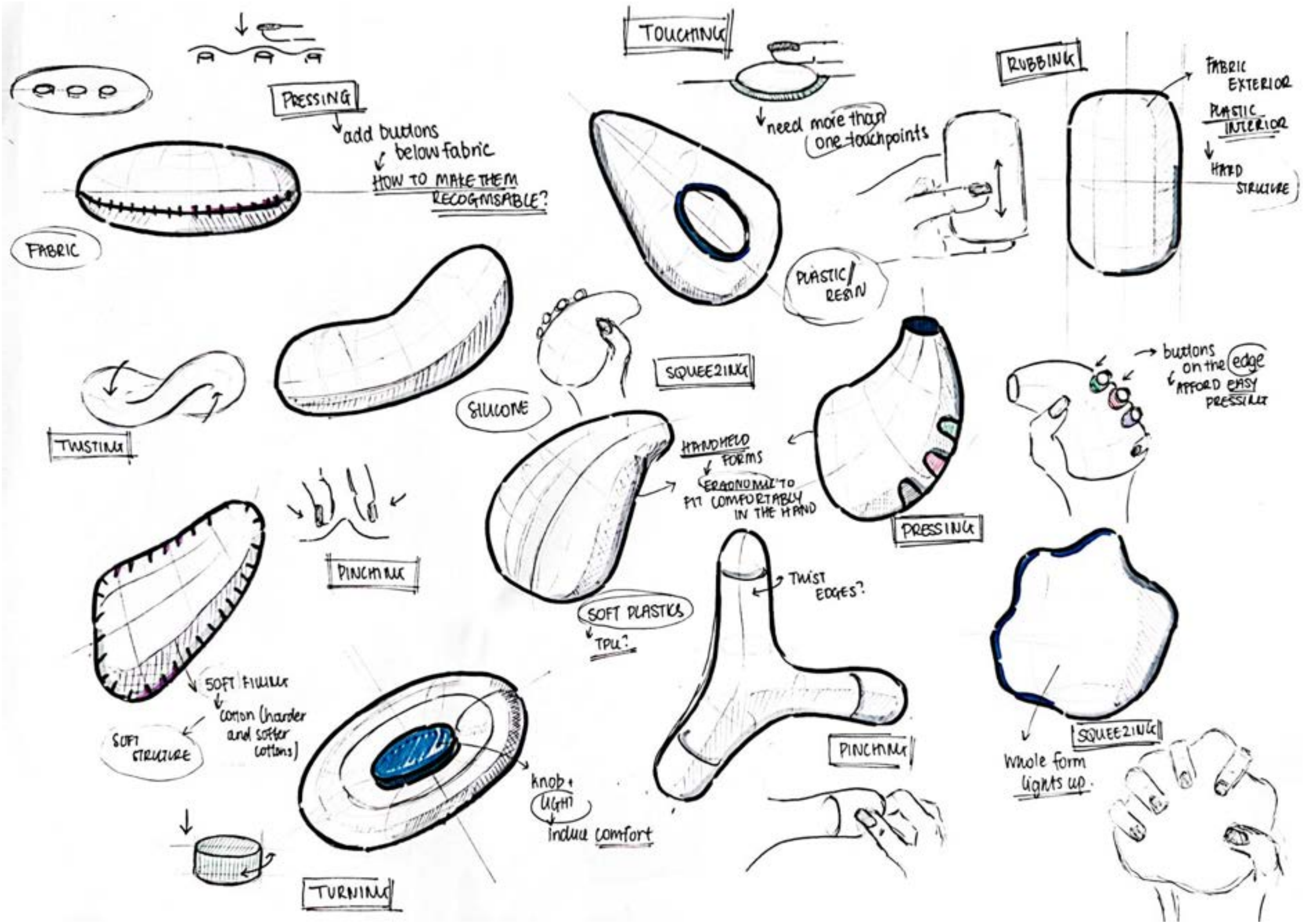
This allows the user to reflect on their experiences and identify possible triggers and patterns that either improve or worsen their condition.

The app will then analyse the user's responses and present insights in the form of simple notes and notifications, making the information easy to understand and act on.

Moodboard: Initial Inspirations

The first step was to start considering what the design of the product might look like. I began by creating a moodboard for inspiration. The idea is to develop **a fidget-like, handheld device that is small, fits comfortably in the hand**, and features **an organic, ergonomic form**. Since I initially wanted the device to have a soft feel, I explored materials such as soft plastics and fabrics, and looked into forms that would naturally afford squeezing. It was also important for the **design to evoke comfort and safety**, while avoiding an overly medicalised appearance.

Design Generation: Sketching



Interactions: Prototyping & Testing

Initially, my idea was to collect data input through different gestures. I envisioned a soft device where squeezing would log pain, twisting would log anxiety, and pinching would log a bowel movement. To test these interactions and how they might work, I created some initial prototypes using fabric and cotton filling.

I gave these prototypes to users and asked how they felt about logging their GI symptoms through these interactions. Although users initially enjoyed squeezing and twisting the device—finding it a very satisfying, anti-stress experience—they expressed **doubts about using these gestures for logging symptoms.** They mentioned a sense of **distrust toward the gestures, as they couldn't be sure whether the action had been registered successfully.** Additionally, they noted that the gestures were so satisfying that they would likely perform them even when they didn't have symptoms to log, leading to inaccurate data collection.

This feedback indicated that **gesture-based interactions might not be the ideal method for passively collecting reliable symptom data from users.**

I decided to further test the feasibility of this idea by experimenting with an FSR force sensor to see how it could detect different gestures. However, this approach quickly proved to be very complex, **particularly due to the difficulty of accurately mapping different gestures using these sensors.** Distinguishing between gestures such as squeezing and twisting was challenging, and the output was often inconsistent and inaccurate.

A discussion with robotics specialist Freddie Hong confirmed these challenges. He advised against relying on gesture recognition for input, noting that even in advanced robotics, accurately detecting and distinguishing gestures remains difficult and prone to errors. Instead, he recommended using buttons or touch sensors, which are simpler to implement and provide clear, reliable outputs. This advice was also well received by users, who had already expressed concerns about gesture-based interactions. They found that **buttons provided more immediate and clear feedback, giving them greater confidence that their input had been successfully recorded.**

Key Tracking Areas: Interview with Gastroenterologists II

To determine the most important tracking areas and understand how to implement them in my device, I contacted the gastroenterologist I spoke with during the previous semester. We had another conversation focusing on the aspects of daily life, habits, and health that are directly linked to gastrointestinal conditions and are vital to track to identify meaningful patterns and connections. According to the expert, the critical areas to track, ranked by importance, are:

Diet and Nutrition
Most critical, as food frequently triggers IBS symptoms. Accurately logging food intake and reactions is essential to identifying problematic ingredients.

GI Symptoms
Tracking symptoms such as pain, bloating, and discomfort allows patterns to be identified and clarifies how lifestyle factors influence these symptoms.

Bowel Movements
Frequency, consistency, and urgency are central to IBS and related conditions. Standardized tools like the Bristol Stool Chart facilitate accurate monitoring.

Stress and Emotional State
Due to the gut-brain axis, emotional fluctuations significantly impact symptom severity, making this a high-priority tracking area for holistic management.

Medications and Supplements
Monitoring medication intake and bodily responses helps evaluate effectiveness, side effects, and interactions with symptoms.

Sleep Quality
Poor sleep exacerbates pain sensitivity, inflammation, and stress, indirectly affecting symptom severity.

Physical Activity
Regular movement improves GI motility and reduces stress. Tracking extremes, such as excessive inactivity or overexertion, is beneficial.

Hydration
Hydration tracking is relevant but typically secondary unless dehydration or excessive caffeine/sugar intake triggers symptoms.

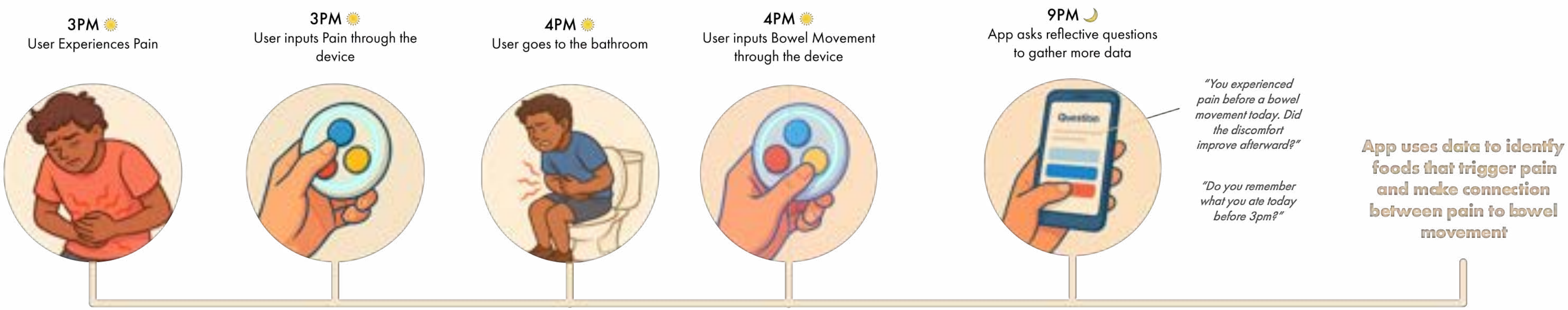
Environmental & Routine Changes
Factors like travel, changes in routine, or even weather variations can affect symptoms. However, these are context-specific and challenging to track consistently

I **selected anxiety, pain, and bowel movements as primary button inputs,** as these symptoms occur spontaneously, and are straightforward to identify. These inputs are also among the most critical symptoms and often result from other tracked factors. Thus, logging these symptoms enables the app to generate targeted questions about preceding events or conditions, thereby effectively monitoring other essential factors.

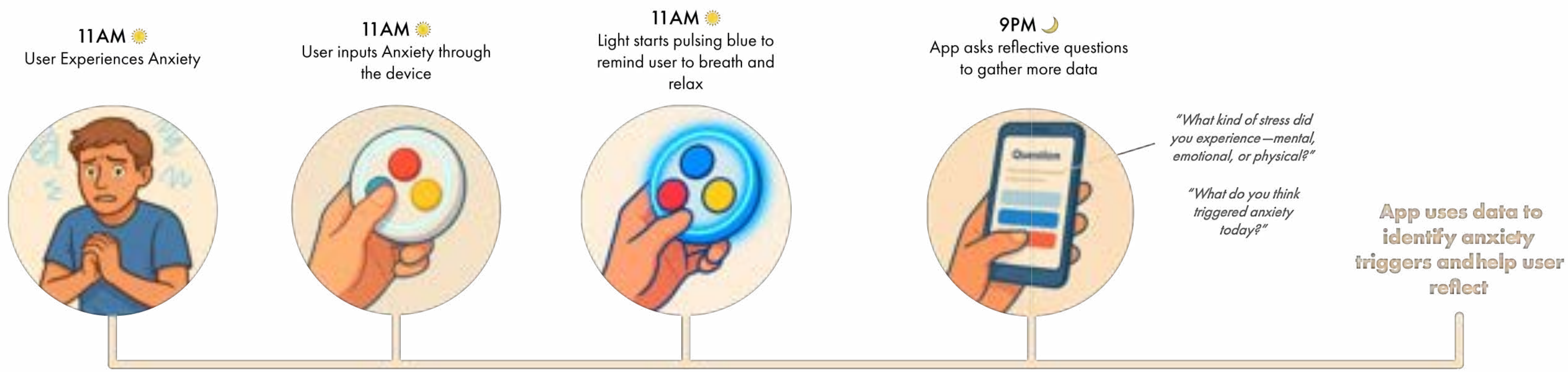
User Map: Product Functionality

At this stage, I created a map to **clearly explain how the device would work.** This was intended to make it easier for others to understand and easier for me to explain the concept to test users from whom I would gather feedback. I developed two different scenarios to illustrate the different ways the device might function and portray the full potential of the idea.

Scenario 1



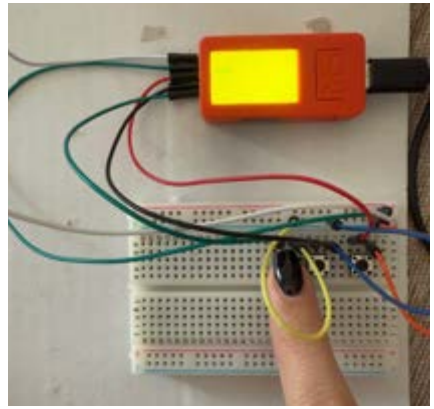
Scenario 2

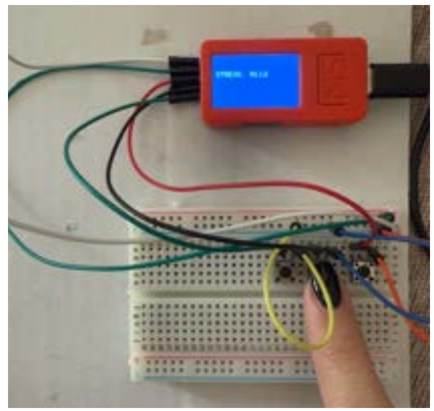


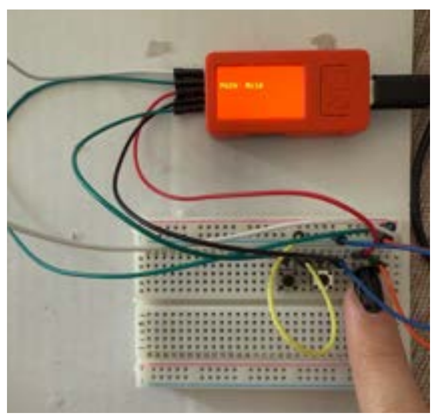
Code Development: Testing & Iteration

Iteration 1: ESP32 & Button Input

```
  if (digitalRead(pinPain) == LOW) {
    Blynk.virtualWrite(V0, 1);  // Pain
    M5.Lcd.fillScreen(RED);
    M5.Lcd.setCursor(10, 40);
    M5.Lcd.print("PAIN");
    delay(1000);
    Blynk.virtualWrite(V0, 0);
  }

  if (digitalRead(pinAnxiety) == LOW) {
    Blynk.virtualWrite(V1, 1);  // Anxiety
    M5.Lcd.fillScreen(BLUE);
    M5.Lcd.setCursor(10, 40);
    M5.Lcd.print("ANXIETY");
    delay(1000);
    Blynk.virtualWrite(V1, 0);
  }

  if (digitalRead(pinBowel) == LOW) {
    Blynk.virtualWrite(V2, 1);  // Bowel
    M5.Lcd.fillScreen(GREEN);
    M5.Lcd.setCursor(10, 40);
    M5.Lcd.print("BOWEL");
    delay(1000);
    Blynk.virtualWrite(V2, 0);
  }
}
```

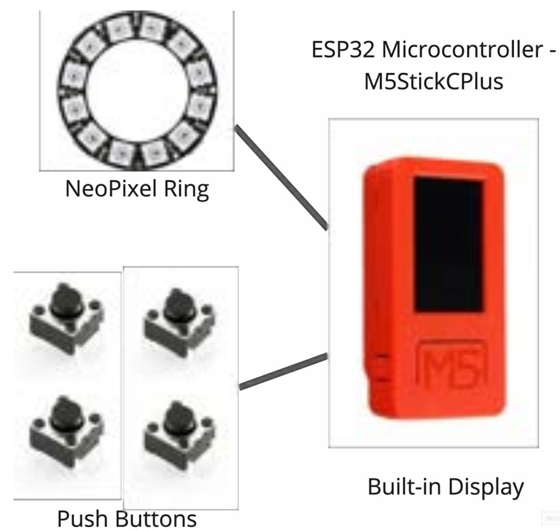




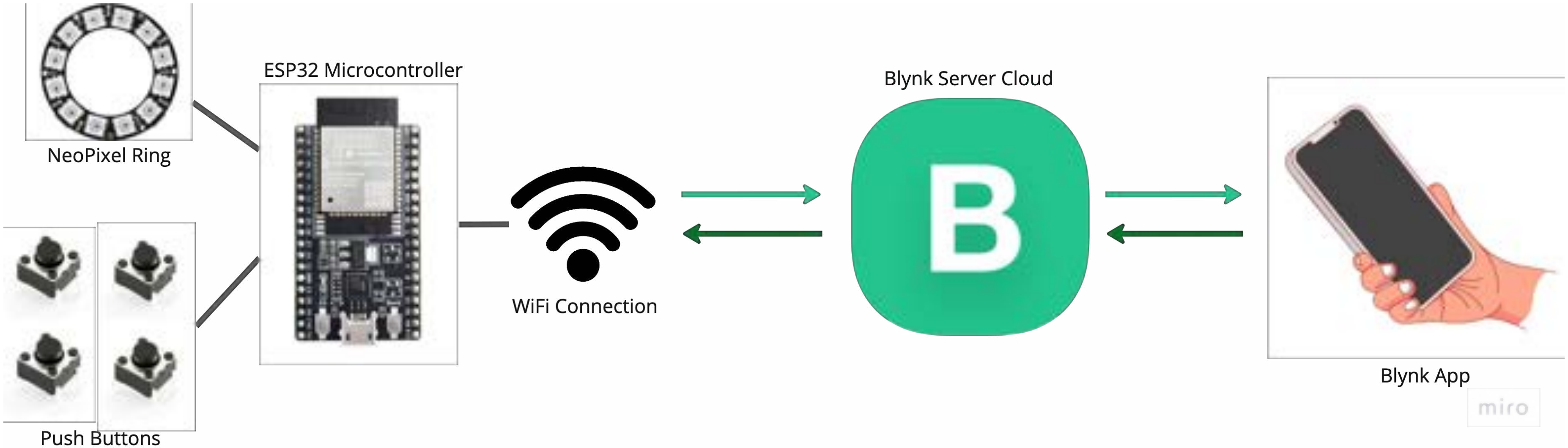


I began working on the functionality of the product by using a microcontroller and a few buttons to test how the system would operate. The first stage involved connecting the buttons to the microcontroller—specifically, the M5StickC Plus—and mapping each button to collect a specific type of input when pressed.

I wrote a cod that reads the input from the buttons and assigns each one to a particular input category: pain, anxiety, or bowel movement. I used the built-in screen on the M5Stick to display the collected input each time a button is pressed. This allowed me to visually confirm that everything was working correctly at this stage of development



Iteration 2: WIFI Connection & Blynk IoT



Since the device needs to connect with an app to process inputs and prompt relevant questions, an important challenge after mapping the buttons was **determining how to send the data to the app.** To achieve this, it was first essential to ensure the use of an ESP32 microcontroller, such as the M5StickC Plus I was already working with, as it includes built-in Wi-Fi and Bluetooth modules—both crucial for data transmission.

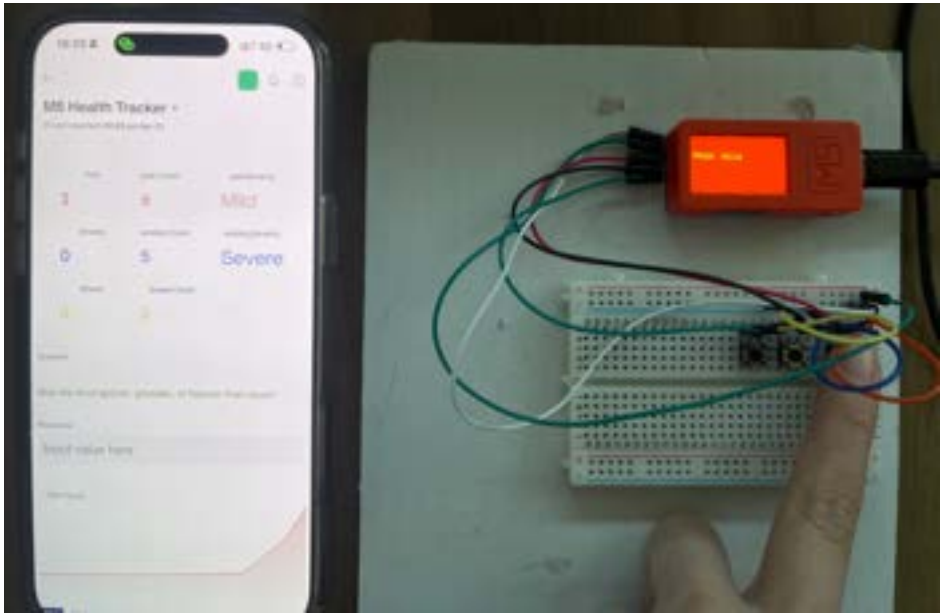
```
#define BLYNK_TEMPLATE_ID "TMPL5rv4RBIB8"
#define BLYNK_TEMPLATE_NAME "M5Stick Buttons"
#define BLYNK_AUTH_TOKEN "OCBQEMEtl1aFfakGg8FqgvPjPtekjgIa"

#include <M5StickCPlus.h>
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

// Wi-Fi credentials
char ssid[] = "Nefeli.";
char pass[] = "agelades";
```

Connecting to Blynk

I used the **Blynk IoT software platform** as the method for sending data. The way it works is that the microcontroller sends the input from the buttons to the Blynk server using specific lines of code and a Wi-Fi connection. Once the data reaches the server, it can be accessed and visualised through the connected app.

By using this platform, I was able to **establish effective communication between the device's hardware and the app interface.**

Iteration 3: Showing Data on App

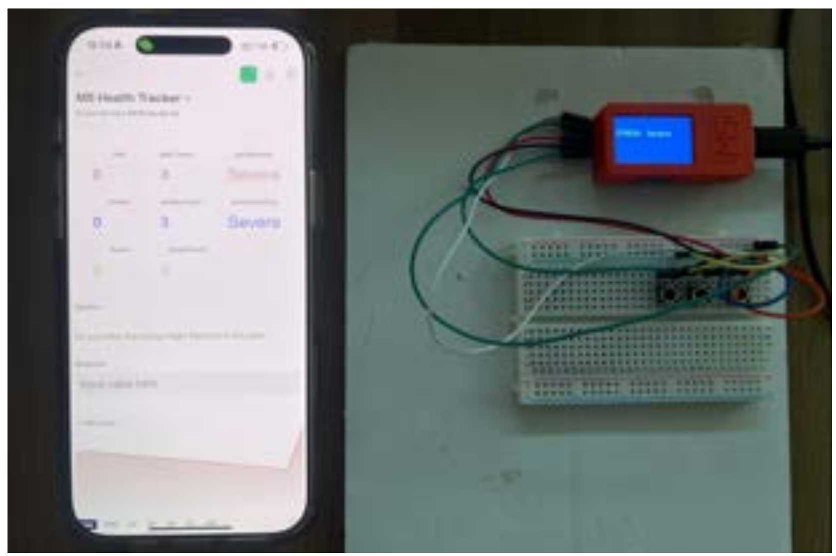


For the first iteration, every time I press one of the three buttons, the number in the first row of the app changes from 0 to 1 while the button is pressed, indicating that the input has been received. Meanwhile, in the second row, the total number of times each button is pressed is displayed, showing how many times per day the user experiences pain, anxiety, or bowel movements. Additionally, there is a graph at the bottom of the screen that tracks how the button presses increase throughout the day, highlighting the specific hours when inputs are logged.

Iteration 5: Tracking Severity of Input

The next step I took was to track the severity of a symptom—for example, whether the pain a user experiences while pressing the button is mild or severe. I achieved this by coding the buttons so that **a normal press logs the input as mild, while a long press logs it as severe.**

I also plan to use the same method for tracking bowel movements, allowing the device to differentiate between normal, constipation, and diarrhoea based on the duration of the button presses.



Reflection Questions: Questions triggered with input

Based on my secondary research and interviews with a doctor, I identified the most important areas of tracking. From this, I developed a series of reflective questions that appear at the end of the day, tailored to the input the user has logged throughout the day. These questions are designed to collect the right kind of data in a simple, user-friendly way that avoids cognitive overload. This list of questions will continue to evolve through further research, user testing, and AI-generated follow-up prompts to ensure the tracking remains comprehensive and holistic.

If the user logged Bowel Movements (V2)

- *"You logged a bowel movement today. How would you describe it in terms of consistency?" (Offer Bristol scale)*
- *"Did you feel urgency or any cramping beforehand?"*
- *"Did you feel better afterward, or was discomfort still present?"*
- *"Did your stool seem different from usual today?"*
- *"How hydrated were you throughout the day?"*

If the user logged Pain (V0)

- *"You experienced some discomfort today. Can you recall what you ate a few hours before the pain started?"*
- *"Did you notice if any meals today were heavier, oilier, or spicier than usual?"*
- *"Was the pain you felt today similar to what you've experienced before?"*
- *"Did any emotional stress or anxious moments happen before the pain occurred?"*
- *"Did you take any medications or supplements today that may have affected your symptoms?"*
- *"Was your sleep last night restful enough to feel recharged today?"*

If the user logged Stress/Anxiety (V1)

- *"You noted stress or anxiety today. Can you recall what might have triggered it?"*
- *"Did you feel the stress in your body—particularly in your gut?"*
- *"Was the stress more emotional, physical, or situational?"*
- *"Did your stress ease over the day, or carry through?"*
- *"How was your overall mood throughout the day?"*

Daily Wrap-Up Prompts

- *"Thinking back on your day, was there anything that seemed to help you feel better?"*
- *"Were there any habits or situations you'd like to try changing tomorrow?"*
- *"Did anything about your routine (sleep, movement, hydration) feel different today?"*
- *"Did you feel supported, in control, or at peace with your health today?"*

Conditional: Pain followed by Bowel Movement

- *"You had pain and later a bowel movement today. Did the pain lessen after going to the toilet?"*
- *"Has this kind of pattern happened before?"*

Conditional: Pain followed by Bowel Movement

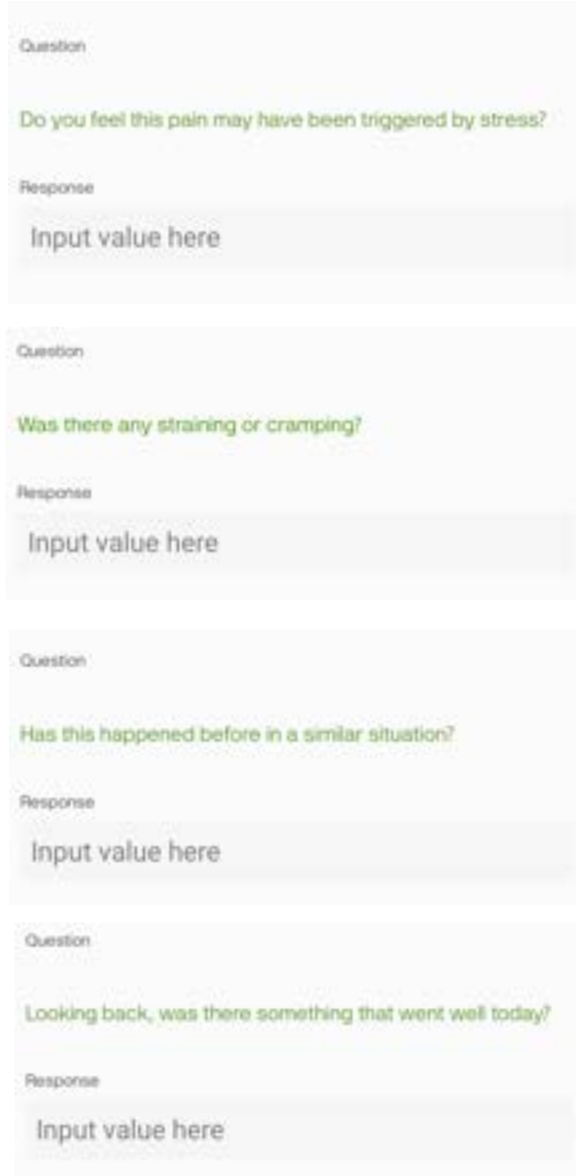
- *"You experienced both stress and pain today. Do you think the discomfort may have been triggered by the stress?"*

Iteration 3: Question Generation

A very important part of developing the code for this device is programming the **questions that appear based on the inputs logged throughout the day**. For the first attempt, I coded the questions to appear immediately after a button is pressed, rather than only at the end of the day. This approach made it easier at this early stage to check that the correct questions were being triggered by the corresponding buttons.

I programmed the device so that, upon pressing a button, a specific question related to that input would immediately appear in the app. Once I confirmed that this was working correctly, I took it a step further **by coding the app to recognise combinations of inputs.** For example, if pain is followed by a bowel movement, a different question would be shown compared to when pain or bowel movement are logged individually.

This stage was a bit tricky, as I needed to incorporate timing logic to distinguish between combinations of inputs and separate, unrelated inputs. This part of the development is still ongoing, as there are currently issues with lags and the device incorrectly recognising all inputs as combinations.




```
// V3: Pain main questions
const int NUM_PAIN_MAIN = 4;
String painMainQuestions[NUM_PAIN_MAIN] = {
  "Do you remember what you ate in the last 2 hours?",
  "Was the food spicier, greasier, or heavier than usual?",
  "Have you felt unusually stressed or anxious today?",
  "Did you sleep well last night?"
};
int painMainIndex = 0;

// Pain conditional if bowel movement (V2) follows pain
const int NUM_PAIN_BOWEL = 2;
String painBowelQuestions[NUM_PAIN_BOWEL] = {
  "Did the pain lessen after the bathroom visit?",
  "Was this a familiar cycle for you?"
};

// Pain conditional if stress event (V1) also occurred
String painStressQuestion = "Do you feel this pain may have been triggered by stress?";

// V2: Bowel main questions
const int NUM_BOWEL_MAIN = 5;
String bowelMainQuestions[NUM_BOWEL_MAIN] = {
  "Did the bowel movement help relieve any discomfort or pain?",
  "Was this bowel movement different than usual?",
  "Was it urgent or unexpected?",
  "Was there any straining or cramping?",
  "Would you describe it as loose, solid, or hard to pass?"
};

// V1: Stress/Anxiety main questions
const int NUM_STRESS_MAIN = 4;
String stressMainQuestions[NUM_STRESS_MAIN] = {
  "Do you feel this stress in your gut?",
  "Was something specific stressing you out today?",
  "Has this happened before in a similar situation?",
  "How rested did you feel this morning?"
};
int stressMainIndex = 0;

// Stress conditional if a pain event follows (from stress side)
String stressPainQuestion = "Do you think this stress might have led to the pain?";

// Daily Optional Check-in (asked once per day if any interaction occurred)
const int NUM_DAILY_CHECK = 2;
String dailyCheckQuestions[NUM_DAILY_CHECK] = {
  "Looking back, was there something that went well today?",
  "Did your gut feel more at peace or more sensitive today?"
};
int dailyCheckIndex = 0;

// ---------- Conditional Timing & Flags ----------
unsigned long lastPainTime   = 0;
unsigned long lastBowelTime  = 0;
unsigned long lastStressTime = 0;
const unsigned long conditionalWindow = 30000;  // 30 seconds window for linking events
```

Design Development: Prototyping & Testing

Exploration 1: Rigid, Medical-like Iterations









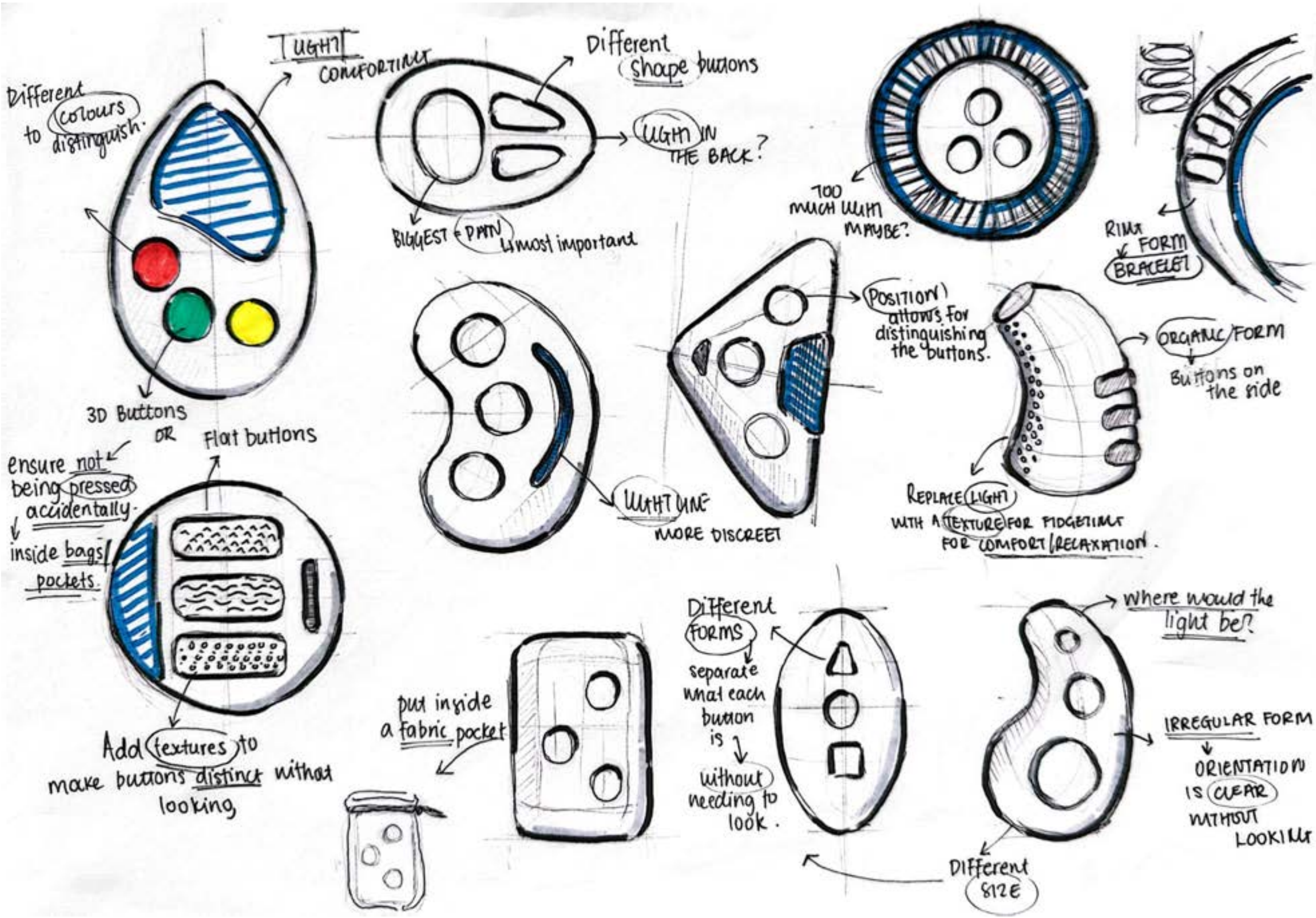
To determine the ideal form, design, and material for my device, I explored different approaches—ranging from hard, rigid, medical-style finishes to softer, fidget toy-like, textured designs. I examined these options by prototyping various device concepts to see which ones resonated most with users and felt most ideal for them.

I started by creating a few quick cardboard prototypes to gather initial user feedback on form and usability. These early prototypes focused more on hard, medicalised designs, which would likely be manufactured from materials like plastic and feature a more industrial finish. By asking users to interact with them, I collected some valuable initial insights that guided the next stages of prototyping.

The key takeaways from this first round of feedback were:

1. **Compact size is essential**—larger prototypes were not preferred.
2. The **form must be ergonomic** to fit comfortably in the hand.
3. Most importantly, button placement and form are critical. Users mentioned that they wanted to be able **to operate the device discreetly, even while it's in their pocket**. This means that the **buttons need to be easily distinguishable—either through their placement, form, or texture—so users can identify them without needing to look.**

With the users' feedback in mind, I began developing new design ideas that would better meet their needs. I created sketches of several new concepts and reworked some of the earlier designs, improving them by adding textures to the buttons and adjusting their forms to enhance usability.



WEEK 28-29

Due to limited access to 3D printers, I used MDF and laser cutting to imitate a plastic finish and created a new series of more refined prototypes for testing. I incorporated the feedback I received regarding size, ergonomics, and the ability to navigate without looking. For these new prototypes, I focused on **adjusting the forms and sizes of the buttons and experimented with their placement on the device to make them more easily distinguishable.**

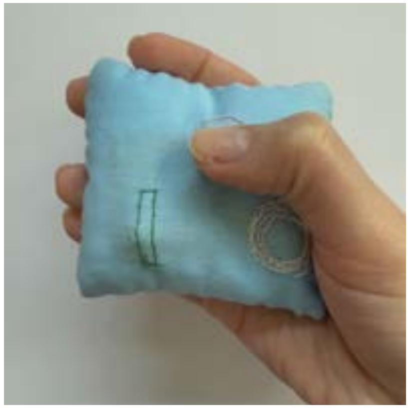
I also plan to add textured surfaces to some of the circular buttons to further enhance their tactile differentiation. At this stage, the prototypes are still in progress; I am planning to add push buttons to some of them, as well as sand and paint the surfaces to achieve a more finished appearance. This will allow for more thorough and accurate user feedback.

Once the refinements are complete, I will distribute the prototypes to users for testing and gather further feedback before moving on to the next iteration of designs.

Exploration 2: Soft, Fidget-like Iterations

Due to the positive feedback I previously received regarding the idea of the device being soft and squeezable, I decided to explore this concept further and investigate how to incorporate buttons while still maintaining a fidget-like, tactile experience. For this iterations, I worked with fabrics, as they are soft materials that naturally afford squeezing and fidgeting gestures. I filled the fabric designs with different types of cotton filling to create various squeezing sensations. Inside, I placed a small cardboard box to imitate the microcontroller and attached the buttons onto it. While these prototypes were well-received and gathered a lot of positive feedback, they also revealed some important challenges. Firstly, **there were concerns about the safety of the electronics**, as fabrics are not waterproof and could leave components vulnerable to damage. Secondly, since this is a handheld device intended for frequent use, **fabric surfaces could easily get dirty and would be difficult to clean effectively.**






To address these challenges, I developed an alternative idea: housing the electronics in a waterproof, durable plastic casing, which would then **be covered with a removable fabric sleeve**. This solution would both protect the internal components and allow the fabric sleeve to be removed, washed, and reattached. Additionally, it would give users the opportunity to customise their device by choosing different sleeve designs based on their preferences.

Peer Review Reflections

Peer review sessions throughout the semester provided essential feedback that helped me refine my design research, proposal, and the development of my proposed product. They also allowed me to see what other people were working on, contribute to their design processes, and exchange valuable ideas with my peers.

The initial feedback I received focused mainly on the functionality of my device. Peers encouraged me to make the functionality very clear and ensure that I was targeting a specific audience. They made me realize that, although existing medicalised app solutions have many shortcomings, some users might still prefer them. Therefore, it needed to be clear that I was designing for users who do not prefer those traditional solutions.

In the beginning, I was very determined to make my device soft, fidget-like, and far removed from anything medicalised—almost resembling a toy. However, my peers advised me not to make that decision too early. They suggested that I test different possibilities, as some users might actually prefer a more rigid, industrial, medical-style device over something playful.

They also helped me significantly with figuring out material choices. I received valuable feedback about how to make the device both safe and squeezable. Materials like silicone and TPU (soft plastics that can afford different gestures but still offer safety and a semi-medical finish) were suggested, and this feedback is something I am still seriously considering. Their input throughout the process has been very valuable and helpful.

Regarding my feedback to my peers:

- Iana's work is very promising; she is quite advanced with her electronics development. I mainly advised her to also consider form early on and not leave it for later stages.
- Ben and Vincent are both tackling very challenging and complex projects. My main concern for them is the feasibility of achieving all their goals by the end of the year, but both projects are very promising, and they seem to be in a strong position at this stage.

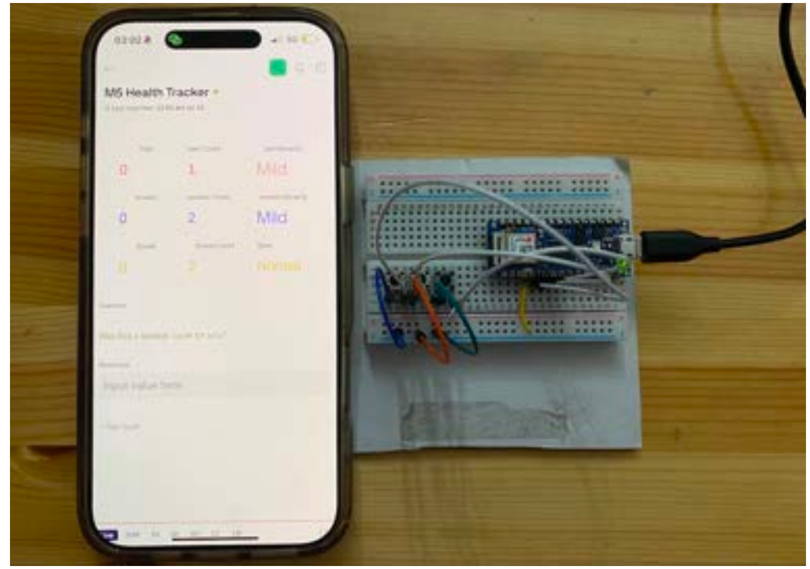
Unit 4: Independent Research Project

Code Development: Further Testing & Iteration

Iteration 6: Changing Microcontroller

Up to this point, I had been using the M5StickC Plus for my device, as it offered both Wi-Fi and Bluetooth connectivity, and its compact size made it ideal for the design. However, I began considering the addition of a NeoPixel ring, and the M5StickC started to become less suitable due to its limited number of GPIO pins, which made it difficult to integrate additional components.

As a result, I decided to switch to the Arduino Nano 33 IoT. Like the M5StickC, it supports both Wi-Fi and Bluetooth, but it also provides more GPIO pins, allowing for greater expandability. The Arduino Nano is also very compact, with the only drawback being the lack of a built-in lithium battery. However, this can be addressed later by wiring an external LiPo battery to the board.



Iteration 7: Neo-Pixel & Bowel Movement Type

After switching to the new microcontroller, I first updated the code to allow bowel movements to be recorded as normal, loose, or hard based on the length of the button press—similar to how pain and anxiety are tracked. A short press indicates a normal bowel movement, a long press indicates loose, and a very long press indicates hard.
Next, I wired the NeoPixel and programmed it to pulse blue whenever the anxiety button is pressed, creating a calming breathing cycle for the user to follow in order to relax. Additionally, I configured the light to turn yellow when the device successfully connects to Wi-Fi, helping me confirm that the product is properly connected.

Design Development: Design Concepts & Renders

Following feedback from last semester's prototypes, I decided to move away from fabric designs and aim for a more solid, industrialised look that **still feels comforting and inviting to the use**r—rather than appearing clinical or stressful. It was also important for the design to remain discreet. With this in mind, I began exploring different design directions in terms of form and interaction and generated some initial renders to visualise possible ideas.

Iteration 1: Ripple Disc

Inspiration: **Water ripples or tree rings.**

Design features:
1.Circular, flat-ish disc with concentric rings.
2.Each ring has a slight tactile distinction and a button subtly embedded (e.g., center = pain, middle = anxiety, outer = bowel movement).
3.Center pulses light (under frosted surface) for anxiety input.
4.Could be worn on a lanyard or kept in pocket.

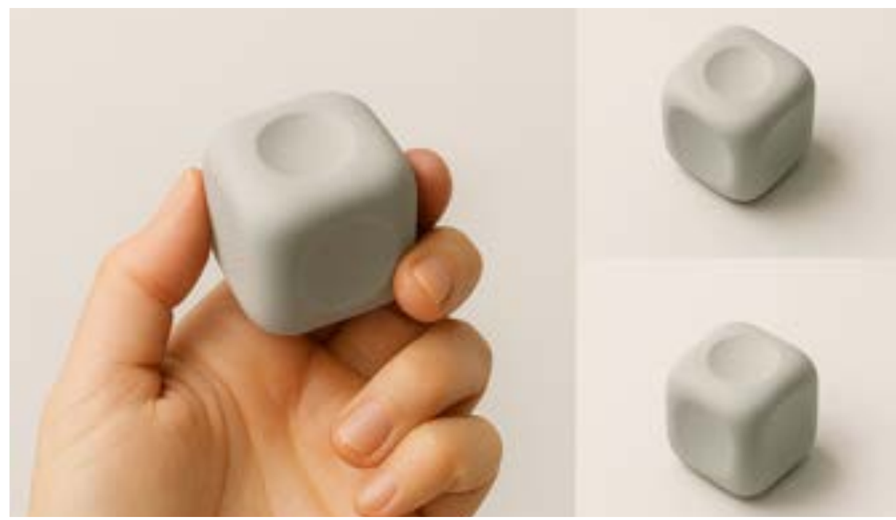
Why? The ripple metaphor resonates with how symptoms radiate through your day. The tactile rings make it easy to navigate without looking.

Iteration 2: Soft Cube with Corners as Buttons

Inspiration: **Modern minimalist fidget cubes.**

Design features:
1. Small cube (~5cm) with rounded edges.
2. Three corners have different button textures: one rubbery, one clicky, one soft-touch.
3. When anxiety is pressed, a soft blue glow emits from inside the cube (through soft silicone sides).
4. Weighted slightly to feel satisfying in hand.

Why? Compact and portable, and the varied textures help create muscle memory and association between symptoms and corners.



Iteration 3: Pebble-Inspired Form

Inspiration: **Natural Pebbles Smoothed by Water**

Design features:
1. Smooth, slightly asymmetrical oval shape fits naturally in the palm.
2. Three concave dimples subtly placed along one side or in a triangular formation — easy to feel and differentiate by touch.
3.Light pulses softly through a translucent base or around the edge (like a halo).
4. Matte silicone outer shell for tactile comfort.

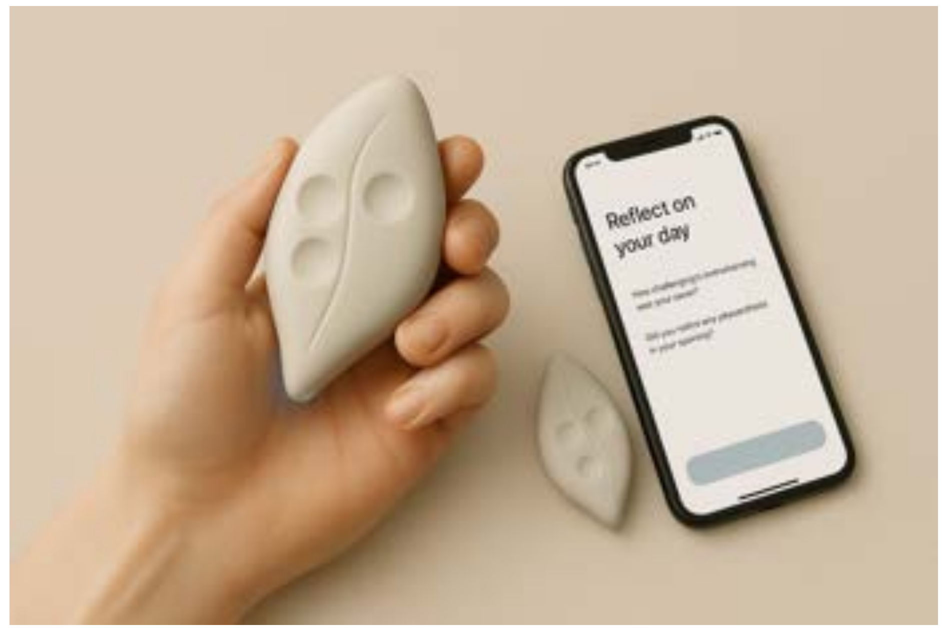
Why? Feels like a grounding object, like a worry stone. Organic shape helps reduce stress and encourages fidgeting as a form of soothing.

Iteration 4: Leaf Pod

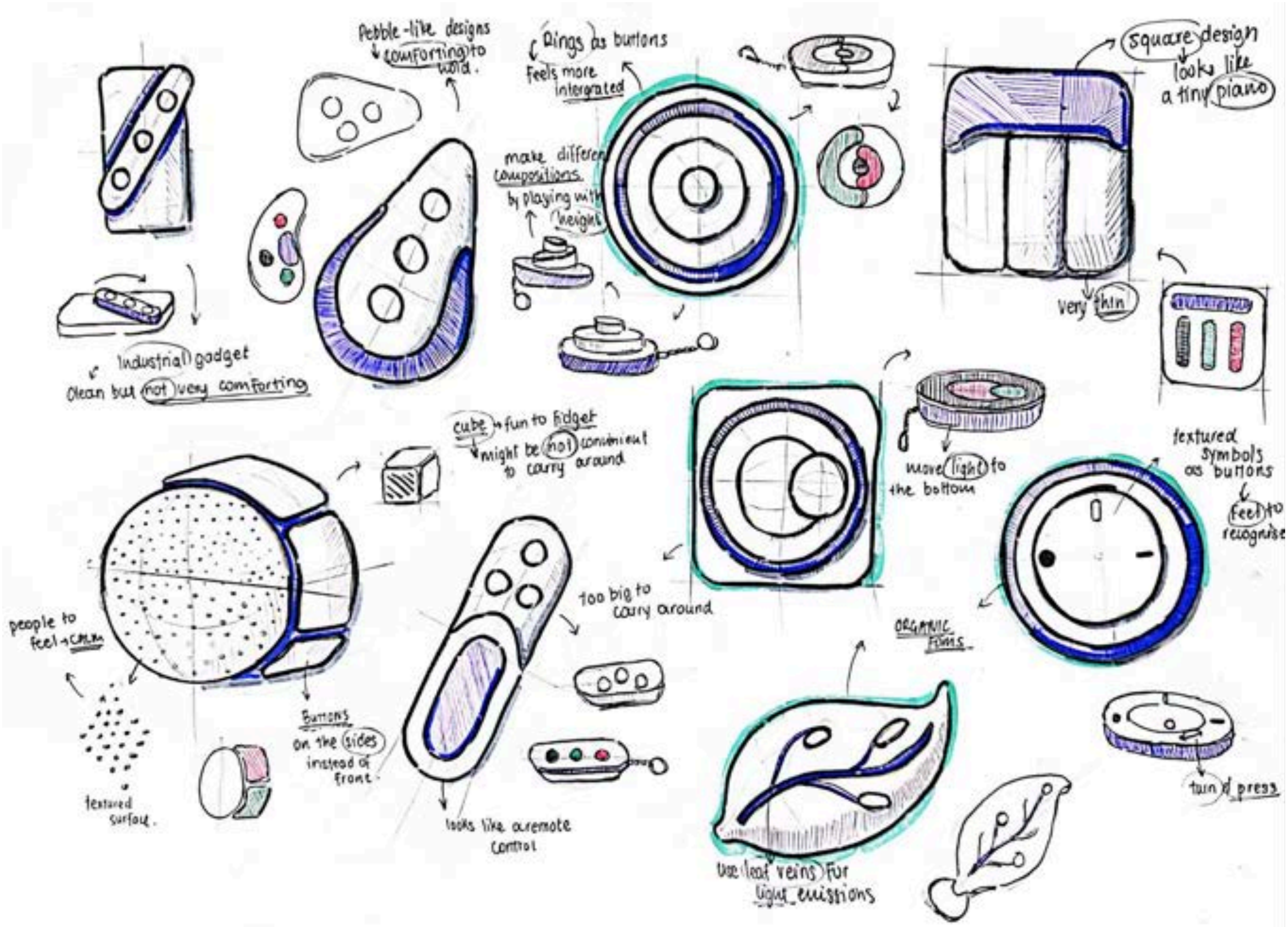
Inspiration: **Nature and Biophilia**

Design features:
1. A slightly curved, leaf-like shape.
2. Three subtle vein-like ridges are the buttons — press anywhere along each to activate.
3. Embedded light follows the contour (e.g., glows down the "central vein" when anxiety is pressed).
4. Could be magnetically attached to a keyring or clipped.

Why? Evokes calmness and life. Organic and gender-neutral, with sensory associations of calm and connection to nature.



Design Development: Sketching

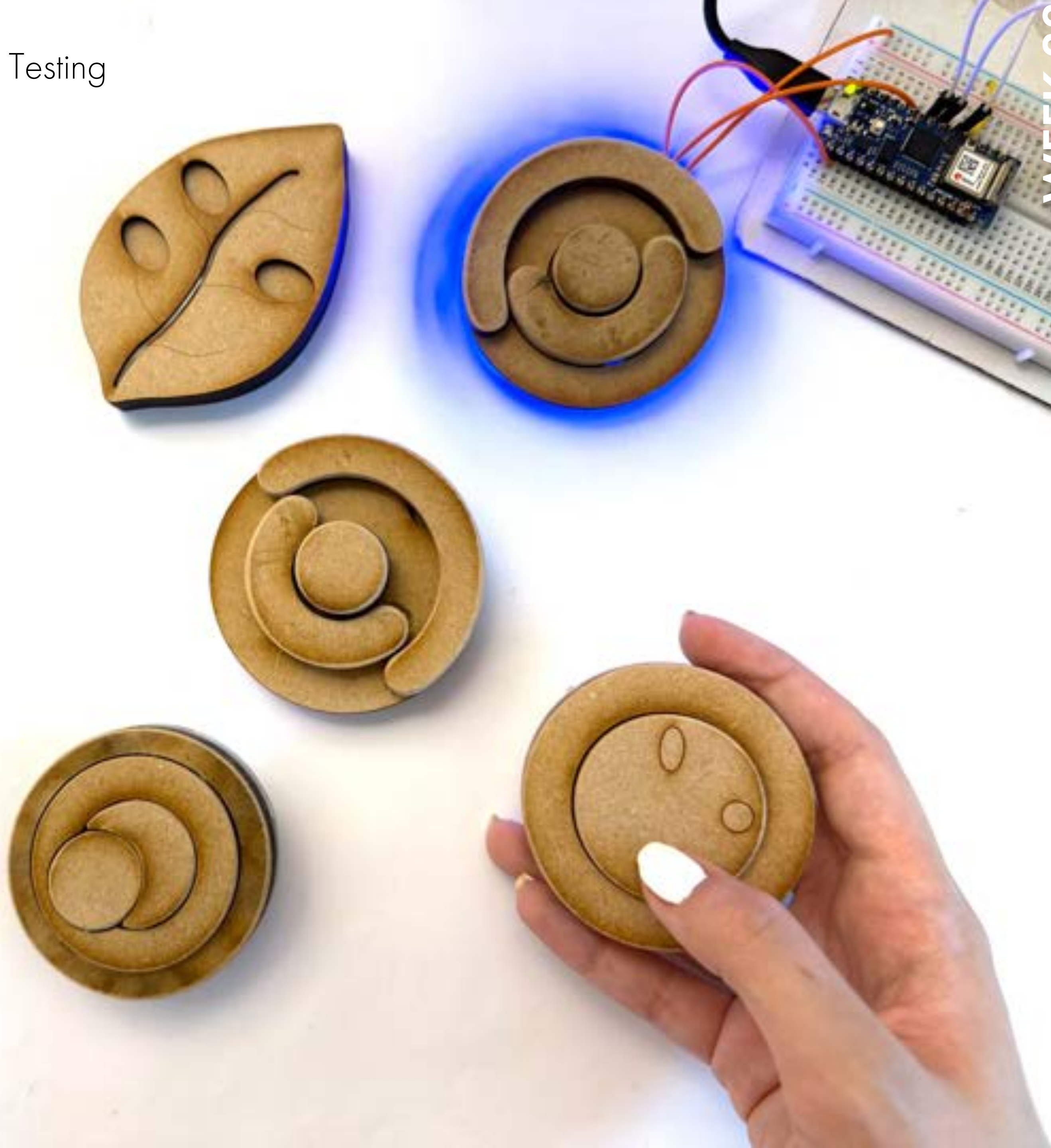


Design Development: Prototyping & Testing

After creating renders with different design concepts, I was inspired to sketch additional potential forms for my device. From those sketches, I selected a few ideas that showed the most promise and developed several prototypes to explore shape, button placement, and overall usability.

The goal was to design a **compact, discreet device** that could be **used without needing to look at it,** while ensuring the **buttons wouldn't be pressed accidentally**. At the same time, the device needed to feel **inviting and comfortable to hold.** A key focus was **integrating the buttons seamlessly into the form,** so they felt intentional—both functionally and aesthetically.

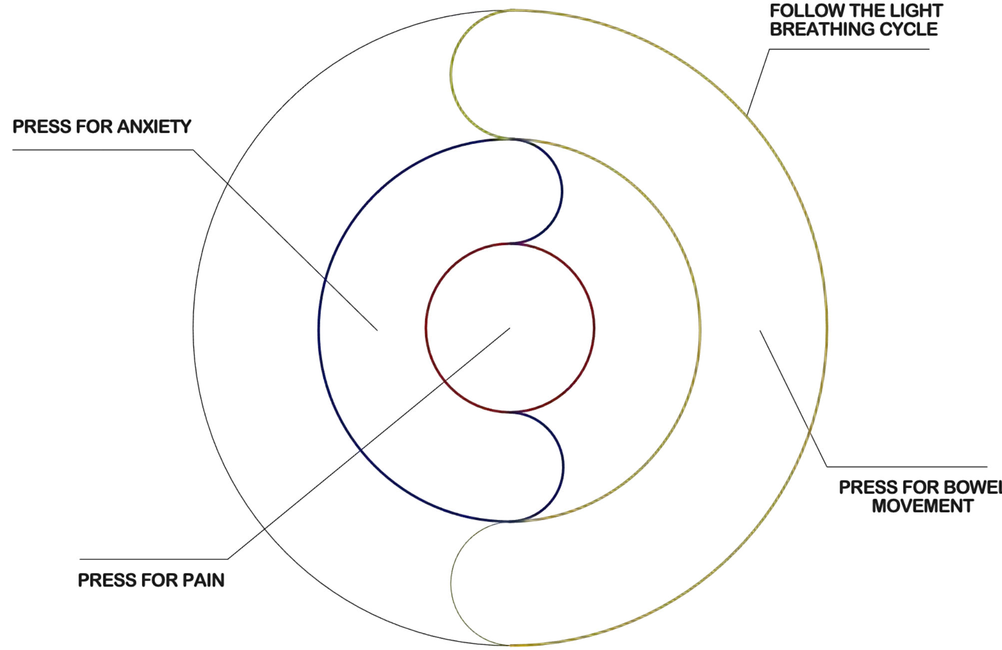
Using MDF and transparent acrylic, I laser-cut testing prototypes and added buttons and lights to assess how they would be used, whether they met the design criteria, and to gather feedback from potential users.

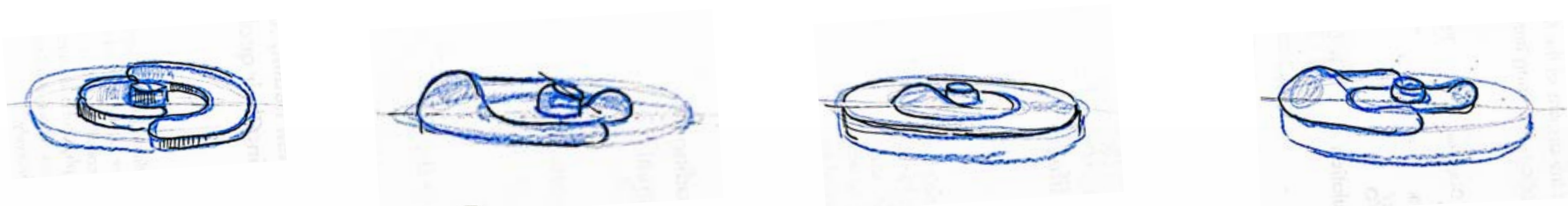


Design Refinement: Resolution through Iterations

After testing multiple forms, the most effective design proved to be a circular device with three concentric rings as buttons and an LED light on the back. Several iterations explored whether the rings should be flat or elevated, and whether the entire ring or just part of it should be pressable. The design needed to ensure the buttons were easy to locate and use. To meet these needs, the final rings were designed with small elevated sections on opposite sides—creating clearly defined, pressable zones that offer tactile feedback and reduce the likelihood of unintentional input.

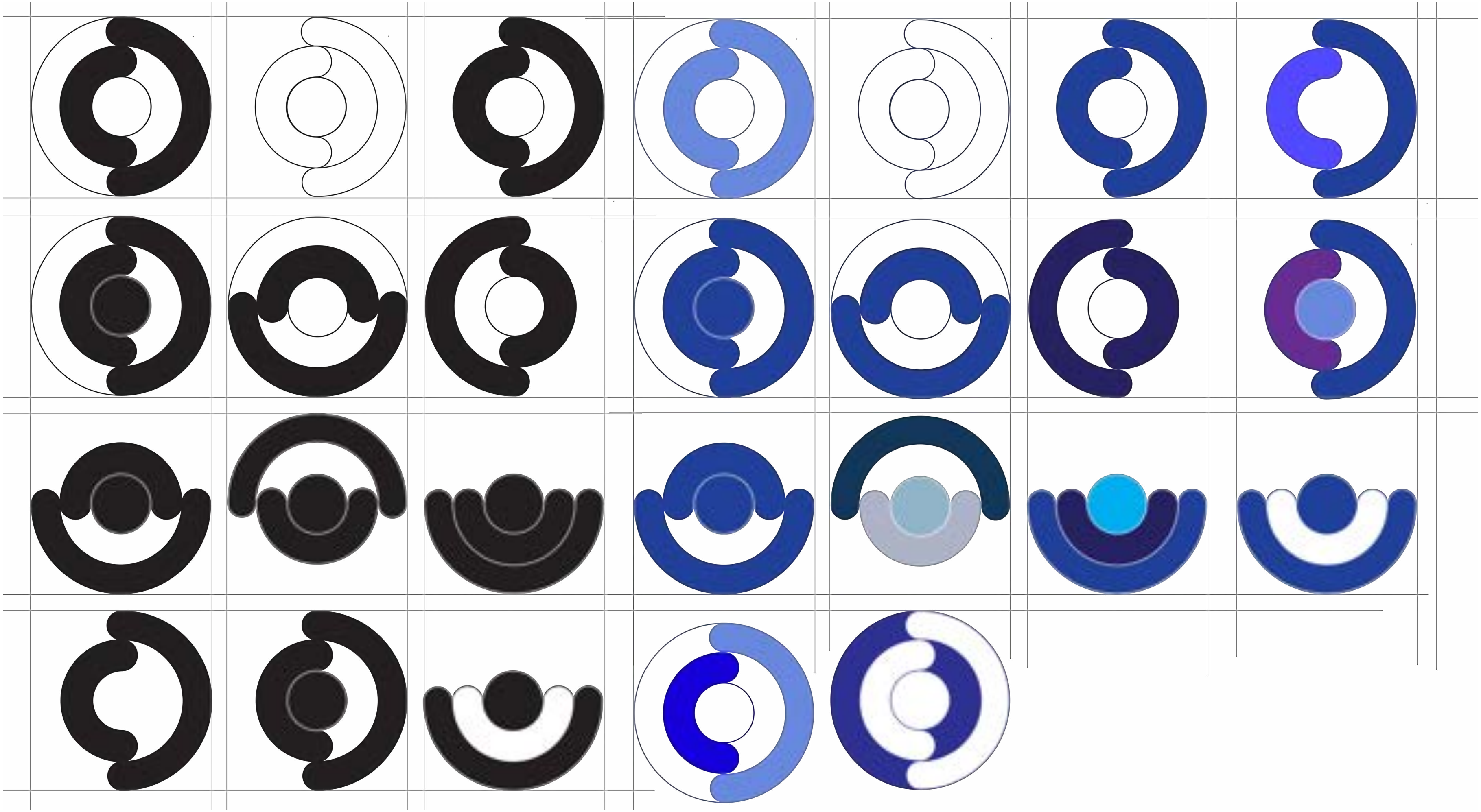
Another important aspect was deciding what to map each button to. I began by asking people what layout felt most intuitive, and the vast majority agreed that **the central button should correspond to the most urgent symptom—pain**. Based on this feedback, I mapped the central ring to pain, the middle ring to anxiety, and the outer ring to bowel movements.





Product Branding: Product Name and Logo

Logo Development: Initial Iterations

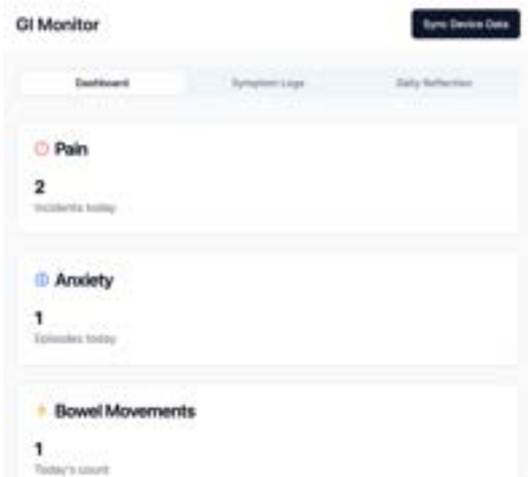


Firstly, I began by working on different logo ideas for my design, thinking that developing a logo might even help me come up with a name. I created several logo concepts inspired by the form of my product. I started with three rings, highlighting the areas where the buttons are located. Then, I experimented by removing some of the ring outlines, adjusting the button positions, emphasizing the negative space, and applying different colours to explore the range of visual outcomes I could achieve.

Name Generation: Further Iterations

Then, I proceeded to brainstorm names for my product. Since my logo resembles the letter "o," I began searching for product names that include an "o" so I could replace it with the logo. I initially explored random names from my Greek heritage that didn't necessarily have a specific meaning but sounded appealing for the product. Later, I was inspired to research words from different cultures that might relate to the nature of my product. Eventually, I came across the word onaka, which means "tummy" in Japanese, and decided it was the perfect name for the product.

App Development: Connection to the Device



Step 1: Set Up App Using Lovable

After completing the hardware electronics, I began developing the companion app that connects to the device. The first step was to create an initial version of the app to display the data coming from the device, which would later be refined into a fully functional application. Using Lovable, an AI tool for app development, I set up the app and created a basic interface where inputs from the device appear once a connection is established. For further development, I continued using Lovable while also working in Figma to design the visuals and create assets such as icons. These assets were then integrated into the app during the build process.

Step 2: Connect to the Device through Blynk

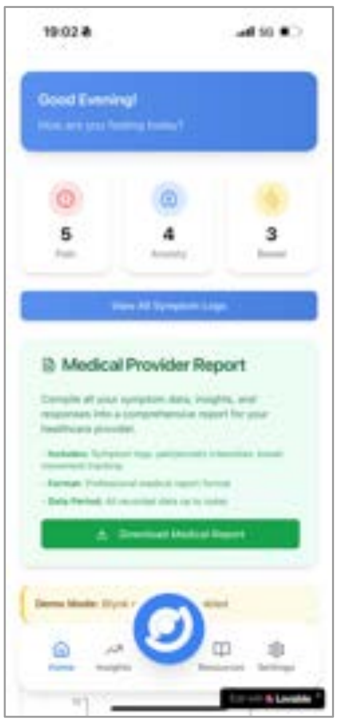
Once the app was set up, a crucial next step was establishing a connection between it and the device. To achieve this, I used Blynk IoT as the intermediary platform. Since I was already using the Blynk cloud to store the data collected from the device, I was able to retrieve that data in the app using the Blynk authentication token. This allowed me to establish a working connection between the device and the app, enabling the symptom logs to display within the application. However, due to limitations in the free version of Blynk, there's a cap on the amount of data that can be stored and retrieved from the cloud each month. To avoid exceeding this limit, I added the option to disconnect the app from the device and instead display a demo version. This version simulates the app's functionality and concept without consuming Blynk credits by pulling live data.

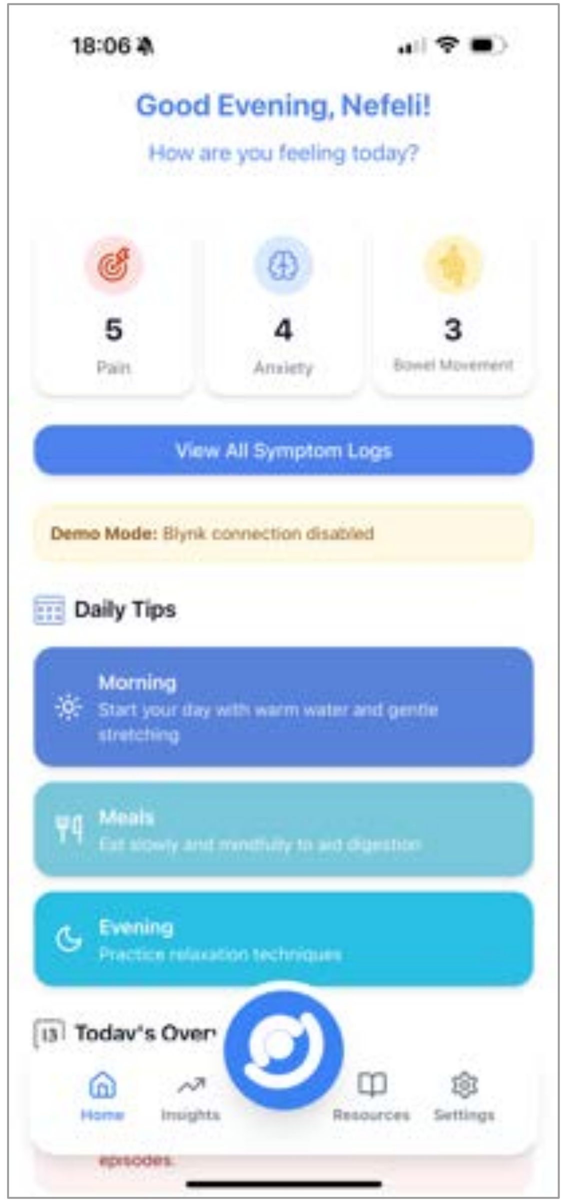
```
#define BLYNK_TEMPLATE_ID        "TMPL5rv4RBIB8"
#define BLYNK_TEMPLATE_NAME      "Nano33IoT Buttons + NeoPixel"
#define BLYNK_AUTH_TOKEN         "OCBQEMEtl1aFfakGg8FqgvPjPtekjgIa"
```

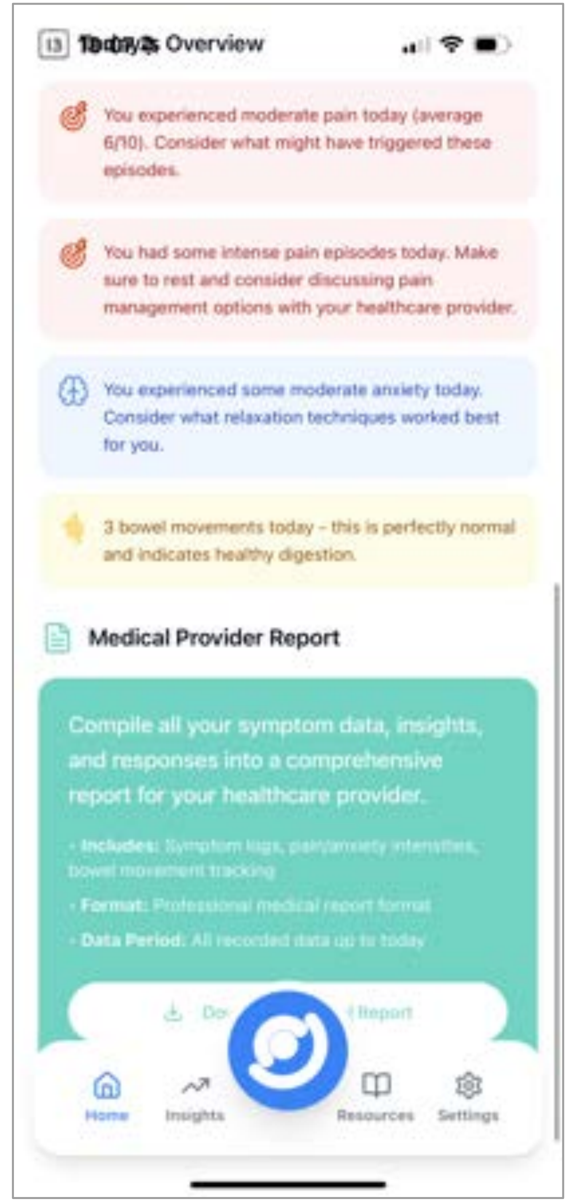
App Development: UI/UX & Features

Step 3: Home Page

The first screen I designed was the home page. This is where the number of symptoms logged is displayed, along with other important information such as a brief user overview, daily tips, and an option to compile the data to share with a doctor. After several iterations, I designed custom icons for the three symptom inputs, as well as for the other sections. The app's colour palette was based on soft shades of blue to create a calming aesthetic, while red, blue, and yellow were used to represent the individual symptoms.

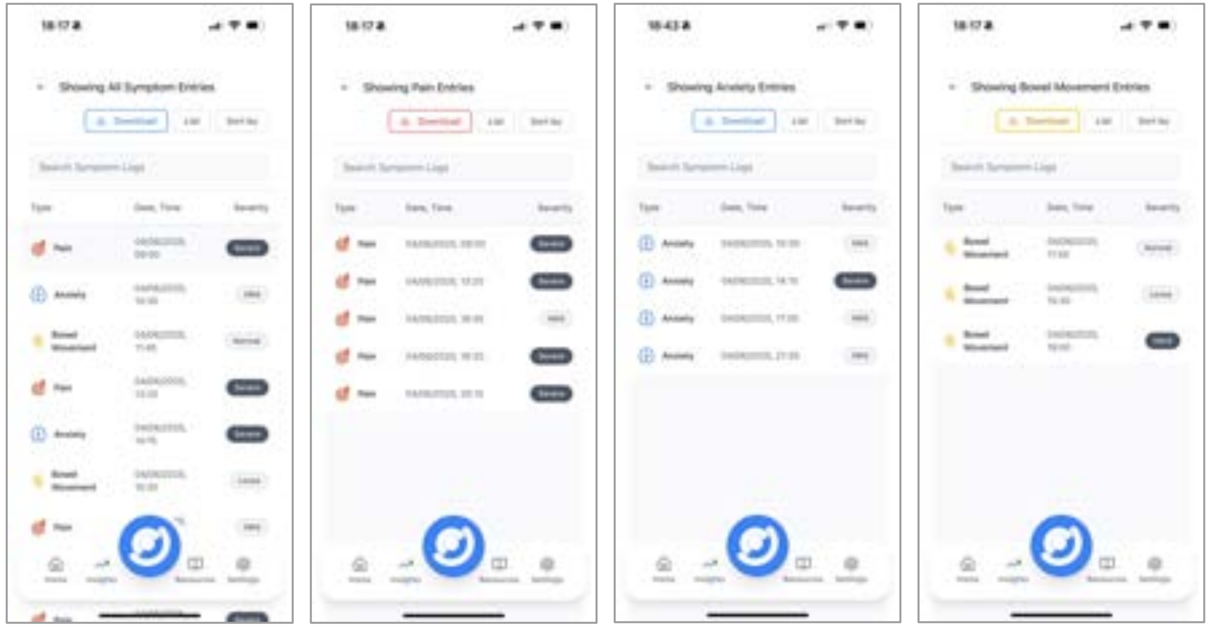


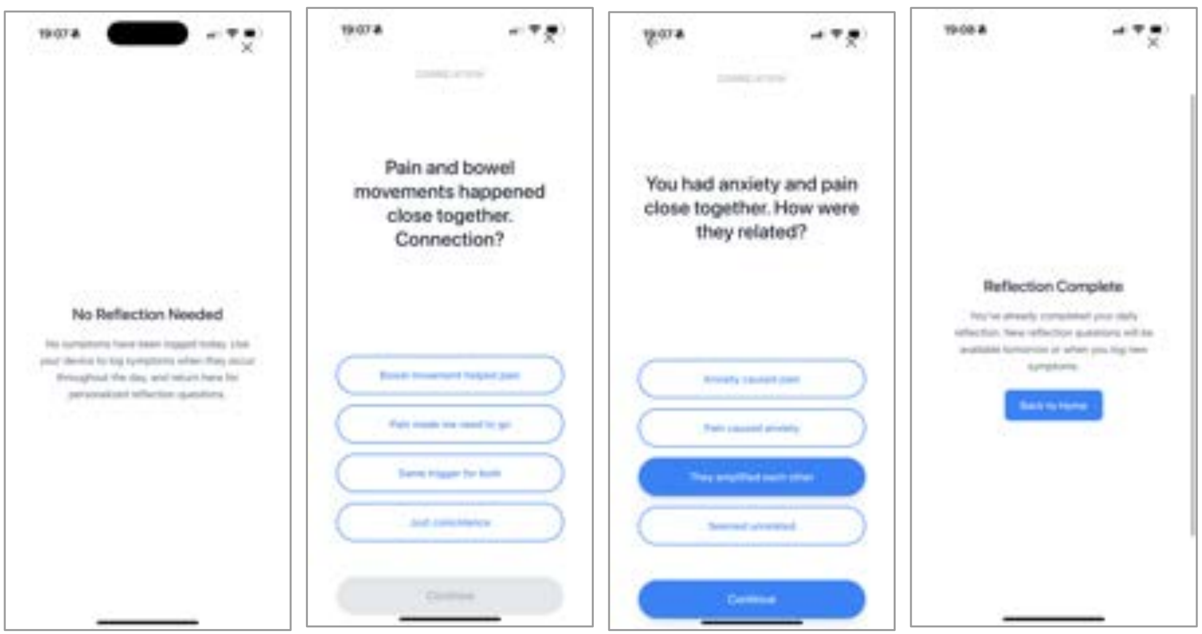




Step 4: Symptom Pages

It was essential to design pages that display all symptom logs in detail, including the exact date, time, and severity of each symptom experienced. Users can view logs for each symptom individually or all symptoms combined, depending on their preference. Another key feature is the option to download the logged symptoms as an Excel file. This allows users to generate a document containing only the symptom data, without the additional information included in the general "compile data" option.



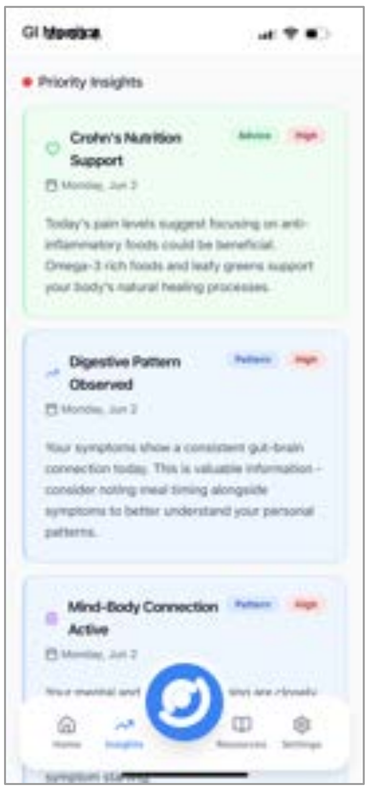


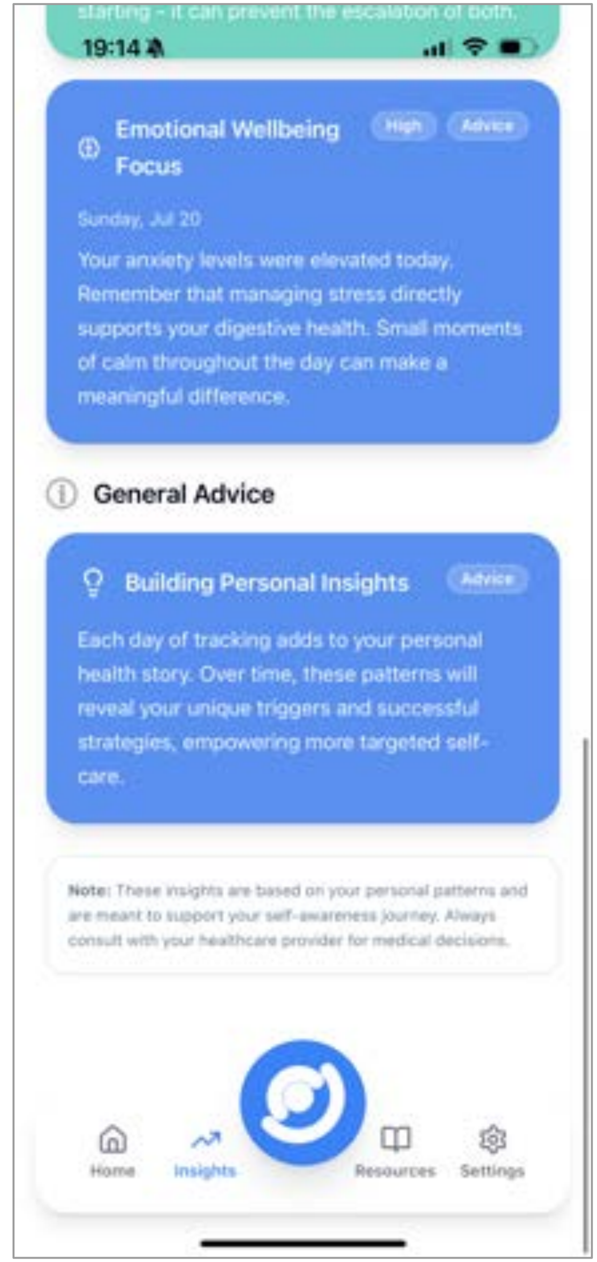
Step 5: Reflective Questions

I made the reflective questions accessible through the middle option in the menu, using the logo as the icon, as it is one of the most vital aspects of the app and the only section that requires user input. I designed the reflective questions page to be a full, non-scrollable screen and removed the menu from this view to avoid distracting the user while answering. The questions are generated only once a day; until then, the page remains blank.

Step 6: Insights Page

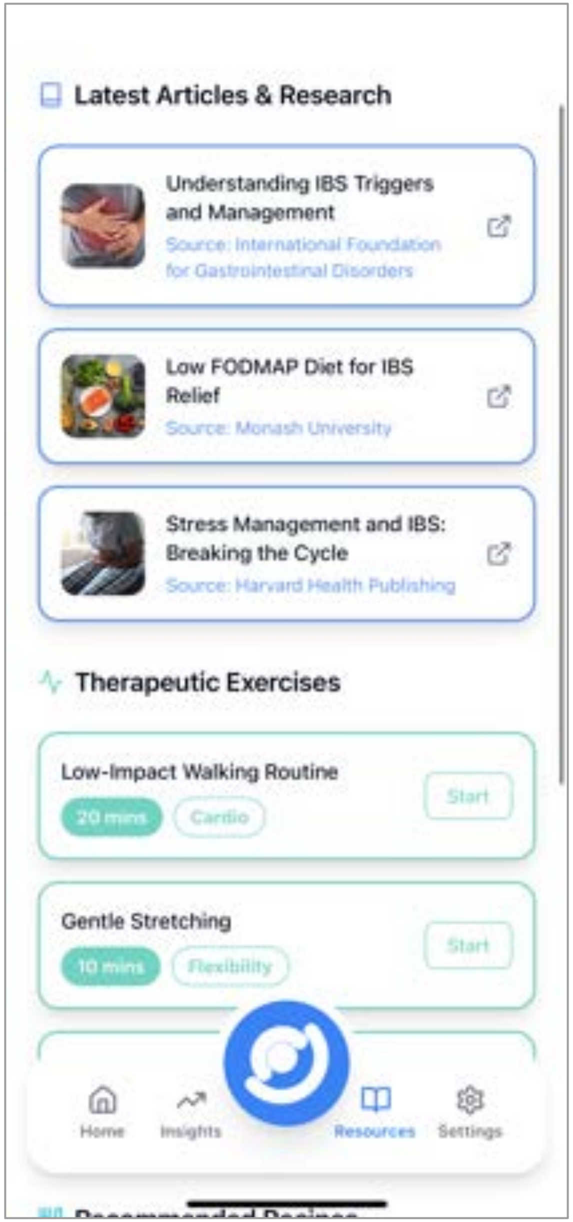
One of the most important pages is the Insights page, where all personalised insights are delivered. This page was carefully designed to make navigation as easy as possible. Insights are organised into categories—such as advice, patterns, and food—with each category using a different shade of blue. Additionally, insights are separated by level of importance: high vs. low, and prioritised insights vs. general advice. Icons are also used to further support intuitive navigation.

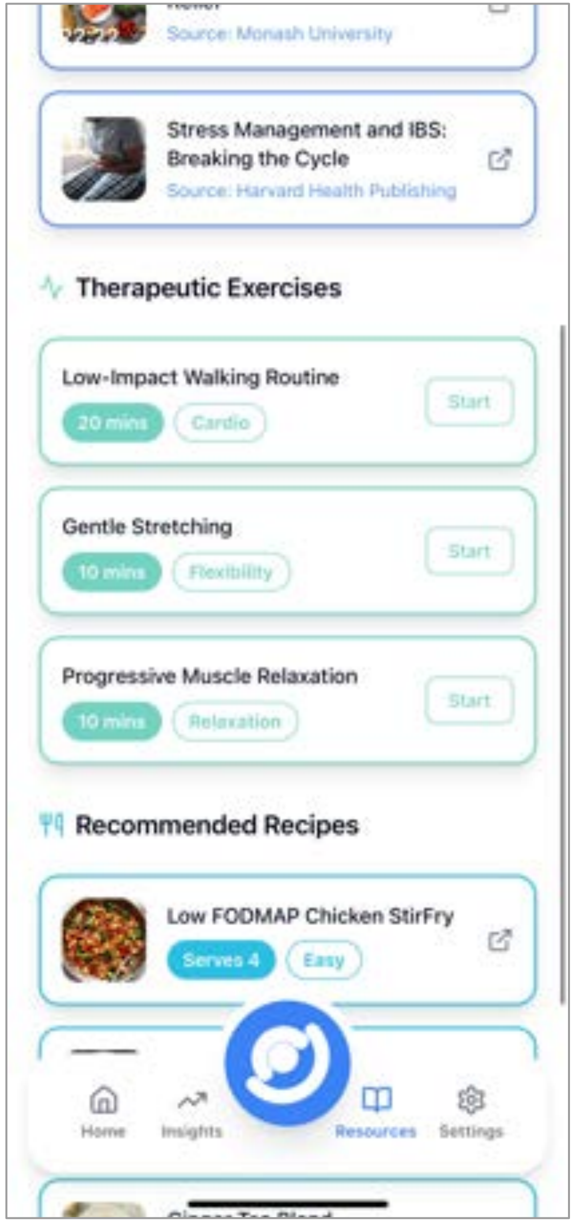


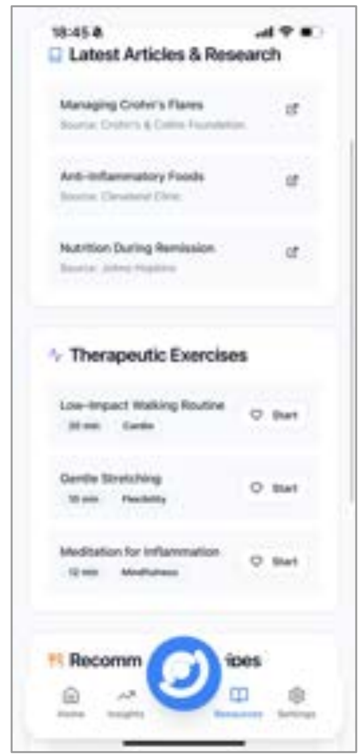


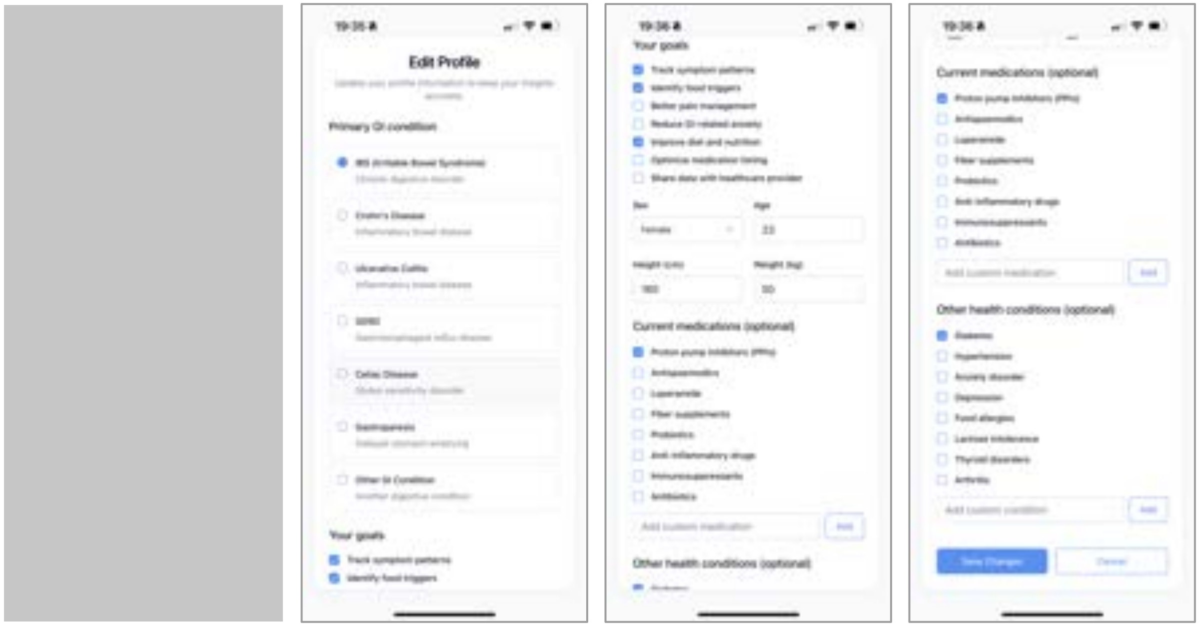
Step 7: Resources Page

I also decided to add a Resources page to provide users with helpful materials relevant to their condition. This includes recent articles and news about their specific GI condition, as well as delicious recommended recipes tailored to their dietary needs. Finally, the page offers a selection of therapeutic exercises customised to the user's daily experiences and needs, helping improve their overall well-being. These resources are updated daily to ensure users have access to as many relevant options as possible.







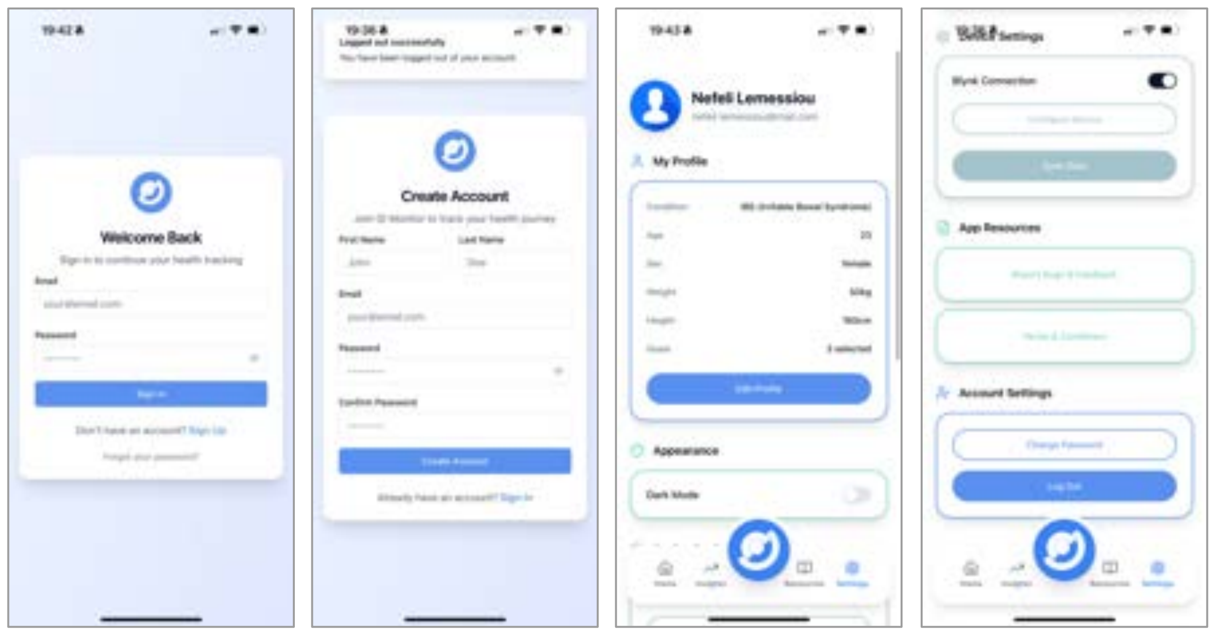


Step 8: Set Up Page

Once the user signs up for the Onaka app, they are asked to answer a series of questions designed to help the app provide them with personalised support. These include questions about their specific GI condition, their goals while using the app, and personal details such as age, gender, height, and weight. Additional information, like current medications or other health conditions, is also collected. This data allows the app to deliver tailored insights and resources based on the user's goals and individual health background.

Step 9: Settings Page

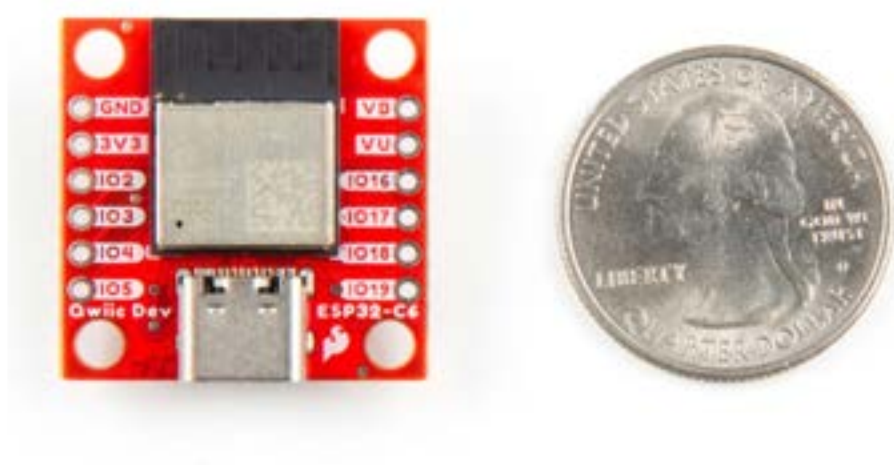
Finally, the settings pages were developed, including the login and sign-up screens. The settings page contains all the supporting app information, such as terms and conditions, the option to log out or change the password, switch to dark mode, and edit the initial setup information provided during sign-up. For the prototype, this page also includes the Blynk connection feature, where users can input the auth token to connect to the device and toggle the connection on or off.



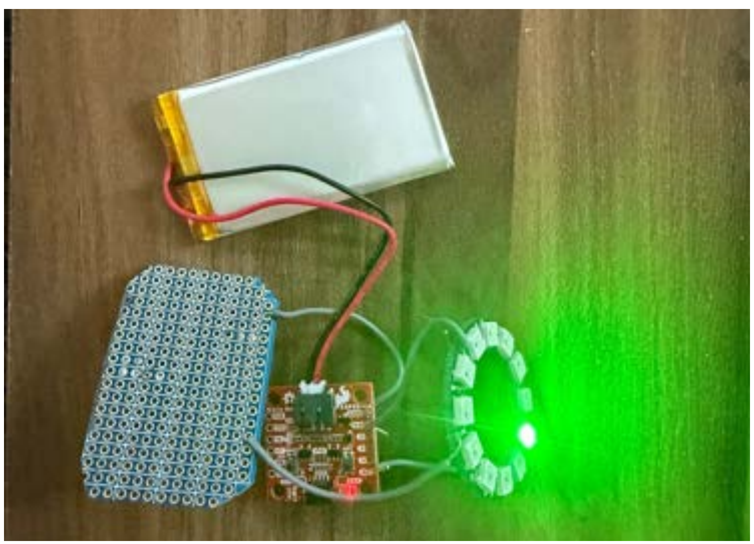
Code Development: Refining & Finalising

Iteration 8: Switching to SparkFun Microcontroller

One more time, I decided to change the microcontroller. As my design developed, I began seriously considering the size constraints and how I would fit the electronics into a small device. I realized that the Arduino Nano was still too large for the compact form I wanted to achieve. Therefore, after a recommendation from Freddie, I decided to switch to the SparkFun Qwiic Pocket Development Board due to its very compact size, which would easily fit into a much smaller base. The only disadvantage of this board was that it provides only a 3V output, in contrast to the Arduino's 5V, which meant the NeoPixel would be less bright—since NeoPixels are typically powered with 5V. However, because a very bright NeoPixel wasn't necessary for this device, the SparkFun board was the perfect choice for my project.

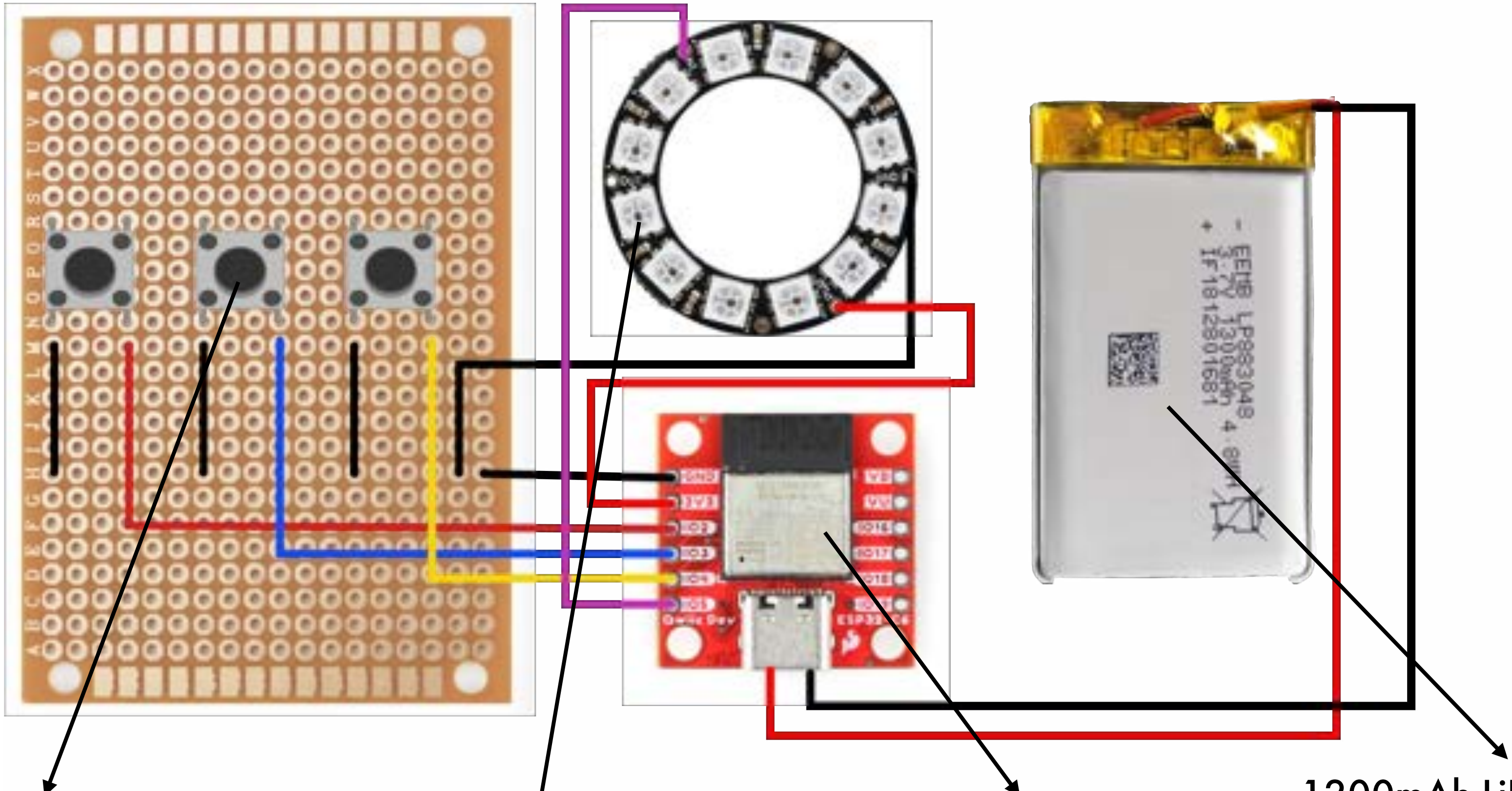


Iteration 9: Adding a LiPo Battery



The final step before completing the electronics was to connect everything to a lithium battery to make the device truly portable. Since I had already switched to the SparkFun microcontroller, this process was much easier, as it includes a built-in 2-pin JST connector for a LiPo battery. Therefore, all I had to do was plug the battery into the microcontroller. However, due to the device's WiFi functionality, a higher-capacity battery was needed, which meant the battery had to be relatively large. This created limitations in fitting it into a small device. As a result, I decided to create two versions: one full-size model without a battery and a larger, fully portable version for the exhibition.

Electronic Components: Why are they used for?



Push Buttons

Three buttons are associated with the three symptom inputs, allowing the user to press one to log their symptom. When a button is pressed, the system recognises it as a symptom input and sends the information to the app.

NeoPixel Ring

The NeoPixel ring is used to produce a blue pulsing light that guides a breathing sequence when anxiety is logged. It also functions as an indicator for things like internet connection and battery status.

SparkFun Qwiic Pocket

This is the microcontroller of the device that processes all the code. Its ability to connect to the internet allows data to be stored and sent to the app. SparkFun was specifically chosen for its compact size, making it ideal for such a small product.

1200mAh LiPo Battery

This battery is essential for making the device fully portable. It connects to the microcontroller using a 2-pin JST connector. Each time the SparkFun board is plugged into a power source via USB-C, the battery charges, allowing the device to continue functioning even after the USB is disconnected.

Product Overview: What?, Why?, Who?, When?, Where?, How?

What?

Onaka is a stress-free symptom-tracking tool designed to support individuals living with chronic gastrointestinal (GI) conditions such as IBS, GERD, and IBD. Combining a discreet, portable tactile device with a simple companion app, onaka enables users to effortlessly log symptoms like pain, anxiety, and bowel movements throughout the day using intuitive button presses.

When anxiety is logged, the device responds immediately with calming light feedback to help users relax in the moment. All data is sent wirelessly to the app, which processes the information and generates reflective questions for users to answer at the end of the day. These responses, together with the logged symptoms, help reveal meaningful patterns and connections, allowing the app to deliver clear, personalised insights and advice.

By reducing cognitive load and anxiety, onaka transforms symptom tracking from a stressful chore into a comforting, empowering daily ritual that helps users better understand and manage their GI health.

Why?

Chronic gastrointestinal (GI) conditions have sharply increased in recent years, driven by factors such as rising anxiety, poor diet, and declining food quality. Millions now suffer not only physical discomfort but also significant emotional and social stress, as the gut and brain are closely linked through the gut-brain axis.

For these individuals, tracking symptoms and diet is essential for improving quality of life. However, most existing tracking tools—usually mobile apps—fall short. They often demand frequent manual input, present complex data, and encourage symptom hyper-awareness, which raises anxiety and can even trigger symptoms. Many users report entering a state of symptom anticipation, constantly waiting for symptoms to appear so they can log them. This hyper-vigilance can itself provoke or worsen the very symptoms they aim to manage.

Onaka recognises these challenges and aims to transform symptom tracking into a stress-free, intuitive experience.

Who?

Onaka is designed for individuals who have been diagnosed with a chronic gastrointestinal (GI) condition and have either been advised by their healthcare provider or have chosen themselves to track their symptoms and habits to help manage their condition. It is particularly suited to users who feel overwhelmed or stressed by traditional tracking methods, such as complicated graphs or excessive data, and who are seeking a more intuitive, comforting, and discreet way to manage their symptoms. Onaka delivers clear, actionable feedback through simple written comments that are easy to interpret.

Additionally, Onaka appeals to healthcare providers by giving patients the option to compile and share their symptom data. This provides clinicians with a more realistic and detailed overview of their patient's experiences, supporting better-informed medical care. Healthcare professionals may also appreciate Onaka as a non-intrusive, supportive tool to recommend for symptom management.

Finally, by helping patients better understand their own condition and identify patterns, Onaka has the potential to reduce unnecessary doctor or hospital visits, benefiting not only the individual but also easing the strain on the wider healthcare system.

When?

The tool is intended for use throughout the day, particularly whenever symptoms occur, enabling immediate and effortless logging. When users experience pain, anxiety, or have a bowel movement, they can simply press the relevant button to record it on the spot. In moments of anxiety, onaka's pulsing blue light offers an on-the-spot calming cue, reminding the user to breathe and relax.

Interaction with the app is only necessary once a day, ideally in the evening. This is when more detailed, reflective interactions take place, encouraging calm, thoughtful consideration of the logged data without creating anxiety through constant monitoring or symptom anticipation.

Beyond this, users can access the app at any time if they wish to check insights, review advice, or explore resources such as articles and recipes. Additionally, the app can be used before healthcare appointments to compile and share their tracked data with their provider.

Where?

Onaka is designed for daily personal use, supporting people wherever they go. The device is intentionally compact and discreet, allowing users to carry it confidently in any setting—public or private. It can easily fit into a pocket, with its recessed button design preventing accidental presses while stored. Alternatively, holes at the top of the design allow it to be attached to a keyring for convenient carrying, or it can be kept in a bag for easy access.

Onaka is made with water-resistant materials, ensuring safe use even in humid environments such as bathrooms, where symptom tracking might often take place. Its comforting, tactile nature enables subtle, discreet interaction in social environments, reducing the embarrassment often associated with managing symptoms in public.

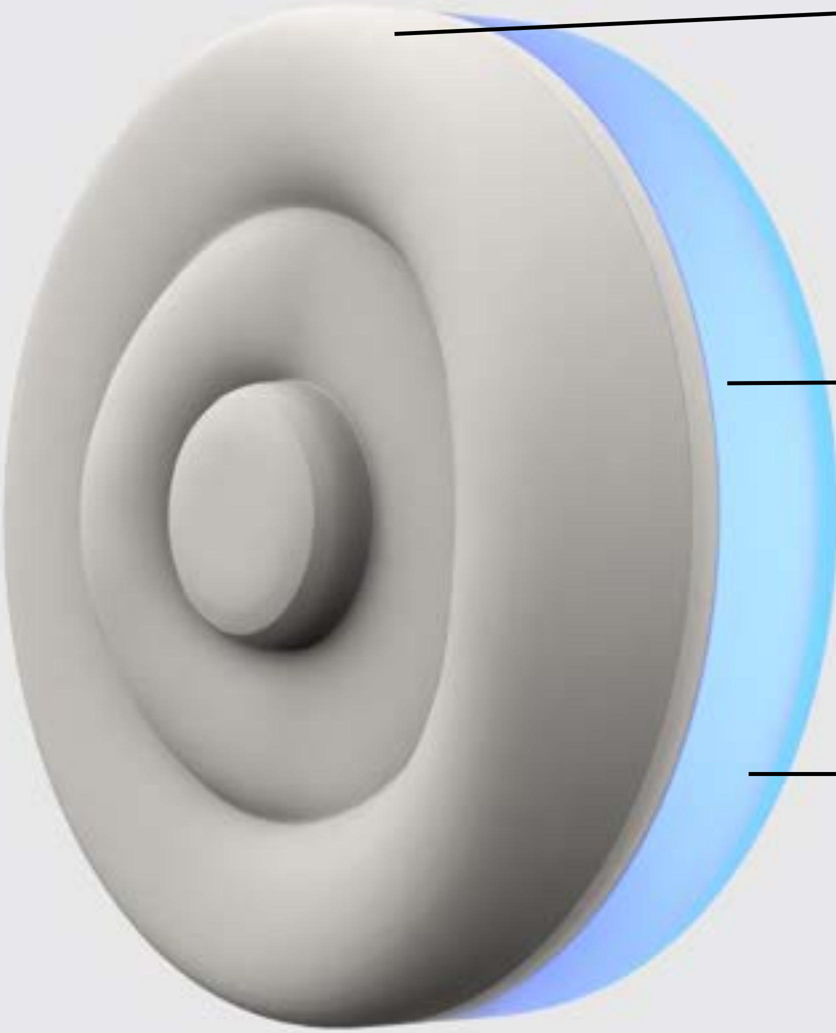
How?

Users log symptoms effortlessly using the concentric ring buttons: the central button represents pain, the middle ring represents anxiety, and the outer ring records bowel movements. A normal press on the pain or anxiety button logs a mild symptom, while a long press marks it as severe. For bowel movements, a normal press indicates a typical movement, a long press records a loose stool, and a very long press indicates a hard stool.

When the anxiety button is pressed, onaka's LED ring activates a calming blue breathing sequence to help the user relax and manage anxiety in the moment.

All logged data is transmitted wirelessly to the companion app through a Wi-Fi–enabled system, with data stored securely in the Blynk cloud server. At the end of the day, the app uses this data to generate reflective questions, helping the user analyse their experiences calmly. Based on their responses and symptoms, onaka provides supportive, easy-to-understand written feedback and notifications. This approach helps users recognise patterns and manage their symptoms proactively, without the stress of interpreting complicated graphs or predictive analytics.

Product Making: Material Choices



Silicone Rubber

Used for the button interface, silicone provides a soft, springy, and satisfying press while being durable enough to withstand millions of uses. Its structure prevents accidental presses and is resistant to water, UV, chemicals, and daily wear—ideal for use in bathrooms or when exposed to sweat, oils, or sanitiser. It's also skin-safe and easily dyeable in different colours.

PLA (to be replaced with PP)

A thin 2mm PLA layer sits beneath the silicone to add structural support during button presses. Without it, the silicone would flex without actuating. While PLA is used for prototyping, production will use injection-moulded polypropylene (PP), a lightweight, cost-effective plastic that's heat- and chemical-resistant—ideal for electronics.

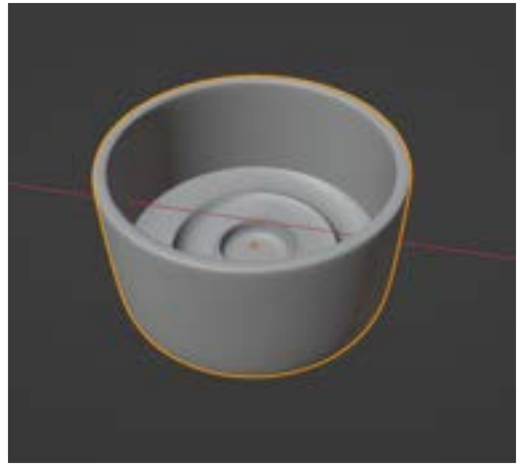
SLA Resin (to be replaced with PP)

The electronics base is made from translucent SLA resin, allowing the LED light to shine through while keeping components discreet. In final production, this will be replaced with translucent PP, which offers the same visual effect with better durability and suitability for mass manufacturing.

Product Making: Silicone Making

Mould Making: Modelling & Printing

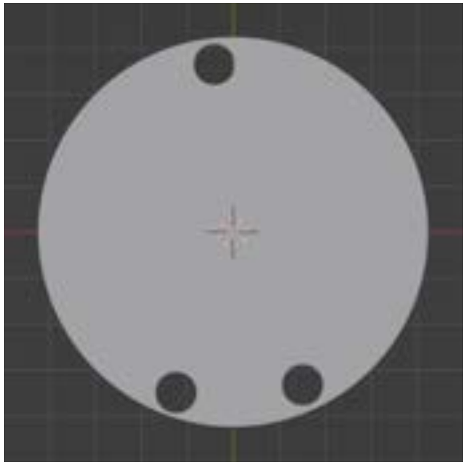
To prototype the silicone button interface, I first needed to create a mould to pour the liquid silicone into so it could take the desired shape. I used Blende to model the mould and give it the appropriate form. After modelling, I printed it using a 3D printer. This stage required several iterations of modelling and printing until I achieved the correct size, form, and print settings. The goal was to ensure a smooth finish that would prevent any unwanted texture from transferring onto the silicone.

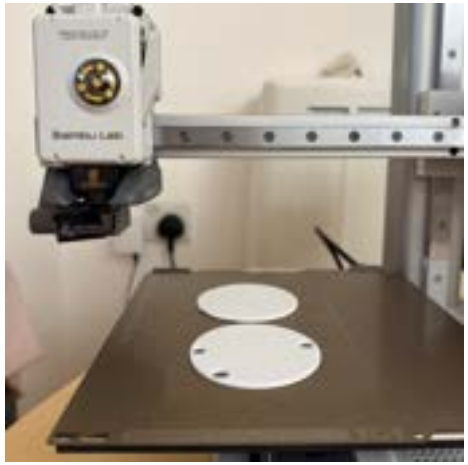


Silicone Casting: Experiments

After I achieved the desired moulds, I began experimenting. First, I mixed the two-part silicone rubber and poured it into the moulds to test whether the results matched expectations. I also experimented with silicone dyes to see if I could achieve different colours. Once I confirmed the results were satisfactory, I worked on embedding the buttons. This was challenging—placing them at the start of curing caused tearing at the bottom, while adding them midway still led to tearing at the back. Eventually, I found a solution: I suspended the buttons using their jumper wires, gluing the wires in place to hang them at the correct height. This prevented contact with the mould and allowed the silicone to cure properly around them.

Product Making: PLA Sheet & SLA Base





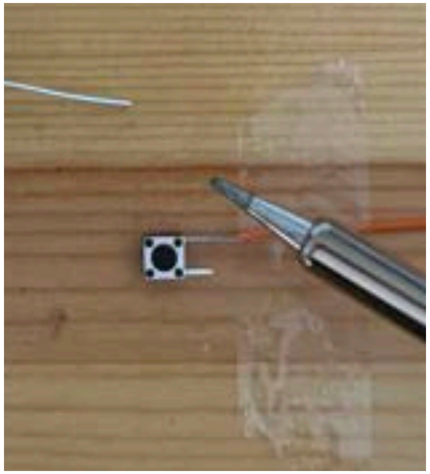
PLA Sheet: Modelling & Printing

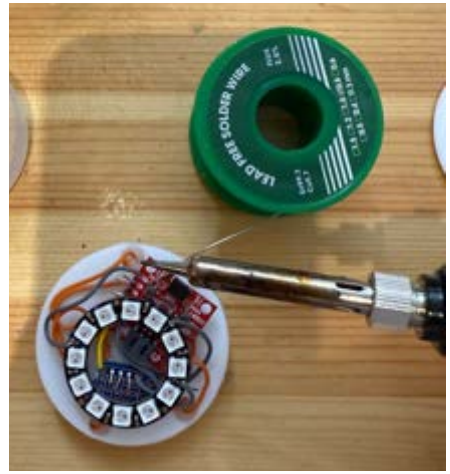
Due to the softness of the silicone button interface, it was difficult to make the buttons pressable unless the silicone was consistently placed on a hard surface. Since the bottom part is hollow, I needed to add a rigid layer in the middle for support. I decided to model, and 3D print a thin PLA layer for this purpose. Its thickness is 2 mm to keep it discreet and help it blend in well. It was also printed in the same colour as the silicone to create a seamless appearance. It was important to correctly place three holes for the button wires to pass through so they could connect to the electronics beneath.

SLA Base : Charging Port & Attachment Mechanism

The last part to design was the base of the product, where all the electronics would be housed. I began by modelling the form in Blender. When designing the base, it was important to consider factors such as how the device would be charged and how it would be attached, to ensure the necessary features were included. For charging, I measured the tip of a USB-C cable and created a small hole in the base so the cable could easily fit for charging. For attachment, I considered multiple methods—such as adding a clip or a band—but ultimately decided to add holes at the top of the product to allow it to be hung using a keychain or hook. Once the modelling was complete, I printed a few test versions in PLA to confirm the dimensions and fit. After validating the design, I sent it to be printed in translucent SLA resin to achieve a clean, high-quality finish. I chose resin because transparent PLA does not produce equally good results, and it was important for the base to be transparent so the light could shine through.

Product Making: Soldering & Assembly





Soldering

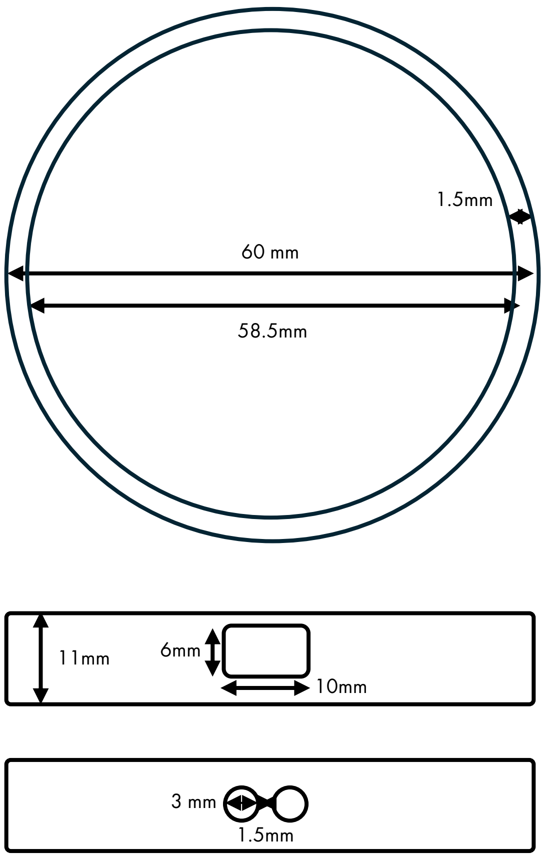
Once all the components were ready, it was time to solder the electronics. The first elements to be soldered were the buttons, which had to be connected before the silicone was moulded so they could be placed in. After that, all the remaining components were soldered together, and a small PCB breadboard was used to connect the ground wires.

Assembly: Gluing Components

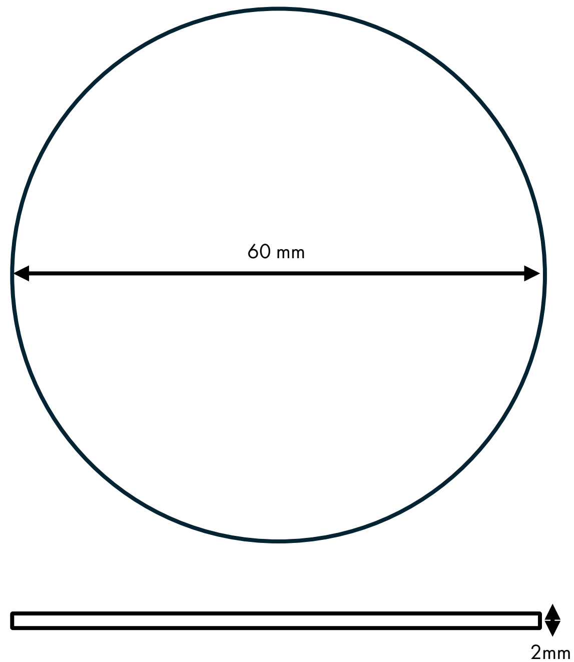
The final step was to glue everything together. After first testing that all the electronics were working correctly and confirming that no further access to them would be necessary, I decided to permanently assemble the parts using epoxy glue, as it offers the most secure bond for plastics. I began by attaching the PLA sheet to the silicone interface. While this worked, the results were not ideal since silicone doesn't adhere easily to PLA. I then moved on to attaching that part to the SLA resin base, which bonded very well. Finally, I cleaned any excess resin using isopropyl alcohol and finished the base by sanding it and applying oil to enhance its appearance.

Final Product: Dimensions

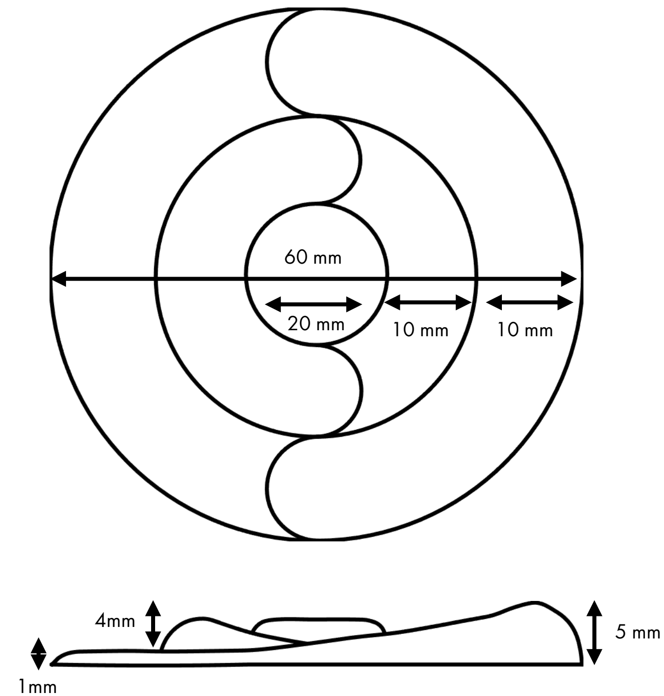
SLA Base



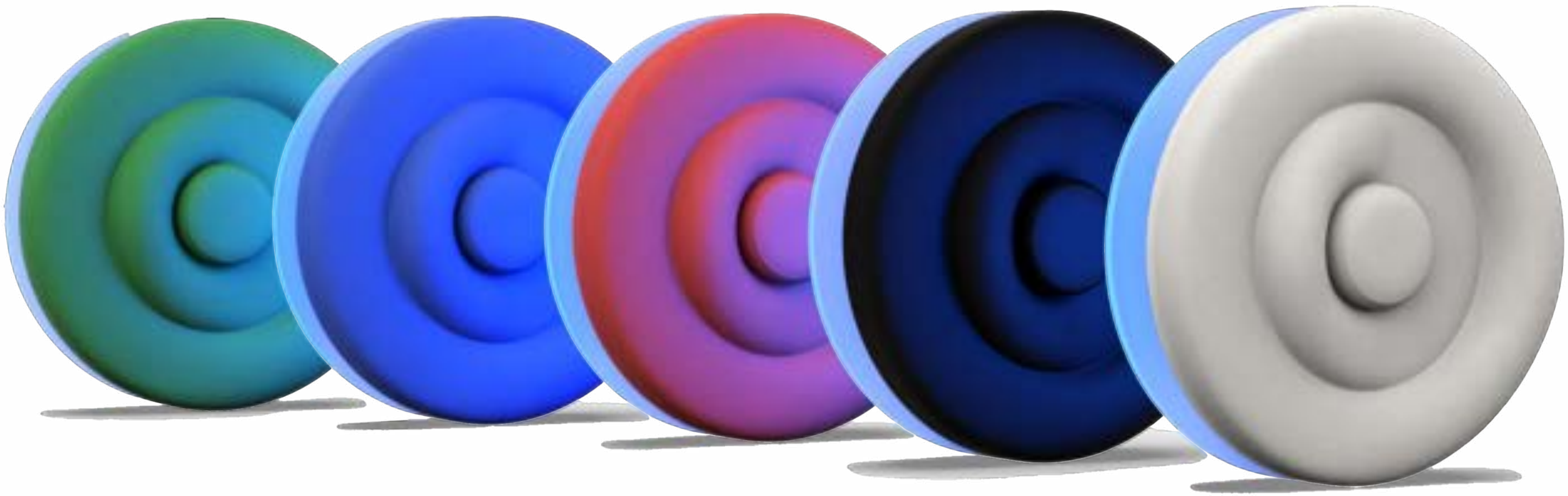
PLA Sheet



Silicone Interface



Final Product: Colour Variations



After finalising Onaka, I decided to create multiple colour variations of the product. Today, users increasingly appreciate having the option to choose between different colours, as it adds a sense of personalisation and helps them feel more connected to the product. Additionally, many users prefer their devices to match—pairing their phones, headphones, and other accessories to create a cohesive set.

With this in mind, I decided to offer more options beyond just white, including black, red, and a selection of other popular primary colours.

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